

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	9
Input Echo:	10
XY Coordinate Calculations:	19
Internal Pressure Calculations:	20
Element: Channel head	21
Element: Channel shell	23
Element: Shell	25
Element: Shell head	28
Shop Pressure Test Results:	31
Field Pressure Test Results:	33
ASME TS Calc: Pressure Test	35
External Pressure Calculations:	52
Element and Detail Weights:	53
Nozzle Flange MAWP:	56
Wind Load Calculation:	57
Earthquake Load Calculation:	60
Center of Gravity Calculation:	62
Saddle Calcs: Bundle Pull New:	63
Saddle Calcs: Bundle Pull Cor:	78
Saddle Calcs: Operating Case:	93
Saddle Calcs: Test Case:	108
Saddle Calcs: Ext. Pressure:	122
Nozzle Calcs.: Manhole	135
Nozzle Calcs.: Tube inlet	140
Nozzle Calcs.: Tube outlet	150
Nozzle Calcs.: Shell Outlet	160
Nozzle Calcs.: Shell Inlet	171
ASME TS Calc: DESIGN CASE	182
Problems/Failures Summary:	220



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\123-C2-NVS.pvdb

Date of Analysis : Apr 10,2026 4:24pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 2 of 220

Vessel Design Summary:

Step: 30 4:24pm Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2600.000 mm ID
Vessel Design Length, Tangent to Tangent 12.42 m
Specified Datum Line Distance 0.00 m
Shell Side Design Temperature 343 °C
Channel Side Design Temperature 370 °C
Shell Side Design Pressure 14.281 N/mm²
Channel Side Design Pressure 18.001 N/mm²
Shop Shell Side Test Pressure 21.247 N/mm²
Field Shell Side Test Pressure 19.216 N/mm²
Shop Channel Side Test Pressure 25.827 N/mm²
Field Channel Side Test Pressure 24.222 N/mm²
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 22	2	...	K21590	
Shell	SA-266 2	...	...	K03506	
Head	SA-387 22	2	...	K21590	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-336 F22	3	...	K21590	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 T22	...	...	K21590	
Tubesheet	SA-336 F22	3	...	K21590	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (N/mm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	18.002	0.00	18.500	3.0000	N/A	No
Channel shell	18.002	0.00	18.300	3.0000	N/A	No
Shell	14.306	0.10	14.600	3.0000	N/A	No
Shell head	14.306	0.10	14.700	3.0000	N/A	No

Liquid Level: 2.60 m Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev m	Element Length m	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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FileName : 123-C2-NVS

Page 3 of 220

Vessel Design Summary: Step: 30 4:24pm Apr 10,2026

Sphere	0.0	0.0	116.0	98.0	95.3	...	1.00	1.00
Cylinder	2.5	2.5	190.0	190.0	187.1	...	1.00	1.00
Cylinder	12.4	9.4	170.0	170.0	166.2	...	1.00	1.00
Sphere	12.4	0.0	92.0	78.0	76.0	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	98	No Calc	...	22.1
20	Tsht	190	No Calc	...	23.3
Tsht	40	190	No Calc	...	No Calc
40	50	170	17.755	0.10131	14.5
50	60	78	6.6625	0.10131	12.9

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.9333	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	9.83336	25400	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube inlet	WRC-297	0.269	Shell	Passed
Tube outlet	WRC-297	0.269	Shell	Passed
Shell Outlet	WRC-297	0.970	Shell	Passed
Shell Inlet	WRC-297	0.970	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	400.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1900.000	mm
Wear Pad Width	300.000	mm
Wear Pad Thickness	30.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1800.000	mm

Baseplate Length	2616.000	mm
Baseplate Thickness	75.000	mm
Baseplate Width	420.000	mm
Number of Ribs (including outside ribs)	7	
Rib Thickness	40.000	mm
Web Thickness	40.000	mm
Height of Center Web	438.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch

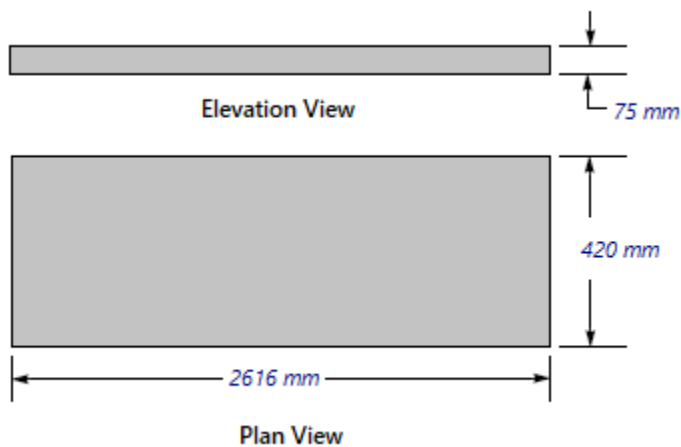
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FileName : 123-C2-NVS

Page 4 of 220

Vessel Design Summary:

Step: 30 4:24pm Apr 10, 2026



Maximum Tensile Bolt Load	27369. N
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Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	4253249.50	N
Maximum Transverse Saddle Shear Load	305847.25	N
Maximum Longitudinal Saddle Shear Load	307588.31	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	4994455.00	N
Maximum Transverse Saddle Shear Load	8420.71	N
Maximum Longitudinal Saddle Shear Load	4118.74	N

Weights:

Fabricated	- Bare w/o removable internals	324918.4	kgm
Shop Test	- Fabricated + water (full)	620116.4	kgm
Shipping	- Fab. + rem. intls.+ shipping app.	324918.4	kgm
Erected	- Fab. + rem. intls.+ insul. (etc)	500919.4	kgm
Empty	- Fab. + intls. + details + wghts.	500919.4	kgm
Operating	- Empty + operating liquid (no ca)	538235.8	kgm
Field Test	- Empty weight + test medium (full)	620116.4	kgm

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FileName : 123-C2-NVS

Page 5 of 220

Nozzle Summary:

Step: 28 4:24pm Apr 10,2026

Nozzle Calculation Summary (N/mm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manhole	4.95	...	...	...		OK	Passed
Tube inlet	4.95	...	...	OK	17.06	OK	Passed
Tube outlet	4.95	...	...	OK	17.06	OK	Passed
Shell Outlet	14.3	OK	...	OK	21.02	OK	Passed
Shell Inlet	14.3	OK	...	OK	21.02	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 14.3 Nozzle : Shell Outlet [Shellside]
 Minimum MAWP Nozzles : 4.95 Nozzle : Manhole [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manhole	0.000	0.000	993.151
20	Tube inlet	1250.000	0.000	1536.000
20	Tube outlet	1250.000	180.000	1536.000
40	Shell Outlet	7515.000	0.000	657.850
40	Shell Inlet	7515.000	180.000	657.850

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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FileName : 123-C2-NVS

Page 6 of 220

Nozzle Schedule:

Step: 27 4:24pm Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Shell Outlet	12.000 in	160	WNF	12.750	33.325	...	...	580.12	1500
Shell Inlet	12.000 in	160	WNF	12.750	33.325	...	...	580.12	1500
Manhole	24.000 in	160	WNF	24.000	59.512	...	...	696.58	1500
Tube inlet	30.000 in Actual		None	36.000	76.200	...	...	988.86	...
Tube outlet	30.000 in Actual		None	36.000	76.200	...	...	910.86	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Shell Outle	SA-266 2	155.000	15.000	...	...	...
Shell Inlet	SA-266 2	155.000	15.000	...	...	...
Manhole	SA-336 F22	50.000	50.000	...	...	...
Tube inlet	SA-336 F22	50.000	50.000	...	...	...
Tube outlet	SA-336 F22	50.000	50.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum m	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Shell Outlet	7.515	0.0	400.00	0.00	Shell
Shell Inlet	7.515	180.0	400.00	0.00	Shell
Manhole	...	0.0	500.00	0.00	Channel head
Tube inlet	1.250	0.0	600.00	0.00	Channel shell
Tube outlet	1.250	180.0	522.00	0.00	Channel shell

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FileName : 123-C2-NVS

Page 7 of 220

Bill of Material:

Step: 26 4:24pm Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 116.0mm THK X 2600.0mm ID X 0.0m	SA-387 22
1	SPHERICAL HEAD: 92.0mm THK X 2600.0mm ID X 0.0m	SA-516 70
16	BAFFLES: 2590.0mm DIA X 19.0mm THK	SA-516 70
16	TIE RODS: 16mm DIA X 9000mm	SA-193 B7
1	TUBESHEET: 2980.0mm X 515.0mm THK	SA-336 F22
5328	TUBES: 9000.0mm X 25.4mm DIA X 2.8mm THK	SA-213 T22
2	SADDLE: 400.0mm X 120 DEG	SA-516 70
1	CLASS 1500 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
2	CYLINDRICAL SEGMENT 190.0mm THK X 3244.5mm X 2.5m	SA-387 22
1	CYLINDRICAL SEGMENT 190.0mm THK X 1679.0mm X 2.5m	SA-387 22
2	CYLINDRICAL SEGMENT 170.0mm THK X 3144.7mm X 9.4m	SA-266 2
1	CYLINDRICAL SEGMENT 170.0mm THK X 1878.6mm X 9.4m	SA-266 2
1	NAMEPLATE	...

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FileName : 123-C2-NVS

Page 8 of 220

MDMT Summary:

Step: 29 4:24pm Apr 10,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell head	[9]	B	26	21		0.908	78.000	1.00	Yes
Left Saddle	[24]	B	-13						No
Right Saddle	[24]	B	-13						No
Shell Outlet	[1]	B	23	2		0.616	29.159	1.00	Yes
Nozzle Flg	[5]	A	-18	-44					
Shell Inlet	[1]	B	23	2		0.616	29.159	1.00	Yes
Nozzle Flg	[5]	A	-18	-44					
Channel head	[9]	A	47	-104		0.247	98.000	1.00	Yes
Manhole	[1]	A	37	-104		0.173	52.073	1.00	Yes
Nozzle Flg	[5]	A	-18	-104					
Warmest MDMT:			47	-104					

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 123-C2-NVS

Page 9 of 220

Warnings and Errors:

Step: 0 4:24pm Apr 10,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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FileName : 123-C2-NVS

Page 10 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	0.3048	m
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	0.0006	m ²
Moment	ft. lbs.	1.3563	N-m
Stress	lbs./sq.in.	6.8946	kPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	N/mm ²
Elast. Modulus	lbs./sq.in.	0.0069	N/mm ²
Pipe Density	lbs./cu.in.	27680.0000	kg/m ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	0.3048	m
Volume	in.3	0.1600E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	14.281	N/mm ²
Channel Side Design Pressure	18.001	N/mm ²
Shell Side Design Temperature	343.0	°C
Channel Side Design Temperature	370.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	14.28	N/mm ²
User defined MAWP, Channel Side	4.95	N/mm ²
User defined MAPnc, Shell Side	0	N/mm ²
User defined MAPnc, Channel Side	0	N/mm ²
User defined Test Pres., Shell Side	0	N/mm ²
User defined Test Pres., Channel Side	0	N/mm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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FileName : 123-C2-NVS

Page 11 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Overall M.A.W.P. + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 12 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 Yes

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4	
Element From Node	10	
Element To Node	20	
Element Type	Spherical	
Description	Channel head	
Distance "FROM" to "TO"	0	m
Inside Diameter	2600	mm
Element Thickness	98	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	116	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18.001	N/mm ²
Design Temperature Internal Pressure	370	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	24	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 22	
Allowable Stress, Ambient	137997	kPa
Allowable Stress, Operating	128860	kPa
Allowable Stress, Hydrotest	278979	kPa
Material Density	7750.4	kg/m ³
P Number Thickness	0	mm
Yield Stress, Operating	245585	kPa
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K21590	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	2.6	m
Liquid Density	43.98	kg/m ³
Element From Node	10	
Detail Type	Nozzle	



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 13 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Detail ID	Manhole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	49412 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F22

Element From Node	10
Detail Type	Weight
Detail ID	davit weight
Dist. from "FROM" Node / Offset dist	-1.3 m
Miscellaneous Weight	2941.8 N
Offset from Element Centerline	0 mm

Element#	2/4
Element From Node	20
Element To Node	40
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	2.5 m
Inside Diameter	2600 mm
Element Thickness	190 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	190 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	370 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	24 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 22
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	2.6 m
Liquid Density	43.98 kg/m ³

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube inlet
Dist. from "FROM" Node / Offset dist	1.25 m
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	40476 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F22

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube outlet
Dist. from "FROM" Node / Offset dist	1.25 m



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FileName : 123-C2-NVS

Page 14 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	39287	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	

Element From Node	20	
Detail Type	Weight	
Detail ID	Pass Partition	
Dist. from "FROM" Node / Offset dist	1.75	m
Miscellaneous Weight	15690	N
Offset from Element Centerline	0	mm

Element#	3/4	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	9.4	m
Inside Diameter	2600	mm
Element Thickness	170	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	170	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	14.281	N/mm ²
Design Temperature Internal Pressure	343	°C
Design External Pressure	0.1013	N/mm ²
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	137990	kPa
Allowable Stress, Operating	122834	kPa
Allowable Stress, Hydrotest	224983	kPa
Material Density	7750.4	kg/m ³
P Number Thickness	32.004	mm
Yield Stress, Operating	184392	kPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1.8	m
Width of Saddle	400	mm
Height of Saddle at Bottom	1900	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	300	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 15 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Tangent to Tangent dist. (optional)	0	m
Element From Node	40	
Detail Type	Saddle	
Detail ID	Right Saddle	
Dist. from "FROM" Node / Offset dist	7.2	m
Width of Saddle	400	mm
Height of Saddle at Bottom	1900	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	300	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	2.6	m
Liquid Density	999.54	kg/m³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell Outlet	
Dist. from "FROM" Node / Offset dist	4.5	m
Nozzle Diameter	12	in.
Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8744.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell Inlet	
Dist. from "FROM" Node / Offset dist	4.5	m
Nozzle Diameter	12	in.
Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8744.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1(ETY HALF	
Dist. from "FROM" Node / Offset dist	1.8	m
Miscellaneous Weight	850185	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1(ETY HALF	
Dist. from "FROM" Node / Offset dist	7.2	m
Miscellaneous Weight	850185	N



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 16 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Log. baffle	
Dist. from "FROM" Node / Offset dist	4.5	m
Miscellaneous Weight	53933	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Fireproofing	
Dist. from "FROM" Node / Offset dist	1.8	m
Miscellaneous Weight	4903	N
Offset from Element Centerline	1900	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Fireproofing 1	
Dist. from "FROM" Node / Offset dist	7.2	m
Miscellaneous Weight	4903	N
Offset from Element Centerline	1900	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1(OPR HALF	
Dist. from "FROM" Node / Offset dist	1.8	m
Miscellaneous Weight	897636	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1(OPR HALF	
Dist. from "FROM" Node / Offset dist	7.2	m
Miscellaneous Weight	897636	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1(HYDHALF)	
Dist. from "FROM" Node / Offset dist	1.8	m
Miscellaneous Weight	1129866	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1(HYD HALF	
Dist. from "FROM" Node / Offset dist	7.2	m
Miscellaneous Weight	1129866	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Upper Saddle wei	
Dist. from "FROM" Node / Offset dist	1.8	m
Miscellaneous Weight	19612	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Upper Saddle wei	
Dist. from "FROM" Node / Offset dist	7.2	m
Miscellaneous Weight	19612	N
Offset from Element Centerline	0	mm



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 17 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Element From Node	40
Detail Type	For./Mom.
Detail ID	Left SD wind &
Dist. from "FROM" Node / Offset dist	1.8 m
Force in X Direction	130238 N
Force in Y Direction	0 N
Force in Z Direction	130238 N
Moment about X Axis	0 N-m
Moment about Y Axis	0 N-m
Moment about Z Axis	0 N-m
Force/Moment Combination Method	SRSS

Element From Node	40
Detail Type	For./Mom.
Detail ID	Right SD wind &
Dist. from "FROM" Node / Offset dist	7.2 m
Force in X Direction	130238 N
Force in Y Direction	0 N
Force in Z Direction	130238 N
Moment about X Axis	0 N-m
Moment about Y Axis	0 N-m
Moment about Z Axis	0 N-m
Force/Moment Combination Method	SRSS

Element From Node	40
Detail Type	For./Mom.
Detail ID	Left SD wind &
Dist. from "FROM" Node / Offset dist	1.8 m
Force in X Direction	6559.4 N
Force in Y Direction	0 N
Force in Z Direction	21814 N
Moment about X Axis	0 N-m
Moment about Y Axis	0 N-m
Moment about Z Axis	0 N-m
Force/Moment Combination Method	SRSS

Element From Node	40
Detail Type	For./Mom.
Detail ID	Right SD wind &
Dist. from "FROM" Node / Offset dist	7.2 m
Force in X Direction	6559.4 N
Force in Y Direction	0 N
Force in Z Direction	21814 N
Moment about X Axis	0 N-m
Moment about Y Axis	0 N-m
Moment about Z Axis	0 N-m
Force/Moment Combination Method	SRSS

Element#	4/4
Element From Node	50
Element To Node	60
Element Type	Spherical
Description	Shell head
Distance "FROM" to "TO"	0 m
Inside Diameter	2600 mm
Element Thickness	78 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	92 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.281 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	200 °C



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FileName : 123-C2-NVS

Page 18 of 220

Input Echo:

Step: 1 4:24pm Apr 10,2026

Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	137990 kPa
Allowable Stress, Operating	129115 kPa
Allowable Stress, Hydrotest	233982 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	32.004 mm
Yield Stress, Operating	194672 kPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	2.6 m
Liquid Density	999.54 kg/m ³

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FileName : 123-C2-NVS

Page 19 of 220

XY Coordinate Calculations: Step: 2 4:24pm Apr 10,2026

XY Coordinate Calculations:

From	To	X (Horiz.) m	Y (Vert.) m	DX (Horiz.) m	DY (Vert.) m
Channel head		...	...	...	...
Channel shell		2.5	...	2.5	...
Shell		12.415	...	9.4	...
Shell head		12.415	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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FileName : 123-C2-NVS

Page 20 of 220

Internal Pressure Calculations: Step: 3 4:24pm Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd N/mm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kPa	Element#
Channel head		18.002	116	3	2600	128860	1
Channel shell		18.002	190	3	2600	138191	2
Shell		14.306	170	3	2600	122834	3
Shell head		14.306	92	3	2600	129115	4

Element Required Thickness and MAWP :

From	To	Design Pressure N/mm ²	M.A.W.P. Corroded N/mm ²	M.A.P. New & Cold N/mm ²	Minimum Thickness mm	Required Thickness mm	Element#
Channel head		18.001	18.5	20.5	98	95.3006	1
Channel shell		18.001	18.3	19.9	190	187.123	2
Shell		14.2808	14.6	16.7	170	166.151	3
Shell head		14.2808	14.7	16.4	78	75.9929	4

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 14.3 N/mm²
 Shell Side MAPnc = 16.4 N/mm²
 Channel Side MAWP = 4.95 N/mm², < design pressure: 18.001 N/mm²
 Channel Side MAPnc = 19.9 N/mm²

Elements Suitable for Design Internal Pressure.

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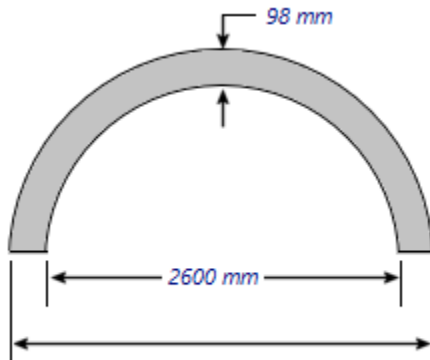
Page 21 of 220

Element: Channel head

Step: 4 4:24pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel head

Spherical Head Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	370 °C
Material	SA-387 22	Ext. Design Temperature	24 °C
Length	0.000 m	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	128860.3 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137996.8 kPa
Inside Diameter Di	2600.000 mm	Actual Stress	125249.9 kPa
Spec. Min. Thk t _e , or s _a	98.0000 mm	Spec. Nominal Thk.	116.000 mm
Surface Area	12.6 m ²	SG of Contents	0.044
Element Volume	4.5 m ³	Weight of Contents	1984.5 N
Empty Weight	102217.0 N	Operating Weight	104201.5 N
		Comp. All. Ope. Stress	99287.1 kPa
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 32.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.250 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (18 \cdot 1303) / (2 \cdot 128860 \cdot 1 - 0.2 \cdot 18)$$

$$= 92.3006 + 3.0000 = 95.3006 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 128860 \cdot 1 \cdot 95) / (1303 + 0.2 \cdot 95)$$

$$= 18.5 - 0.0011 = 18.5 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$



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FileName : 123-C2-NVS

Page 22 of 220

Element: Channel head

Step: 4 4:24pm Apr 10,2026

$$= (2 \cdot 137997 \cdot 1 \cdot 98) / (1300 + 0.2 \cdot 98)$$

$$= 20.5 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (18 \cdot (1303 + 0.2 \cdot 95)) / (2 \cdot 1 \cdot 95)$$

$$= 125249.938 \text{ kPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.406 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govern. thk, $t_g = 98$, $t_r = 23.5$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 47 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4.950 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	98.000 mm	Basic MDMT	47 °C
Stress/Thk. Ratio	0.247	Temperature Difference	151 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 32.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.250 mm

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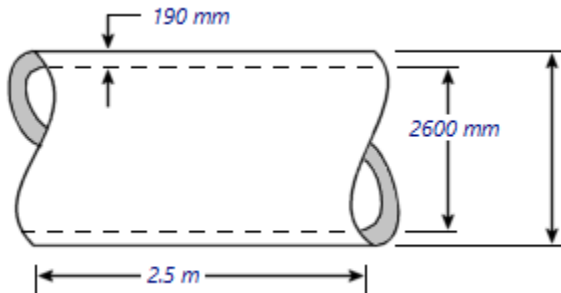
Page 23 of 220

Element: Channel shell

Step: 5 4:24pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	370 °C
Material	SA-387 22	Ext. Design Temperature	24 °C
Length	2.500 m	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138190.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2600.000 mm	Actual Stress	136231.0 kPa
Spec. Min. Thk t _e , or sa	190.0000 mm	Spec. Nominal Thk.	190.000 mm
Surface Area	23.4 m ²	SG of Contents	0.044
Element Volume	13.0 m ³	Weight of Contents	5724.5 N
Empty Weight	316424.0 N	Operating Weight	322148.5 N
		Comp. All. Ope. Stress	108270.3 kPa
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 1303) / (138191 \cdot 1 - 0.6 \cdot 18) \\
 &= 184.1232 + 3.0000 = 187.1232 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138191 \cdot 1 \cdot 187) / (1303 + 0.6 \cdot 187) \\
 &= 18.3 - 0.0011 = 18.3 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147989 \cdot 1 \cdot 190) / (1300 + 0.6 \cdot 190) \\
 &= 19.9 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (1303 + 0.6 \cdot 187)) / (1 \cdot 187) \\
 &= 136231.000 \text{ kPa}
 \end{aligned}$$



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FileName : 123-C2-NVS

Page 24 of 220

Element: Channel shell

Step: 5 4:24pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.810 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4.950 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	190.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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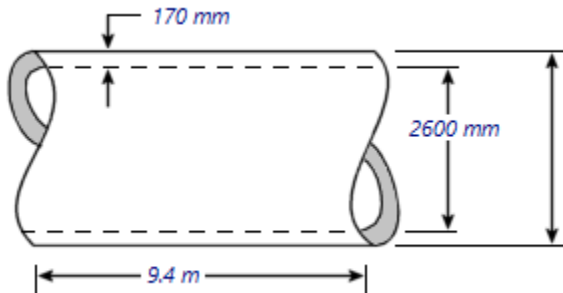
Page 25 of 220

Element: Shell

Step: 6 4:24pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	14.306 N/mm ²	Int. Design Temperature	343 °C
Material	SA-266 2	Ext. Design Temperature	200 °C
Length	9.400 m	External Pressure Chart	CS-2
		UNS Number	K03506
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	122834.1 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	2600.000 mm	Actual Stress	120200.6 kPa
Spec. Min. Thk t,e, or sa	170.0000 mm	Spec. Nominal Thk.	170.000 mm
Surface Area	86.8 m ²	SG of Contents	1.000
Element Volume	21.0 m ³	Weight of Contents	210342.9 N
Empty Weight	1056886.0 N	Operating Weight	1267228.9 N
		Comp. All. Ope. Stress	112187.7 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (14.3 \cdot 1303) / (122834 \cdot 1 - 0.6 \cdot 14.3) \\
 &= 163.1506 + 3.0000 = 166.1506 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.026 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (122834 \cdot 1 \cdot 167) / (1303 + 0.6 \cdot 167) \\
 &= 14.6 - 0.026 = 14.6 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 170) / (1300 + 0.6 \cdot 170) \\
 &= 16.7 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (14.3 \cdot (1303 + 0.6 \cdot 167)) / (1 \cdot 167) \\
 &= 120200.602 \text{ kPa}
 \end{aligned}$$



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FileName : 123-C2-NVS

Page 26 of 220

Element: Shell

Step: 6 4:24pm Apr 10,2026

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	14.280 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	170.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 9833 / \sqrt{(1470 \cdot 167)}$$

$$= 19.847$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 13 \text{ and } Mx < 2 \cdot (Do/t)^{0.94}, Ch = 1.12 \cdot Mx^{-1.058}$$

$$= 1.12 \cdot 19.8^{-1.058}$$

$$= 0.0475$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.047 \cdot 191213 (167/2940)$$

$$= 824609.688 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 824609.7/212995.9$$

$$= 3.871$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = Fy$$

$$= 212995.906 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [212996] \geq Fy [212996], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = Fy/FS$$

$$= 212995.906/1.667$$

$$= 127771.977 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 128 (167/2940)$$

$$= 14.516 \text{ N/mm}^2$$



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FileName : 123-C2-NVS

Page 27 of 220

Element: Shell

Step: 6 4:24pm Apr 10,2026

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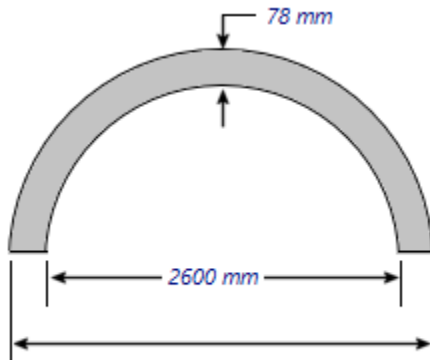
Page 28 of 220

Element: Shell head

Step: 7 4:24pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell head

Spherical Head Design Information:			
Design Pressure P	14.306 N/mm ²	Int. Design Temperature	343 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	0.000 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	129114.8 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	2600.000 mm	Actual Stress	125697.8 kPa
Spec. Min. Thk t _e , or sa	78.0000 mm	Spec. Nominal Thk.	92.000 mm
Surface Area	12.2 m ²	SG of Contents	1.000
Element Volume	4.5 m ³	Weight of Contents	45101.7 N
Empty Weight	79624.9 N	Operating Weight	124726.6 N
		Comp. All. Ope. Stress	112187.7 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 32.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.250 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (14.3 \cdot 1303) / (2 \cdot 129115 \cdot 1 - 0.2 \cdot 14.3)$$

$$= 72.9929 + 3.0000 = 75.9929 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 N/mm²

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 129115 \cdot 1 \cdot 75) / (1303 + 0.2 \cdot 75)$$

$$= 14.7 - 0.026 = 14.7 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$



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FileName : 123-C2-NVS

Page 29 of 220

Element: Shell head

Step: 7 4:24pm Apr 10,2026

$$= (2 \cdot 137990 \cdot 1 \cdot 78) / (1300 + 0.2 \cdot 78)$$

$$= 16.4 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (14.3 \cdot (1303 + 0.2 \cdot 75)) / (2 \cdot 1 \cdot 75)$$

$$= 125697.797 \text{ kPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.126 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govern. thk, $t_g = 78$, $t_r = 68.1$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.91$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 26 °C

Min Metal Temp. at Required thickness (UCS 66.1) 21 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	14.280 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	78.000 mm	Basic MDMT	26 °C
Stress/Thk. Ratio	0.908	Temperature Difference	5 °C
UCS-66(c) red.	No	Computed Min. Temp.	21 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 32.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.250 mm

EXTERNAL PRESSURE RESULTS for: Shell head

Spherical Head From 50 to 60 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per Code Case 2286:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 191213 (75 / 1378)$$

$$= 780486.688 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 780486.7 / 224995.5$$

$$= 3.469$$

Predicted Buckling Stress [Fic]:

$$\text{Since } F_r > 1.6 \text{ and } F_r < 6.25, F_{ic} = 1.31 \cdot F_y / (1.15 + S_y / F_{he})$$

$$= 1.31 \cdot 224996 / (1.15 + 224996 / 780487)$$

$$= 204928.781 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$



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FileName : 123-C2-NVS

Page 30 of 220

Element: Shell head

Step: 7 4:24pm Apr 10,2026

$$\begin{aligned} &= 2.407 - 0.741 \cdot 204929/224996 \\ &= 1.732 \end{aligned}$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned} &\text{Since } Fr > 1.6 \text{ and } Fr < 6.25, Fha = 1.31 \cdot Fy / (FS \cdot (1.15 + Sy/Fhe)) \\ &= 1.31 \cdot 224996 / (1.73 \cdot (1.15 + 224996/780487)) \\ &= 118313.164 \text{ kPa} \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$\begin{aligned} &= 2 \cdot Fha[t/Ro] \\ &= 2 \cdot 118313 [75/1378] \\ &= 12.880 \text{ N/mm}^2 \end{aligned}$$

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FileName : 123-C2-NVS

Page 31 of 220

Shop Pressure Test Results: Step: 8 4:24pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	19.215 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	19.216 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	21.247 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	16.259 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	20.422 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.422 N/mm ²
App 27-4	= M.A.W.P.	14.280 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.661 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	24.222 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.827 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	5.636 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	25.741 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.741 N/mm ²
App 27-4	= M.A.W.P.	4.950 N/mm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel head	174045.4	278978.8	0.624	25.85
Channel shell	192385.4	278978.8	0.690	25.85
Shell	175424.1	224982.9	0.780	21.27
Shell head	179386.4	233982.2	0.767	21.27

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	147988.73	138190.73	1.071
Tube inlet	Nozzle	147988.73	138190.73	1.071
Tube outlet	Nozzle	147988.73	138190.73	1.071
Shell Outlet	Nozzle	137989.50	122834.11	1.123
Shell Inlet	Nozzle	137989.50	122834.11	1.123

Minimum 1.071

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
Channel head	137996.80	128860.33	1.071
Channel shell	147988.73	138190.73	1.071
Shell	137989.50	122834.11	1.123
Shell head	137989.50	129114.78	1.069

Minimum 1.069

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
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FileName : 123-C2-NVS

Page 32 of 220

Shop Pressure Test Results: Step: 8 4:24pm Apr 10, 2026

Tube Material	117991.02	113991.31	1.035
Tubesheet Material	147988.73	138190.73	1.071

Minimum 1.035

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manhole	122058.77	278978.75	0.438
Tube inlet	144765.20	278978.75	0.519
Tube outlet	144765.20	278978.75	0.519
Shell Outlet	94847.73	224982.88	0.422
Shell Inlet	94847.73	224982.88	0.422

Elements Suitable for Test Pressure.

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FileName : 123-C2-NVS

Page 33 of 220

Field Pressure Test Results: Step: 9 4:24pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	19.215 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	19.216 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	21.247 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	16.259 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	20.422 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.422 N/mm ²
App 27-4	= M.A.W.P.	14.280 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.661 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	24.222 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.827 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	5.636 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	25.741 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.741 N/mm ²
App 27-4	= M.A.W.P.	4.950 N/mm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 14.3 • 1.04

= 19.216 N/mm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 18 • 1.04

= 24.222 N/mm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel head	168704.9	278978.8	0.605	24.25
Channel shell	183495.8	278978.8	0.658	24.25
Shell	161669.3	224982.9	0.719	19.24
Shell head	169063.0	233982.2	0.723	19.24

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	147988.73	138190.73	1.071
Tube inlet	Nozzle	147988.73	138190.73	1.071
Tube outlet	Nozzle	147988.73	138190.73	1.071
Shell Outlet	Nozzle	137989.50	122834.11	1.123
Shell Inlet	Nozzle	137989.50	122834.11	1.123

Minimum

1.071

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



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FileName : 123-C2-NVS

Page 34 of 220

Field Pressure Test Results: Step: 9 4:24pm Apr 10, 2026

Channel head	137996.80	128860.33	1.071
Channel shell	147988.73	138190.73	1.071
Shell	137989.50	122834.11	1.123
Shell head	137989.50	129114.78	1.069

Minimum 1.069

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	117991.02	113991.31	1.035
Tubesheet Material	147988.73	138190.73	1.071

Minimum 1.035

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manhole	114483.41	278978.75	0.410
Tube inlet	135780.64	278978.75	0.487
Tube outlet	135780.64	278978.75	0.487
Shell Outlet	85794.85	224982.88	0.381
Shell Inlet	85794.85	224982.88	0.381

Elements Suitable for Test Pressure.

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FileName : 123-C2-NVS

Page 35 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		19.22 N/mm ²
Shell Thickness	ts	170.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2600.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03506
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Test Pressure		24.22 N/mm ²
Channel Thickness	tc	190.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2600.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	5328
Tube Wall Thickness	et	2.7690 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	9000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8485.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		117991.02 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	204984.41 kPa
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1255.670 mm
Largest Center-to-Center Tube Distance	U1	101.6000 mm
Length of Expanded Portion of Tube	ltx	300.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	4.8 m ²
Perimeter of Tube Layout	Cp	8067.5 mm



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FileName : 123-C2-NVS

Page 36 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K21590
Tubesheet Allowable Stress at Temperature S 147988.73 kPa
Tubesheet Allowable Stress at Ambient Tt 147988.73 kPa
Thickness of Tubesheet h 515.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 2980.000 mm

Area of the Untubed Lanes AL 0.198 m²

Junction Stress Reduction option Perform Plastic Calculation

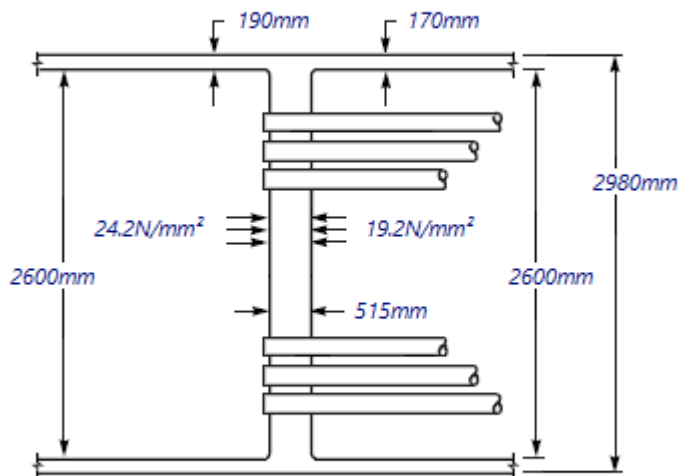
Perform Differential Pressure Design No

Run Multiple Load Cases Yes

Tubesheet Integral with Both

Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20133E+06 N/mm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20133E+06 N/mm²

Channel - TM-1 Material Group D



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FileName : 123-C2-NVS

Page 37 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10,2026

Elastic Mod. at Design Temperature	21.1 °C	0.21099E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.21099E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 °C	0.21099E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(24.2 \cdot (25.4/2 - 0)) / (117991 \cdot 1 + 0.4 \cdot 24.2)$

= $2.4092 + 0.0000 = 2.4092$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$

= $9741 / \sqrt{(12.7 \cdot 2.77)}$

= 1642.576

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)0.94$, $Ch = 0.55 / (Do/t)$

= $0.55 \cdot (25.4/2.77)$

= 0.0600

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$

= $1.6 \cdot 0.06 \cdot 210987 (2.77/25.4)$

= 2206438.500 kPa

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy

= $2206438.5/204984.4$

= 10.764

Predicted Buckling Stress [Fic]:

Since $Fr \geq 2.439$, $Fic = Fy$

= 204984.406 kPa

Factor of Safety [FS]:

Since $Fic [204984] \geq Fy[204984]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = Fy/FS$

= $204984.406/1.667$

= 122966.039 kPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha(t / Do)$

= $2 \cdot 123 (2.77/25.4)$

= 26.812 N/mm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.4092 mm
Allowable Internal Pressure at Corroded thickness	28.19 N/mm ²
Required Internal Design Pressure	24.22 N/mm ²
Allowable External Pressure at Corroded thickness	26.81 N/mm ²



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FileName : 123-C2-NVS

Page 38 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Required External Design Pressure	19.22 N/mm ²
Required Thickness due to Shell Side pressure	1.9846 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1256 + 25.4$$

$$= 2536.740 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 300 / 515$$

$$= 0.583 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 2.77 (210987/210987) \cdot (117991/147989) \cdot (0.58); 25.4 - 2 \cdot 2.77)$$

$$= 22.8279 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.8 / \sqrt{ 1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2) }$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.8) / 32.4$$

$$= 0.2952$$

$h/p = 16.220472$; $\mu^* = 0.295244$; $E^* = 58005 \text{ N/mm}^2$

$E^*/E = 0.274924$; $v^* = 0.357158$

Note: As h/p (16.220) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 39 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2600 + 170)170} \right)$$

$$= 0.0026 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0026 \cdot 201334 \cdot 170^3 / (6(1 - 0.3^2))$$

$$= 479912992.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2600 + 170) \cdot 479912992 / 515^3 \cdot (1 + 515 \cdot 0.0026 + (515 \cdot 0.0026)^2 / 2)$$

$$= 192402.4 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2600^2 / (4 \cdot 201334 \cdot 170) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 170)))$$

$$= 0.42427\text{E-}04 \text{ mm/kPa}$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 479912992 \cdot 0.0026 \cdot 0.42427\text{E-}04 (1 + 515 \cdot 0.0026)$$

$$= 130696.8438 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2600 + 190)190} \right)$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0025 \cdot 210987 \cdot 190^3 / (6(1 - 0.3^2))$$

$$= 661760256.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2600 + 190 / \cos(0)) \cdot 661760256 / 515^3 \cdot (1 + 515 \cdot 0.0025 \cdot (1 + 1) / 2 + (515 \cdot 0.0025)^2 / 2 \cdot 1)$$

$$= 252432.7 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2600^2 / (4 \cdot 210987 \cdot 190) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 190)))$$

$$= 0.36267\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 661760256 [0.36267\text{E-}04 (1 + 0.0025 \cdot 1 \cdot 515) - 0.000 (1/515 + 1 \cdot 0.0025/2)]$$

$$= 140379.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 40 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

$$= 2980/2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 58005 \cdot (192402 + 252433 + 210987 \cdot \ln(1.17))$$

$$= 5.3064$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 2537^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 0 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 24.2]$$

$$= -498152.7188 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -498153 + 140379 \cdot 24.2 - 130697 \cdot 0$$

$$= 2902182.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - Do^2/32 \cdot F(P_s - P_t) / (1 + F)$$

$$= (2902182) - 2537^2/32 \cdot 5.31(0 - 24.2) / (1 + 5.31)$$

$$= 4558827.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + Do^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 4558827 + 2537^2/64 \cdot (3 + 0.36) \cdot (0 - 24.2)$$

$$= -3617571.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|4558827|; |-3617571|)$$

$$= 4558827.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 4558827 / (0.3 (515 - 5)^2)$$

$$= 356170.2812 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 309976$$

$$= 588955.12 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 4615607 / (0.32 (388 - 5)^2)$$

$$= 588438.8125 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 388 + 0 + 0$$

$$= 387.7015 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(388; 305)$$

$$= 387.7015 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 41 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |0 - 24.2|$$

$$= 24.222 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.2 \cdot 515/2537 \cdot 0.8 \cdot 0.95 \cdot 309976$$

$$= 38261.625 \text{ kPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1/(4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p/C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1/(4 \cdot 0.2 \cdot 235582) \cdot (4 \cdot 4.78/8067) \cdot |0 - 24.2| + 0$$

$$= 304.5701 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 170.000mm

Min. Shell len. adjacent to the tubesheet is: 1196.695 mm

Note: For a given Channel thickness of: 190.000 mm

Min. Channel len. adjacent to the tubesheet is: 1265.132mm

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 194735$$

$$= 38336.223 \text{ N}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 194735$$

$$= 38336.2227 \text{ N}$$

Weld Strength Factor [f_w]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 194735/194735$$

$$= 1.000$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 194735$$

$$= 27667.8594 \text{ N}$$

Minimum Length of Weld [l_w]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.371 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 42 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10,2026

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm
Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 38336 is >= tube strength 38336 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	356170.3 <=	588955.1 kPa	Ok
Tubesheet shear stress	139322.8 <=	235582.0 kPa	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	387.701	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D2 un-corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 1256 + 25.4$$

$$= 2536.740 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 300 / 515$$

$$= 0.583 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(d_t - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (p); d_t - 2t_t)$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 43 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

$$= \max(25.4 - 2 \cdot 2.77 (210987/210987) \cdot (117991/147989) \cdot (0.58); 25.4 - 2 \cdot 2.77) \\ = 22.8279 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\ = 31.8 / \sqrt{(1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2))} \\ = 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^* \\ = (32.4 - 22.8) / 32.4 \\ = 0.2952$$

$$h/p = 16.220472; \mu^* = 0.295244; E^* = 58005 \text{ N/mm}^2 \\ E^*/E = 0.274924; \nu^* = 0.357158$$

Note: As h/p (16.220) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4} / \sqrt{(Ds + ts)ts}) \\ = ((12(1 - 0.3^2))^{1/4} / \sqrt{(2600 + 170)170}) \\ = 0.0026 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2)) \\ = 0.0026 \cdot 201334 \cdot 170^3 / (6(1 - 0.3^2)) \\ = 479912992.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2) \\ = 6(2600 + 170) \cdot 479912992/515^3 \cdot (1 + 515 \cdot 0.0026 + (515 \cdot 0.0026)^2/2) \\ = 192402.4 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts))) \\ = 2600^2 / (4 \cdot 201334 \cdot 170) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 170))) \\ = 0.42427\text{E-}04 \text{ mm/kPa}$$

Parameter [ws]:

$$= \rho_s \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ = 1.02 \cdot 479912992 \cdot 0.0026 \cdot 0.42427\text{E-}04 (1 + 515 \cdot 0.0026) \\ = 130696.8438 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(Dc + tc)tc}) \\ = ((12(1 - 0.3^2))^{1/4} / \sqrt{(2600 + 190)190}) \\ = 0.0025 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_s^2)) \\ = 0.0025 \cdot 210987 \cdot 190^3 / (6(1 - 0.3^2)) \\ = 661760256.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(Dc + tc/\cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ = 6(2600 + 190/\cos(0)) \cdot 661760256/515^3 \cdot (1 + 515 \cdot 0.0025 \cdot (1 + 1)/2 + (515 \cdot 0.0025)^2/2 \cdot 1) \\ = 252432.7 \text{ N/mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 44 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2600^2 / (4 \cdot 210987 \cdot 190) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 190)))$$

$$= 0.36267E-04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ωc]:

$$= \rho c \cdot k c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 661760256 [0.36267E-04 (1 + 0.0025 \cdot 1 \cdot 515) - 0.000 (1/515 + 1 \cdot 0.0025/2)]$$

$$= 140379.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980/2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 58005 \cdot (192402 + 252433 + 210987 \cdot \ln(1.17))$$

$$= 5.3064$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [M_{ts}]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2537^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 19.2 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 0]$$

$$= 395201.0938 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M^*]:

$$= M_{ts} + \omega c \cdot P_t - \omega s \cdot P_s$$

$$= 395201 + 140379 \cdot 0 - 130697 \cdot 19.2$$

$$= -2116332.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [M_p]:

$$= (M^* - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= (-2116332) - 2537^2 / 32 \cdot 5.31 (19.2 - 0) / (1 + 5.31)$$

$$= -3587165.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= -3587165 + 2537^2 / 64 \cdot (3 + 0.36) \cdot (19.2 - 0)$$

$$= 2899443.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|-3587165|; |2899443|)$$

$$= 3587165.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 3587165 / (0.3 (515 - 5)^2)$$

$$= 280256.6562 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 45 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 309976$$

$$= 588955.12 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 3675585 / (0.34 \cdot (339 - 5)^2)$$

$$= 588105.7500 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 339 + 0 + 0$$

$$= 338.7935 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(339; 242)$$

$$= 338.7935 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |19.2 - 0|$$

$$= 19.216 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.2 \cdot 515 / 2537 \cdot 0.8 \cdot 0.95 \cdot 309976$$

$$= 38261.625 \text{ kPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 235582) \cdot (4 \cdot 4.78 / 8067) \cdot |19.2 - 0| + 0$$

$$= 241.6256 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 170.000mm

Min. Shell len. adjacent to the tubesheet is: 1196.695 mm

Note: For a given Channel thickness of: 190.000 mm

Min. Channel len. adjacent to the tubesheet is: 1265.132mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 194735$$

$$= 38336.223 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 194735$$

$$= 38336.2227 \text{ N}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 46 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Weld Strength Factor [fw]:

= S / S_w where $S_w = \min(S \text{ and } S_t)$

= 194735/194735

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > F_t)

= $0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 194735$

= 27667.8594 N

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 3.371 mm

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 38336 is >= tube strength 38336 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 un-corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	280256.7 <=	588955.1 kPa	Ok
Tubesheet shear stress	110529.4 <=	235582.0 kPa	Ok

Thickness results for loadcase D2 un-corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	338.793	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + d_t$

= $2 \cdot 1256 + 25.4$

= 2536.740 mm

Basic ligament efficiency for shear [μ]:

= $(p - d_t) / p$

= $(31.8 - 25.4) / 31.8$

= 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 47 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2600/2537$$

$$= 1.0249$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2600/2537$$

$$= 1.0249$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 300/515$$

$$= 0.583 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.77 (210987/210987) \cdot (117991/147989) \cdot (0.58); 25.4 - 2 \cdot 2.77)$$

$$= 22.8279 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2)) }$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.8) / 32.4$$

$$= 0.2952$$

$h/p = 16.220472$; $\mu^* = 0.295244$; $E^* = 58005 \text{ N/mm}^2$

$E^*/E = 0.274924$; $\nu^* = 0.357158$

Note: As h/p (16.220) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (2600 + 170) 170 }$$

$$= 0.0026 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0026 \cdot 201334 \cdot 170^3 / (6 (1 - 0.3^2))$$

$$= 479912992.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (2600 + 170) \cdot 479912992/515^3 \cdot (1 + 515 \cdot 0.0026 + (515 \cdot 0.0026)^2 / 2)$$

$$= 192402.4 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 2600^2 / (4 \cdot 201334 \cdot 170) \cdot (1 - 2600 \cdot 0.3 / (2 (2600 + 170)))$$

$$= 0.42427 \text{E-04 mm/kPa}$$

Parameter [ωs]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 479912992 \cdot 0.0026 \cdot 0.42427 \text{E-04} (1 + 515 \cdot 0.0026)$$

$$= 130696.8438 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 48 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} \right) / \sqrt{\left((D_c + t_c) t_c \right)}$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{\left((2600 + 190) 190 \right)}$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / \left(6(1 - v_s^2) \right)$$

$$= 0.0025 \cdot 210987 \cdot 190^3 / \left(6(1 - 0.3^2) \right)$$

$$= 661760256.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2600 + 190 / \cos(0)) \cdot 661760256 / 515^3 \cdot (1 + 515 \cdot 0.0025 \cdot (1 + 1) / 2 + (515 \cdot 0.0025)^2 / 2 \cdot 1)$$

$$= 252432.7 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2600^2 / (4 \cdot 210987 \cdot 190) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 190)))$$

$$= 0.36267\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 661760256 [0.36267\text{E-}04 (1 + 0.0025 \cdot 1 \cdot 515) - 0.000 (1/515 + 1 \cdot 0.0025/2)]$$

$$= 140379.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 58005 \cdot (192402 + 252433 + 210987 \cdot \ln(1.17))$$

$$= 5.3064$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2537^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 19.2 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 24.2]$$

$$= -102951.6328 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -102952 + 140379 \cdot 24.2 - 130697 \cdot 19.2$$

$$= 785849.9375 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 49 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

$$= ((M \cdot) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((785850) - 2537^2/32 \cdot 5.31 (19.2 - 24.2)) / (1 + 5.31)$$

$$= 971662.3125 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu \cdot) \cdot (Ps - Pt)$$

$$= 971662 + 2537^2/64 \cdot (3 + 0.36) \cdot (19.2 - 24.2)$$

$$= -718127.6875 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 971662 | ; | -718128 |)$$

$$= 971662.3125 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 971662 / (0.3 (515 - 5)^2)$$

$$= 75913.6562 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 309976$$

$$= 588955.12 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 996492 / (0.35 (175 - 5)^2)$$

$$= 588841.1875 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 175 + 0 + 0$$

$$= 174.7991 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(175 ; 62.9)$$

$$= 174.7991 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:

$$= | Ps - Pt |$$

$$= | 19.2 - 24.2 |$$

$$= 5.006 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 515/2537 \cdot 0.8 \cdot 0.95 \cdot 309976$$

$$= 38261.625 \text{ kPa}$$

$| Ps - Pt |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 235582) \cdot (4 \cdot 4.78/8067) \cdot | 19.2 - 24.2 | + 0$$

$$= 62.9445 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 170.000mm

Min. Shell len. adjacent to the tubesheet is: 1196.695 mm



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FileName : 123-C2-NVS

Page 50 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10,2026

Note: For a given Channel thickness of: 190.000 mm
Min. Channel len. adjacent to the tubesheet is: 1265.132mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 194735$$

$$= 38336.223 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 194735$$

$$= 38336.2227 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 194735 / 194735$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 194735$$

$$= 27667.8594 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 3.371 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 38336 is >= tube strength 38336 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	75913.7 <=	588955.1 kPa	Ok
Tubesheet shear stress	28793.4 <=	235582.0 kPa	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	174.799	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025



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FileName : 123-C2-NVS

Page 51 of 220

ASME TS Calc: Pressure Test

Case: 1 4:24p Apr 10, 2026

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	387.701 ...	356170	588955	139323 235582	Ps+Pt D1	Ok
D2uc	338.793 ...	280257	588955	110529 235582	Ps+Pt D2	Ok
D3uc	174.799 ...	75914	588955	28793 235582	Ps+Pt D3	Ok
Max:	387.7015 ... mm	0.605	0.591	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		...	356223	...	441716	Ok
D2uc		...	356223	...	441716	Ok
D3uc		...	356223	...	441716	Ok
Max		...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	387.701	515.000	Ok
Tube thickness :	2.409	2.769	Ok
Tube-tubesheet fillet weld leg :	1.685	3.000	Ok
Tube-tubesheet groove weld leg :	1.685	3.000	Ok

Given Shell Thickness: 170.0000 mm

Given Channel Thickness: 190.0000 mm

Min Shell length of thk, (170.000) adj. to tubesheet: 1196.695 mm

Min Channel length of thk, (190.000) adj. to tubesheet: 1265.132 mm

Note: This is a full strength Tube to Tubesheet Joint.



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FileName : 123-C2-NVS

Page 52 of 220

External Pressure Calculations: Step: 10 4:24pm Apr 10,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length m	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kPa
10	20	No Calc	2796	95	No Calc	No Calc
20	Tsht	2.9333	2980	187	No Calc	No Calc
Tsht	40	No Calc	...	187	No Calc	No Calc
40	50	9.83336	2940	167	No Calc	No Calc
50	60	No Calc	2756	75	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	98	No Calc	...	22.1
20	Tsht	190	No Calc	...	23.3
Tsht	40	190	No Calc	...	No Calc
40	50	170	17.755	0.10131	14.5
50	60	78	6.6625	0.10131	12.9

Minimum

12.9

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.9333	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	9.83336	25400	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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FileName : 123-C2-NVS

Page 53 of 220

Element and Detail Weights:

Step: 11

4:24pm Apr 10,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume m ³	Corroded Metal Wgt. kgm	Corroded ID Volume m ³	Extra due Misc % kgm
10	20	10423.9	4.4927	10176.5	4.52388	1042.39
20	40	32268.4	12.9597	31793.1	13.0196	3226.84
40	50	107780	20.9528	105992	21.178	10778
50	60	8120.02	4.4927	7872.55	4.52388	812.002
Total		158591	42.90	155834	43.25	15859



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. m	Y Offset, Dtl. Cent. m	Z Offset Dtl. Cent. m	Description
10	Liqd	202.374	-0.43333	0.14534E-06	...	Liquid: 10
10	Nozl	5542.8	-1.3	...	...	Manhole
10	Wght	300	-1.3	...	...	davit weight
20	Liqd	583.771	1.25	...	...	Liquid: 20
20	Nozl	4540.46	1.25	1.681	...	Tube inlet
20	Nozl	4407.03	1.25	-1.681	...	Tube outlet
20	Wght	1600	1.75	...	...	Pass Partition
40	Sadl	2388.6	1.8	1.5	...	Left Saddle
40	Sadl	2388.6	7.2	1.5	...	Right Saddle
40	Liqd	21450.4	4.7	...	...	Liquid: 40
40	Nozl	980.884	4.5	1.46193	...	Shell Outlet
40	Nozl	980.884	4.5	-1.46193	...	Shell Inlet
40	Wght	86700.5	1.8	2.5	...	103-C1(ETY HALF
40	Wght	86700.5	7.19997	2.5	...	103-C1(ETY HALF
40	Wght	5500	4.5	...	...	Log. baffle
40	Wght	500	1.8	-1.9	...	Fireproofing
40	Wght	500	7.19997	-1.9	...	Fireproofing 1
40	Wght	91539.5	1.8	2.5	...	103-C1(OPR HALF
40	Wght	91539.5	7.19997	2.5	...	103-C1(OPR HALF
40	Wght	115222	1.8	2.5	...	103-C1(HYDHALF)
40	Wght	115222	7.19997	2.5	...	103-C1(HYD HALF
40	Wght	2000	1.8	...	...	Upper Saddle wei
40	Wght	2000	7.19997	...	...	Upper Saddle wei
40	Forc	...	1.8	...	...	Left SD wind &
40	Forc	...	7.19997	...	...	Right SD wind &
40	Forc	...	1.8	...	...	Left SD wind &
40	Forc	...	7.19997	...	...	Right SD wind &
50	Liqd	4599.4	0.43333	0.14534E-06	...	Liquid: 50
20	FTsh	18769.7	2.7575	...	...	Tubesheet
20	Tube	94264.8	7	...	...	Tubes
40	Bafl	400.168	0.53	...	...	Baffles
40	Bafl	400.168	1.06	...	...	Baffles
40	Bafl	400.168	1.59	...	...	Baffles
40	Bafl	400.168	2.12	...	...	Baffles
40	Bafl	400.168	2.65	...	...	Baffles
40	Bafl	400.168	3.18	...	...	Baffles
40	Bafl	400.168	3.71	...	...	Baffles
40	Bafl	400.168	4.24	...	...	Baffles
40	Bafl	400.168	4.77	...	...	Baffles
40	Bafl	400.168	5.3	...	...	Baffles



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 54 of 220

Element and Detail Weights: Step: 11 4:24pm Apr 10,2026

40 Baf1	400.168	5.83	...	...	Baffles
40 Baf1	400.168	6.36	...	...	Baffles
40 Baf1	400.168	6.89	...	...	Baffles
40 Baf1	400.168	7.42	...	...	Baffles
40 Baf1	400.168	7.95	...	...	Baffles
40 Baf1	401.017	8.48	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	4777.2
Liquid	26836.0
Nozzles	16452.1
Weights	599323.9
Exchanger Components	119438.0
Liquid in Tubes	802.4

Sum of the Detail Weights 767629.5 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	174451.1	174451.1	174451.1	174451.1	174451.1	174451.1
Saddles	4777.2	4777.2	4777.2	4777.2	4777.2	4777.2
Nozzles	16452.1	16452.1	16452.1	16452.1	16452.1	16452.1
Wld Weights	9800.0	9800.0	9800.0	9800.0	9800.0	9800.0
Exchanger	119438.0	119438.0	119438.0	119438.0	119438.0	119438.0
Empty Weights	...	...	...	176001.0	176001.0	...
Test Weights	...	233044.0	...	...	...	...
Ope Weights	...	...	...	...	...	185679.0
Ope. Liquid	...	...	...	...	...	26836.0
Tube Ope Lqd	...	...	...	...	...	802.4
Test Liquid	...	43916.7	...	...	...	...
Tube Tst Lqd	...	18237.3	...	...	...	...
Totals	324918.4	620116.4	324918.4	500919.4	500919.4	538235.8

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	324918.4 kgm
Shop Test Wt.	- Fabricated weight + water (full)	620116.4 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	324918.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	500919.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	500919.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	538235.8 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	535202.4 kgm
Field Test Wt.	- Empty weight + test medium (full)	620116.4 kgm

Exchanger Tube Data

Volume of Exchanger tubes :	17.8 m ³
Weight of Ope Liq in tubes :	802.4 kgm
Weight of Water in tubes :	18237.3 kgm

Surface Areas of Elements:

From	To	Outside Surface Area m ²	Inside Surface Area m ²
------	----	--	---------------------------------------



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FileName : 123-C2-NVS

Page 55 of 220

Element and Detail Weights: Step: 11 4:24pm Apr 10, 2026

10	20	12.595	10.616
20	40	23.3991	20.4153
40	50	86.7995	76.7615
50	60	12.1717	10.616

Total	135.0	118.4 m ²
Total	1453.1	1274.9 ft ²

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FileName : 123-C2-NVS

Page 56 of 220

Nozzle Flange MAWP:

Step: 12 4:24pm Apr 10,2026

Nozzle Flange MAWP Results: (N/mm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Manhole	18.33	25.51	370	1500	GR 1.1	...	...	...	...
Shell Outlet	18.97	25.51	343	1500	GR 1.1	...	...	...	...
Shell Inlet	18.97	25.51	343	1500	GR 1.1	...	...	...	...

Shellside Flange RatingLowest Flange Pressure Rating was (Ope)[ShellSide]: 18.967 N/mm²Lowest Flange Pressure Rating was (Amb)[ShellSide]: 25.511 N/mm²**Channelside Flange Rating**Lowest Flange Pressure Rating was (Ope)[TubeSide]: 18.333 N/mm²Lowest Flange Pressure Rating was (Amb)[TubeSide]: 25.511 N/mm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 57 of 220

Wind Load Calculation:

Step: 13 4:24pm Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.91)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.567
Structure Height to Diameter ratio	4.997

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
α = 9.5 : Zg = 274 m

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1.9 + 0.3 = 2.2 \text{ m}$$

$$= 7.22 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (7.22 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Zg)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 58 of 220

Wind Load Calculation:

Step: 13 4:24pm Apr 10,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [0.77] kPa

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.57 • 33

= 255 lbs. [1133] N

Element	Hgt (z) m	K1	K2	K3	Kz	Kzt	qz kPa
Channel head	2.2	0.000	0.000	0.000	0.849	1.000	0.766
Channel shell	2.2	0.000	0.000	0.000	0.849	1.000	0.766
Shell	2.2	0.000	0.000	0.000	0.849	1.000	0.766
Shell head	2.2	0.000	0.000	0.000	0.849	1.000	0.766

Wind Loads on Masses/Equipment/Piping

ID	Wind Area m ²	Elevation m	Pressure kPa	Force N
davit weight	0.00	2.20	0.77	0.00
Pass Partitio	0.00	2.20	0.77	0.00
103-C1(ETY HA	0.00	4.70	0.77	0.00
103-C1(ETY HA	0.00	4.70	0.77	0.00
Log. baffle	0.00	2.20	0.77	0.00
Fireproofing	0.00	0.30	0.77	0.00
Fireproofing	0.00	0.30	0.77	0.00
103-C1(OPR HA	8.39	4.70	0.77	6426.76
103-C1(OPR HA	8.39	4.70	0.77	6426.76
103-C1(HYDHAL	0.00	4.70	0.77	0.00
103-C1(HYD HA	0.00	4.70	0.77	0.00
Upper Saddle	0.00	2.20	0.77	0.00
Upper Saddle	0.00	2.20	0.77	0.00

Wind Load Calculation:

From	To	Wind Height m	Wind Diameter m	Wind Area m ²	Wind Pressure kPa	Element Wind Load N
10	20	2.2	3.3552	3.06931	0.76606	1245.98
20	40	2.2	3.576	8.93778	0.76606	3628.26
40	50	2.2	3.528	33.155	0.76606	27598
50	60	2.2	3.3072	2.98212	0.76606	1210.58

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.



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FileName : 123-C2-NVS

Page 59 of 220

Wind Load Calculation:

Step: 13 4:24pm Apr 10, 2026

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(10.50.75)$
 = 0.117 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.12$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 5231093 \cdot 1.25$
 = 307588.312 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 5231093$
 = 164047.1 N

Earthquake Load Calculation:

From	To	Earthquake Height m	Earthquake Weight N	Element Ope Load N
10	20	1.3	871849	51264.7



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FileName : 123-C2-NVS

Page 61 of 220

Earthquake Load Calculation: Step: 14 4:24pm Apr 10, 2026

20	40		1.3	871849	51264.7
40	Sad1		1.3	871849	51264.7
Sad1	50		1.3	871849	51264.7
40	50		1.3	871849	51264.7
50	60		1.3	871849	51264.7

The Earthquake Weight does not include the weight of the saddles.

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FileName : 123-C2-NVS

Page 62 of 220

Center of Gravity Calculation: Step: 15 4:24pm Apr 10,2026

Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	7.515 m
Center of Gravity of the Liquid	8.393 m
Center of Gravity of the Nozzles	1.138 m
Center of Gravity of the Added Weights (Operating)	7.454 m
Center of Gravity of the Added Weights (Empty)	7.451 m
Center of Gravity of the Tubesheet(s)	2.757 m
Center of Gravity of the Tubes	7.000 m
Center of Gravity of the Baffles	7.521 m
Center of Gravity of Bare Shell New and Cold	6.123 m
Center of Gravity of Bare Shell Corroded	6.119 m
Vessel CG in the Operating Condition	6.636 m
Vessel CG in the Fabricated (Shop/Empty) Condition	6.522 m
Vessel CG in the Test Condition	6.454 m

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 63 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

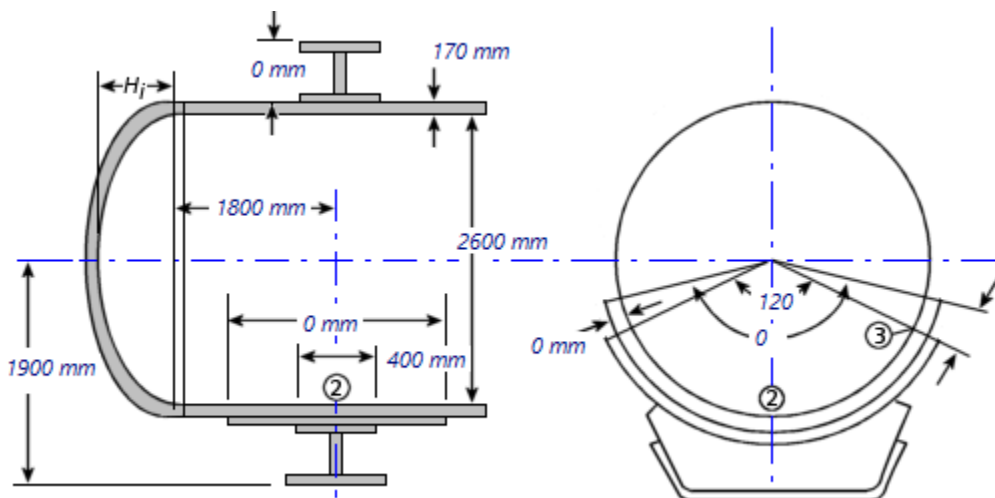
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1385 \cdot 170}, 2 \cdot 1800)$$

$$= 1156.9617 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1385.00	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		128860.33	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.40	
Tube Bundle Pull out Force		1171208.62	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		No	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		3427504.25	N
Pressure used in Analysis	P	0.000	N/mm²

Horizontal Vessel Analysis Results:	Actual	Allowable
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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 64 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

	kPa	kPa
Long. Stress at Top of Midspan	-592.06	-224982.88
Long. Stress at Top of Midspan	592.06	122834.11
Long. Stress at Bottom of Midspan	592.06	122834.11
Long. Stress at Top of Saddles	18014.54	122834.11
Long. Stress at Bottom of Saddles	-9984.77	-224982.88
Long. Stress at Bottom of Saddles	9984.77	122834.11
Tangential Shear in Shell	8826.68	98267.29
Circ. Stress at Horn of Saddle	15439.62	184251.16
Circ. Compressive Stress in Shell	1324.90	122834.11

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

$$= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$$

$$= 1171208 \cdot 1.5 \cdot 1900/5400$$

$$= 618137.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(0, 1756812) \cdot 1900/5400$$

$$= 91649.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (0/2 + 152052) \cdot 1900/2598$$

$$= 0.0 \text{ N}$$

Load Combination Results for Tube Bundle Pull Out [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 2809366 + \max(618138, 0, 91649, 0)$$

$$= 3427504.2 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3450926.75	N
Transverse Shear Load	0.00	N
Longitudinal Shear Load	1756812.88	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2600/2 + 170 + 30) \sin(120/2), 2 \cdot 1385)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -3427504 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.35^2) / (2 \cdot 1.8 \cdot 9.4)) / (1 + (4 \cdot 1.35)/(3 \cdot 9.4))]$$

$$= -1968287.6 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 3427504 \cdot 9.4/4 ((1 + 2(1.39^2 - 1.35^2)/(9.4^2)) / (1 + (4 \cdot 1.35)/(3 \cdot 9.4)) - 4 \cdot 1.8/9.4)$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 65 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

= 606771.4 N-m

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1385 / (2 \cdot 170) - 606771 / (\pi \cdot 1385^2 \cdot 170)$$

$$= -592.06 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1385 / (2 \cdot 170) + 606771 / (\pi \cdot 1385^2 \cdot 170)$$

$$= 592.06 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1385 / (2 \cdot 170) - 1968287 / (0.11 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= 18014.54 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1385 / (2 \cdot 170) + 1968287 / (0.19 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= -9984.77 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3427504 (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.35 / 3))$$

$$= 1775173.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1775173 / (1385 \cdot 170)$$

$$= 8826.68 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1385 \cdot 170}$$

$$= 378.481 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 3427504 \cdot 0.1 / (170(400 + 378 + 378))$$

$$= -1324.90 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -3427504 / (4 \cdot 170(400 + 378 + 378)) - 12 \cdot 0.053 \cdot 3427504 \cdot 1385 / (9.4 \cdot 170^2)$$

$$= -15439.62 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1385 \cdot 170}, 2 \cdot 1.8)$$

$$= 1156.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.400	



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 66 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1300 \cdot 170) + 300})^3) / 12$$

$$= 423076832.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 423076832 + 0.18 (170 - 85)^2$$

$$= 1701329024.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	1033.4	170.0	85.0	0.2	14932138.0	0.170E+10
Wearplate	300.0	30.0	185.0	0.0	1665000.0	0.262E+07
Web	40.0	325.0	362.5	0.0	4712499.5	0.595E+09
BasePlate	420.0	75.0	562.5	0.0	17718748.0	0.486E+10
Totals	...	...	...	0.2	39028384.0	0.716E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 39028384 / 0.23$$

$$= 170.302 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3427504$$

$$= 697571.4375 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 13039.0645 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1900 - 1300 \cdot \sin(120 / 2) / 1.05$$

$$= 824.909 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 575665.9375 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13689.6299 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(3427504 + 23423) \cdot 420) / (4 \cdot 2616 \cdot 173320)}$$

$$= 48.965 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 67 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3450926 / (2616 \cdot 420)$$

$$= 3141640.000 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.091 \cdot 3141640$$

$$= 285606.875 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 285607 / 0.023$$

$$= 12385.579 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(213333120 / 0.023)}$$

$$= 96.183 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900 - \cos(120/2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 68 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1075 / 96.2$$

$$= 11.177$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1756812 / (2 \cdot 2616) \cdot 433 \cdot 1075 / 2$$

$$= 78183.109 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ ($11.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152247 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12386 / 152247 + (78183 \cdot 200 / 213333328) / 173320$$

$$= 0.504$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1900 - \cos(1500/2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(215429184 / 0.032)}$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325 / 82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1756812 / (2 \cdot 2616)$$

$$= 335.782 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 69 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 336 \cdot 433 \cdot 325 \\
 &= 47273.504 \text{ N-m}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B , base, c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 47274 / 1077146 \\
 &= 43871.102 \text{ kPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($3.94 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (3.94)^2 / (2 \cdot 123^2)) 259980 / \\
 &\quad (5/3 + 3 \cdot (3.94) / (8 \cdot 123) - (3.94^3) / (8 \cdot 123^3)) \\
 &= 154794 \text{ kPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base, c} \cdot C_1 / I) / S_{ba} \\
 &= 18132 / 154794 + (47274 \cdot 200 / 215429184) / 173320 \\
 &= 0.370
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load Q0 (Weight)	Q0	3359277.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	1756812.9 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	1756812.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b , l]:

$$\begin{aligned}
 &= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 1756812 / (0.0014 \cdot 8) \\
 &= 156034.5 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b , t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 0 / (0.0014 \cdot 8) \\
 &= 0.0 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction) }$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 1756812 / (103192 \cdot 8) \\
 &= 0.0021 \text{ m}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 1900 \cdot 0 + 0 \\
 &= 0.00 \text{ N-m}
 \end{aligned}$$

Eccentricity (e):



PV Elite 28 Licensee: SL User
 FileName : 123-C2-NVS Page 70 of 220
 Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

= Rmom / QO
 = 0/3359277
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.0021, 0]
 = 0.0021 m²[Failed]

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(400 + 1.56 \cdot \sqrt{1385 \cdot 170}, 2 \cdot 1800)$
 = 1156.9617 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1385.00	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.34	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		129114.78	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1171208.62	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		No	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		1620961.75	N
Pressure used in Analysis	P	0.000	N/mm²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	-285.63	-224982.88
Long. Stress at Top	of Midspan	285.63	122834.11
Long. Stress at Bottom	of Midspan	285.63	122834.11
Long. Stress at Top	of Saddles	8480.05	122834.11
Long. Stress at Bottom	of Saddles	-4700.17	-224982.88
Long. Stress at Bottom	of Saddles	4700.17	122834.11
Tangential Shear in Shell		4179.36	98267.29
Circ. Stress at Horn of Saddle		7301.83	184251.16



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 71 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

Circ. Compressive Stress in Shell	626.58	122834.11
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Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 1171208 • 1.5 • 1900/5400
 = 618137.9 N

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 = max(Fl,eq; Sum of X Forces) • B / Ls
 = max(0, 1756812) • 1900/5400
 = 91649.0 N

Saddle Reaction Force due to Earthquake Ft [Fst]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (0/2 + 152052) • 1900/2598
 = 0.0 N

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 1002823 + max(618138, 0, 91649, 0)
 = 1620961.8 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1644384.25	N
Transverse Shear Load	0.00	N
Longitudinal Shear Load	1756812.88	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(2600/2 + 170 + 30)sin(120/2), 2•1385)
 = 2598.076 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -1620961•1.8[1-(1-1.8/9.4+(1.39²-1.34²)/(2•1.8•9.4))/(1+(4•1.34)/(3•9.4))]
 = -926539.4 N-m

Moment per Equation 4.15.2 [M2]:
 = Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 1620961•9.4/4((1+2(1.39²-1.34²)/(9.4²))/(1+(4•1.34)/(3•9.4))-4•1.8/9.4)
 = 292729.1 N-m

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:
 = P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 1385/(2•170) - 292729/(π•1385²•170)
 = -285.63 kPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 1385/(2 • 170) + 292729/(π • 1385² • 170)
 = 285.63 kPa



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 72 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 1385 / (2 \cdot 170) - 926539 / (0.11 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= 8480.05 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1385 / (2 \cdot 170) + 926539 / (0.19 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= -4700.17 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1620961 (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.34 / 3))$$

$$= 840529.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 840529 / (1385 \cdot 170)$$

$$= 4179.36 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1385 \cdot 170}$$

$$= 378.481 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1620961 \cdot 0.1 / (170 (400 + 378 + 378))$$

$$= -626.58 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1620961 / (4 \cdot 170 (400 + 378 + 378)) - 12 \cdot 0.053 \cdot 1620961 \cdot 1385 / (9.4 \cdot 170^2)$$

$$= -7301.83 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1385 \cdot 170}, 2 \cdot 1.8)$$

$$= 1156.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$

$$= ((1.56 \cdot \sqrt{1300 \cdot 170}) + 300)^3 / 12$$

$$= 423076832.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 423076832 + 0.18 (170 - 85)^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 73 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

= 1701329024.000 mm⁴

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	1033.4	170.0	85.0	0.2	14932138.0	0.170E+10
Wearplate	300.0	30.0	185.0	0.0	1665000.0	0.262E+07
Web	40.0	325.0	362.5	0.0	4712499.5	0.595E+09
BasePlate	420.0	75.0	562.5	0.0	17718748.0	0.486E+10
Totals	...	...	...	0.2	39028384.0	0.716E+10

Distance to Centroid [C1]:

= AY / A

= 39028384/0.23

= 170.302 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.2 • 1620961

= 329900.8438 N

Tension Stress, St = (Fh/As) = 6166.5352 kPa

Allowed Stress, Sa = 0.6 * Yield Str = 155988.1250 kPa

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1900 - 1300 • sin(120/2)/1.05

= 824.909 mm

Bending Moment, M = Fh * d = 272248.3750 N-m

Bending Stress, Sb = (M * C1 / I) = 6474.2051 kPa

Allowed Stress, Sa = 2/3 * Yield Str = 173320.1406 kPa

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(1620961 + 23423)420/(4 • 2616 • 173320))

= 33.800 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120/2) (1300 + 170 + 30)

= 2598.076 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2598/(7 - 1)

= 433.013 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 433 • 420/2

= 0.091 m²



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 74 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1644384 / (2616 \cdot 420)$$

$$= 1497007.625 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.091 \cdot 1497007$$

$$= 136093.141 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 136093 / 0.023$$

$$= 5901.792 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(213333120 / 0.023)}$$

$$= 96.183 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900 - \cos(120/2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1075 / 96.2$$

$$= 11.177$$

Bending Moment [Rm]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 75 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1756812 / (2 \cdot 2616) \cdot 433 \cdot 1075 / 2$$

$$= 78183.109 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ (11.2 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152247 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5902 / 152247 + (78183 \cdot 200 / 213333328) / 173320$$

$$= 0.462$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1900 - \cos(1500 / 2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (215429184 / 0.032) }$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 325 / 82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 1756812 / (2 \cdot 2616)$$

$$= 335.782 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 336 \cdot 433 \cdot 325$$

$$= 47273.504 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 47274 / 1077146$$

$$= 43871.102 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (3.94 < 123) per AISC E2-1 [Sca]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 76 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10,2026

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (3.94)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (3.94) / (8 \cdot 123) - (3.94^3) / (8 \cdot 123^3))$$

$$= 154794 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 8640/154794 + (47274 \cdot 200/215429184) / 173320$$

$$= 0.309$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load Q0 (Weight)	Q0	1552735.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	1756812.9 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	1756812.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1756812 / (0.0014 \cdot 8)$$

$$= 156034.5 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (0.0014 \cdot 8)$$

$$= 0.0 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (1756812 - 1552735) / (171987 \cdot 8)$$

$$= 0.0001 \text{ m}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1756812 / (103192 \cdot 8)$$

$$= 0.0021 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1900 \cdot 0 + 0$$

$$= 0.00 \text{ N-m}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0/1552735$$

$$= 0.00 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 77 of 220

Saddle Calcs: Bundle Pull New: Step: 16 4:24pm Apr 10, 2026

= max[Bltarearl, Bltarears, Bltareart]

= max[0.00015, 0.0021, 0]

= **0.0021 m²[Failed]**

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 78 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

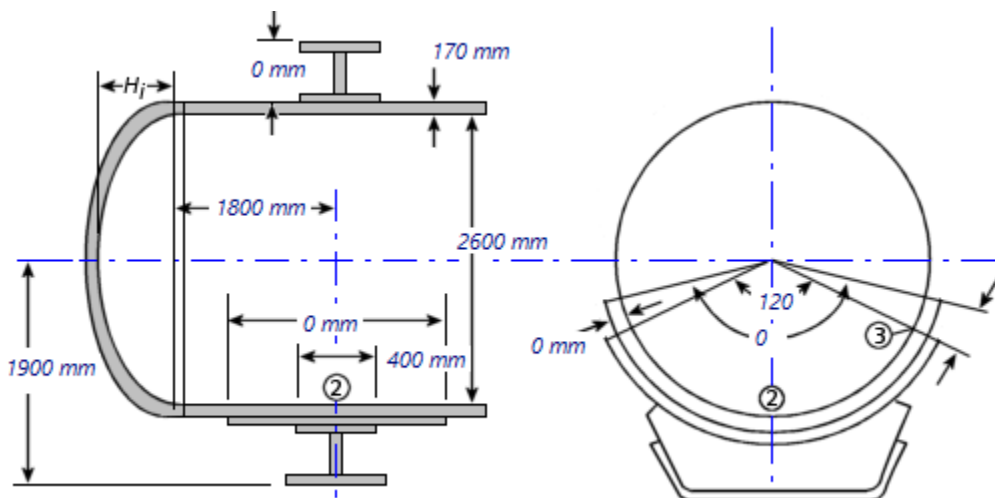
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1800)$$

$$= 1150.6591 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1386.50	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		128860.33	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.40	
Tube Bundle Pull out Force		1171208.62	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		No	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	3412631.25	N
Pressure used in Analysis	P	0.000 N/mm²

Horizontal Vessel Analysis Results:	Actual	Allowable
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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 79 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10, 2026

	kPa	kPa
Long. Stress at Top of Midspan	-597.60	-224982.88
Long. Stress at Top of Midspan	597.60	122834.11
Long. Stress at Bottom of Midspan	597.60	122834.11
Long. Stress at Top of Saddles	18225.90	122834.11
Long. Stress at Bottom of Saddles	-10101.92	-224982.88
Long. Stress at Bottom of Saddles	10101.92	122834.11
Tangential Shear in Shell	8934.98	98267.29
Circ. Stress at Horn of Saddle	15887.14	184251.16
Circ. Compressive Stress in Shell	1350.20	122834.11

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

$$= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$$

$$= 1171208 \cdot 1.5 \cdot 1900/5400$$

$$= 618137.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(0, 1756812) \cdot 1900/5400$$

$$= 91649.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (0/2 + 152052) \cdot 1900/2598$$

$$= 0.0 \text{ N}$$

Load Combination Results for Tube Bundle Pull Out [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 2794493 + \max(618138, 0, 91649, 0)$$

$$= 3412631.2 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3436053.75	N
Transverse Shear Load	0.00	N
Longitudinal Shear Load	1756812.88	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2600/2 + 170 + 30) \sin(120/2), 2 \cdot 1386)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -3412631 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.35^2) / (2 \cdot 1.8 \cdot 9.4)) / (1 + (4 \cdot 1.35)/(3 \cdot 9.4))]$$

$$= -1960477.8 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 3412631 \cdot 9.4/4 ((1 + 2(1.39^2 - 1.35^2)/(9.4^2)) / (1 + (4 \cdot 1.35)/(3 \cdot 9.4)) - 4 \cdot 1.8/9.4)$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 80 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

= 602949.1 N-m

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$

= $0 \cdot 1386 / (2 \cdot 167) - 602949 / (\pi \cdot 1386^2 \cdot 167)$

= -597.60 kPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$

= $0 \cdot 1386 / (2 \cdot 167) + 602949 / (\pi \cdot 1386^2 \cdot 167)$

= 597.60 kPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= $P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$

= $0 \cdot 1386 / (2 \cdot 167) - 1960477 / (0.11 \cdot \pi \cdot 1386^2 \cdot 167)$

= 18225.90 kPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= $P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$

= $0 \cdot 1386 / (2 \cdot 167) + 1960477 / (0.19 \cdot \pi \cdot 1386^2 \cdot 167)$

= -10101.92 kPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L-2a) / (L + (4 \cdot h_m / 3))$

= $3412631 \cdot (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.35 / 3))$

= 1767154.8 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K_2 \cdot T / (R_m \cdot t)$

= $1.17 \cdot 1767154 / (1386 \cdot 167)$

= 8934.98 kPa

Decay Length (4.15.20) [x_1, x_2]:

= $0.78 \cdot \sqrt{(R_m \cdot t)}$

= $0.78 \cdot \sqrt{(1386 \cdot 167)}$

= 375.330 mm

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

= $-K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$

= $-0.76 \cdot 3412631 \cdot 0.1 / (167(400 + 375 + 375))$

= -1350.20 kPa

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

= $-Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$

= $-3412631 / (4 \cdot 167(400 + 375 + 375)) -$

$12 \cdot 0.053 \cdot 3412631 \cdot 1386 / (9.4 \cdot 167^2)$

= -15887.14 kPa

Effective reinforcing plate width (4.15.24) [B1]:

= $\min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$

= $\min(400 + 1.56 \cdot \sqrt{(1386 \cdot 167)}, 2 \cdot 1.8)$

= 1150.66 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.400	



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 81 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1303 \cdot 167) + 300})^3) / 12$$

$$= 398874272.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 398874272 + 0.17 (170 - 83.5)^2$$

$$= 1682372864.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	1027.7	167.0	83.5	0.2	14330825.0	0.168E+10
Wearplate	300.0	30.0	182.0	0.0	1638000.0	0.198E+07
Web	40.0	325.0	359.5	0.0	4673499.5	0.581E+09
BasePlate	420.0	75.0	559.5	0.0	17624248.0	0.479E+10
Totals	...	...	...	0.2	38266576.0	0.706E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 38266576 / 0.23$$

$$= 169.978 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3412631$$

$$= 694544.4375 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 12982.4834 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1900 - 1303 \cdot \sin(120 / 2) / 1.05$$

$$= 822.428 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 571444.1250 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13753.1992 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3412631 + 23423) \cdot 420) / (4 \cdot 2616 \cdot 173320)}$$

$$= 48.859 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 82 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3436053 / (2616 \cdot 420)$$

$$= 3128100.000 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.091 \cdot 3128100$$

$$= 284375.969 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 284376 / 0.023$$

$$= 12332.199 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(213333120 / 0.023)}$$

$$= 96.183 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900 - \cos(120/2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 83 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1075/96.2$$

$$= 11.177$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 1756812 / (2 \cdot 2616) \cdot 433 \cdot 1075/2$$

$$= 78183.109 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($11.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152247 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 12332/152247 + (78183 \cdot 200/21333328) / 173320$$

$$= 0.504$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1900 - \cos(1500/2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(215429184/0.032)}$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325/82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1756812 / (2 \cdot 2616)$$

$$= 335.782 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 84 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10, 2026

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 336 \cdot 433 \cdot 325 \\
 &= 47273.504 \text{ N-m}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 47274 / 1077146 \\
 &= 43871.102 \text{ kPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($3.94 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (3.94)^2 / (2 \cdot 123^2)) 259980 / \\
 &\quad (5/3 + 3 \cdot (3.94) / (8 \cdot 123) - (3.94^3) / (8 \cdot 123^3)) \\
 &= 154794 \text{ kPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba} \\
 &= 18054 / 154794 + (47274 \cdot 200 / 215429184) / 173320 \\
 &= 0.370
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load Q0 (Weight)	Q0	3344405.0 N
Saddle Load QL (Wind/Seismic contribution)	QL	1756812.9 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	1756812.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$\begin{aligned}
 &= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 1756812 / (0.0014 \cdot 8) \\
 &= 156034.5 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 0 / (0.0014 \cdot 8) \\
 &= 0.0 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction) }$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 1756812 / (103192 \cdot 8) \\
 &= 0.0021 \text{ m}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 1900 \cdot 0 + 0 \\
 &= 0.00 \text{ N-m}
 \end{aligned}$$

Eccentricity (e):



PV Elite 28 Licensee: SL User
 FileName : 123-C2-NVS Page 85 of 220
 Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

= Rmom / QO
 = 0/3344405
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.0021, 0]
 = 0.0021 m²[Failed]

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1800)$
 = 1150.6591 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1386.50	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.34	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		129114.78	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1171208.62	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		No	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		1606089.12	N
Pressure used in Analysis	P	0.000	N/mm²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	-286.92	-224982.88
Long. Stress at Top	of Midspan	286.92	122834.11
Long. Stress at Bottom	of Midspan	286.92	122834.11
Long. Stress at Top	of Saddles	8537.88	122834.11
Long. Stress at Bottom	of Saddles	-4732.22	-224982.88
Long. Stress at Bottom	of Saddles	4732.22	122834.11
Tangential Shear in Shell		4210.09	98267.29
Circ. Stress at Horn of Saddle		7476.98	184251.16



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 86 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10, 2026

Circ. Compressive Stress in Shell	635.45	122834.11
-----------------------------------	--------	-----------

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

$$= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$$

$$= 1171208 \cdot 1.5 \cdot 1900/5400$$

$$= 618137.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,eq}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(0, 1756812) \cdot 1900/5400$$

$$= 91649.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (0/2 + 152052) \cdot 1900/2598$$

$$= 0.0 \text{ N}$$

Load Combination Results for Tube Bundle Pull Out [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 987951 + \max(618138, 0, 91649, 0)$$

$$= 1606089.1 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1629511.75	N
Transverse Shear Load	0.00	N
Longitudinal Shear Load	1756812.88	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2600/2 + 170 + 30)\sin(120/2), 2 \cdot 1386)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1606089 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.34^2)/(2 \cdot 1.8 \cdot 9.4))/(1 + (4 \cdot 1.34)/(3 \cdot 9.4))]$$

$$= -918381.3 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 1606089 \cdot 9.4/4 [(1 + 2(1.39^2 - 1.34^2)/(9.4^2))/(1 + (4 \cdot 1.34)/(3 \cdot 9.4)) - 4 \cdot 1.8/9.4]$$

$$= 289484.2 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 0 \cdot 1386/(2 \cdot 167) - 289484/(\pi \cdot 1386^2 \cdot 167)$$

$$= -286.92 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1386/(2 \cdot 167) + 289484/(\pi \cdot 1386^2 \cdot 167)$$

$$= 286.92 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 87 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10, 2026

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 1386 / (2 \cdot 167) - 918381 / (0.11 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= 8537.88 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1386 / (2 \cdot 167) + 918381 / (0.19 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= -4732.22 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1606089 (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.34 / 3))$$

$$= 832668.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 832669 / (1386 \cdot 167)$$

$$= 4210.09 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1386 \cdot 167}$$

$$= 375.330 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 1606089 \cdot 0.1 / (167 (400 + 375 + 375))$$

$$= -635.45 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1606089 / (4 \cdot 167 (400 + 375 + 375)) - 12 \cdot 0.053 \cdot 1606089 \cdot 1386 / (9.4 \cdot 167^2)$$

$$= -7476.98 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1.8)$$

$$= 1150.66 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw, c	438.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1303 \cdot 167} + 300)^3) / 12$$

$$= 398874272.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 398874272 + 0.17 (170 - 83.5)^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 88 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

= 1682372864.000 mm⁴

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	1027.7	167.0	83.5	0.2	14330825.0	0.168E+10
Wearplate	300.0	30.0	182.0	0.0	1638000.0	0.198E+07
Web	40.0	325.0	359.5	0.0	4673499.5	0.581E+09
BasePlate	420.0	75.0	559.5	0.0	17624248.0	0.479E+10
Totals	...	...	...	0.2	38266576.0	0.706E+10

Distance to Centroid [C1]:

= AY / A

= 38266576/0.23

= 169.978 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.2 • 1606089

= 326873.9688 N

Tension Stress, St = (Fh/As) = 6109.9561 kPa

Allowed Stress, Sa = 0.6 • Yield Str = 155988.1250 kPa

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1900 - 1303 • sin(120/2)/1.05

= 822.428 mm

Bending Moment, M = Fh • d = 268939.1875 N-m

Bending Stress, Sb = (M • C1 / I) = 6472.6782 kPa

Allowed Stress, Sa = 2/3 • Yield Str = 173320.1406 kPa

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(1606089 + 23423)420/(4 • 2616 • 173320))

= 33.647 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120/2) (1300 + 170 + 30)

= 2598.076 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2598/(7 - 1)

= 433.013 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 433 • 420/2

= 0.091 m²



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 89 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1629511 / (2616 \cdot 420)$$

$$= 1483468.000 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.091 \cdot 1483468$$

$$= 134862.250 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 134862 / 0.023$$

$$= 5848.413 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(213333120 / 0.023)}$$

$$= 96.183 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900 - \cos(120/2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1075 / 96.2$$

$$= 11.177$$

Bending Moment [Rm]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 90 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10,2026

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 1756812 / (2 \cdot 2616) \cdot 433 \cdot 1075 / 2$$

$$= 78183.109 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ (11.2 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152247 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5848 / 152247 + (78183 \cdot 200 / 213333328) / 173320$$

$$= 0.461$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1900 - \cos(1500 / 2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (215429184 / 0.032) }$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 325 / 82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 1756812 / (2 \cdot 2616)$$

$$= 335.782 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 336 \cdot 433 \cdot 325$$

$$= 47273.504 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 47274 / 1077146$$

$$= 43871.102 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (3.94 < 123) per AISC E2-1 [Sca]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 91 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10, 2026

$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (3.94)^2 / (2 \cdot 123^2)) 259980 / \\
 &\quad (5/3 + 3 \cdot (3.94) / (8 \cdot 123) - (3.94^3) / (8 \cdot 123^3)) \\
 &= 154794 \text{ kPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Mbase,c \cdot C1/I) / Sba \\
 &= 8562/154794 + (47274 \cdot 200/215429184) / 173320 \\
 &= 0.308
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load QO (Weight)	QO	1537862.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	1756812.9 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	1756812.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$\begin{aligned}
 &= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 1756812 / (0.0014 \cdot 8) \\
 &= 156034.5 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 0 / (0.0014 \cdot 8) \\
 &= 0.0 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$\begin{aligned}
 &= (QL - QO) / (Stba \cdot Rnbolts) \\
 &= (1756812 - 1537862) / (171987 \cdot 8) \\
 &= 0.0002 \text{ m}^2
 \end{aligned}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 1756812 / (103192 \cdot 8) \\
 &= 0.0021 \text{ m}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 1900 \cdot 0 + 0 \\
 &= 0.00 \text{ N-m}
 \end{aligned}$$

Eccentricity (e):

$$\begin{aligned}
 &= Rmom / QO \\
 &= 0 / 1537862 \\
 &= 0.00 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}
 \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 92 of 220

Saddle Calcs: Bundle Pull Cor: Step: 17 4:24pm Apr 10, 2026

= max[Bltarearl, Bltarears, Bltareart]

= max[0.00016, 0.0021, 0]

= **0.0021 m²[Failed]**

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 93 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

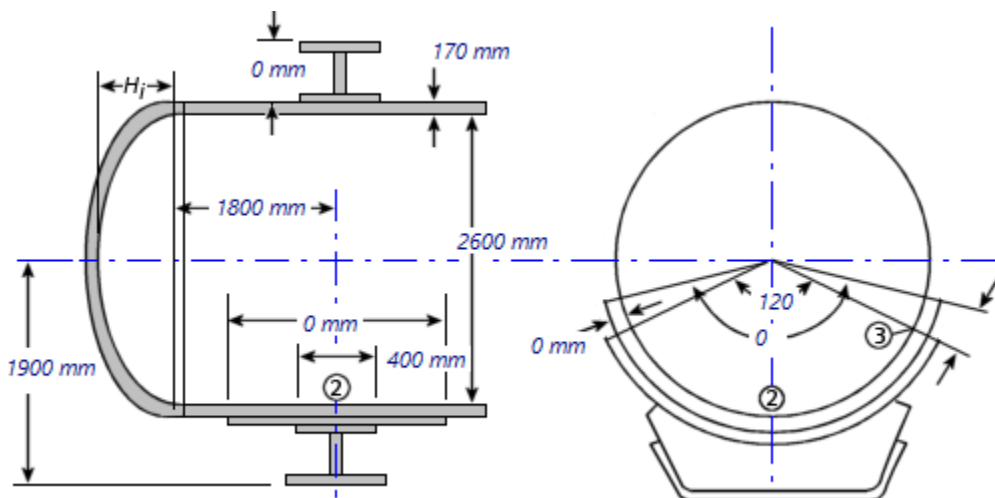
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1800)$$

$$= 1150.6591 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1386.50	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		128860.33	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	4229826.50	N
Pressure used in Analysis	P	14.294 N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	58591.30	122834.11
Long. Stress at Bottom	of Midspan	60072.71	122834.11



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 94 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

Long. Stress at Top of Saddles	81922.31	122834.11
Long. Stress at Bottom of Saddles	46811.06	122834.11
Tangential Shear in Shell	11074.57	98267.29
Circ. Stress at Horn of Saddle	19691.51	153542.62
Circ. Compressive Stress in Shell	1673.52	122834.11

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (33683/2 + 152053) \cdot 1900/2598$$

$$= 370542.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(8237, 260476) \cdot 1900/5400$$

$$= 4615.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(307588, 260476) \cdot 1900/5400$$

$$= 108225.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (307588/2 + 152053) \cdot 1900/2598$$

$$= 671007.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 3558819 + \max(4616, 370543, 108226, 671008)$$

$$= 4229826.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.94 \cdot 1.2)^2)$$

$$= 8237.911 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 4253249.50 N

Transverse Shear Load 305847.25 N

Longitudinal Shear Load 307588.31 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2600/2 + 170 + 30) \sin(120/2), 2 \cdot 1386)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$$

$$= -4229826 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.35^2) /$$

$$(2 \cdot 1.8 \cdot 9.4) / (1 + (4 \cdot 1.35) / (3 \cdot 9.4))]$$

$$= -2429937.8 \text{ N-m}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 95 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L)$$

$$= 4229826 \cdot 9.4 / 4 \cdot ((1 + 2(1.39^2 - 1.35^2) / (9.4^2)) / (1 + (4 \cdot 1.35) / (3 \cdot 9.4)) - 4 \cdot 1.8 / 9.4)$$

$$= 747332.5 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) - 747332 / (\pi \cdot 1386^2 \cdot 167)$$

$$= 58591.30 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) + 747332 / (\pi \cdot 1386^2 \cdot 167)$$

$$= 60072.71 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) - 2429937 / (0.11 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= 81922.31 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) + 2429937 / (0.19 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= 46811.06 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 4229826 (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.35 / 3))$$

$$= 2190321.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 2190321 / (1386 \cdot 167)$$

$$= 11074.57 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1386 \cdot 167)}$$

$$= 375.330 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 4229826 \cdot 0.1 / (167(400 + 375 + 375))$$

$$= -1673.52 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -4229826 / (4 \cdot 167(400 + 375 + 375)) - 12 \cdot 0.053 \cdot 4229826 \cdot 1386 / (9.4 \cdot 167^2)$$

$$= -19691.51 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1386 \cdot 167)}, 2 \cdot 1.8)$$

$$= 1150.66 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 5400.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13497\text{E-}04 \cdot 5400(343 - 21.1)$$

$$= 23.460 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 96 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1303 \cdot 167)} + 300)167^3)/12$$

$$= 398874272.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 398874272 + 0.17 (170 - 83.5)^2$$

$$= 1682372864.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	1027.7	167.0	83.5	0.2	14330825.0	0.168E+10
Wearplate	300.0	30.0	182.0	0.0	1638000.0	0.198E+07
Web	40.0	325.0	359.5	0.0	4673499.5	0.581E+09
BasePlate	420.0	75.0	559.5	0.0	17624248.0	0.479E+10
Totals	...	...	...	0.2	38266576.0	0.706E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 38266576/0.23$$

$$= 169.978 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 4229826$$

$$= 860861.4375 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 16091.2959 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1900 - 1303 \cdot \sin(120/2) / 1.05$$

$$= 822.428 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 708283.3125 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 17046.5684 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 97 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(4229826 + 23423)420}{4 \cdot 2616 \cdot 129781}}$$

$$= 62.820 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4253249 / (2616 \cdot 420)$$

$$= 3872054.750 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.091 \cdot 3872054$$

$$= 352008.969 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 352009 / 0.023$$

$$= 15265.161 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 98 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

= 200.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{213333120 / 0.023}$

= 96.183 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 1900 - $\cos(120/2)$ (1300 + 170 + 30) - 75

= 1075.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955 / 194672}$

= 142.386

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1075/96.2

= 11.177

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= 307588 / (2 • 2616) • 433 • 1075/2

= 13688.544 N-m

Compressive Allowable, $KL/r < Cc$ (11.2 < 142) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.2)^2 / (2 \cdot 142^2)) 194672 /$

$(5/3 + 3 \cdot (11.2) / (8 \cdot 142) - (11.2^3) / (8 \cdot 142^3))$

= 114427 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$

= 15265/114427 + (13689 • 200/21333328) / 129781

= 0.232

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/0.032

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 400/2

= 200.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1900 - $\cos(1500/2)$ (75 + 0 + 0) - 0

= 325.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{215429184 / 0.032}$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 99 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

= 82.410 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 325/82.4

= 3.944

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 307588/(2 • 2616)

= 58.790 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 58.8 • 433 • 325

= 8276.794 N-m

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus

= 8277/1077146

= 7681.089 kPa

Compressive Allowable, KL/r < Cc (3.94 < 142) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))

= (1-(3.94)²/(2 • 142²))194672/

(5/3+3•(3.94)/(8• 142)-(3.94³)/(8•142³))

= 116036 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 22347/116036 + (8277 • 200/215429184)/129781

= 0.252

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	283983.50 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	129114.48 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load QO (Weight)	QO	3582241.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	108225.5 N
Maximum Transverse Force	Ft	305847.2 N
Maximum Longitudinal Force	F1	307588.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= FI / (Bolt Area • Number of Bolts)

= 307588/(0.0014 • 8)

= 27319.0 kPa. Must be less than 170390.1 kPa.

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 305847/(0.0014 • 8)

= 27164.4 kPa. Must be less than 170390.1 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 100 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 307588 / (170390 \cdot 8)$
 $= 0.0002 \text{ m}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1900 \cdot 305847 + 0$
 $= 581345.38 \text{ N-m}$

Eccentricity (e):
 $= Rmom / QO$
 $= 581345 / 3582241$
 $= 162.22 \text{ mm} < Bplen / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 0.00023, 0]$
 $= 0.0002 \text{ m}^2$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $tg = 30$, $c = 3 \text{ mm}$, $E* = 1$
Thickness Ratio = $tr \cdot E* / (tg - c) = -1$, Temp. Reduction = $0 \text{ }^\circ\text{C}$
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B $4 \text{ }^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit $-13 \text{ }^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1800)$
 $= 1150.6591 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1386.50	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.34	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		129114.78	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 101 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

Distance from Saddle Base to Centerline B 1900.00 mm
Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 2507326.00 N
Pressure used in Analysis P 14.294 N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	58884.09	122834.11
Long. Stress at Bottom of Midspan	59779.92	122834.11
Long. Stress at Top of Saddles	72660.80	122834.11
Long. Stress at Bottom of Saddles	51944.36	122834.11
Tangential Shear in Shell	6572.53	98267.29
Circ. Stress at Horn of Saddle	11672.59	153542.62
Circ. Compressive Stress in Shell	992.02	122834.11

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (33683/2 + 152052) • 1900/2598
= 370540.2 N

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(8237, 260476) • 1900/5400
= 4615.9 N

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(307588, 260476) • 1900/5400
= 108225.5 N

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (307588/2 + 152052) • 1900/2598
= 671005.1 N

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 1836320 + max(4616, 370540, 108226, 671005)
= 2507326.0 N

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + (π/4(OD • WindDiaMult)²))
= 1.1 • 766 (0 + π/4(2.94 • 1.2)²)
= 8237.911 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2530748.50 N
Transverse Shear Load 305846.03 N
Longitudinal Shear Load 307588.31 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
= min(2(2600/2 + 170 + 30)sin(120/2), 2•1386)
= 2598.076 mm



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 102 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -2507326 \cdot 1.8 \left[1 - \left(1 - \frac{1.8}{9.4} + \frac{(1.39^2 - 1.34^2)}{(2 \cdot 1.8 \cdot 9.4)} \right) / \left(1 + \frac{(4 \cdot 1.34)}{(3 \cdot 9.4)} \right) \right]$$

$$= -1433719.5 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 9.4)} \right) - 4 \cdot 1.8 / 9.4$$

$$= 2507326 \cdot 9.4 / 4 \left(\frac{(1 + 2(1.39^2 - 1.34^2)/(9.4^2)) / (1 + (4 \cdot 1.34)/(3 \cdot 9.4)) - 4 \cdot 1.8 / 9.4}{(3 \cdot 9.4)} \right) - 4 \cdot 1.8 / 9.4$$

$$= 451924.6 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) - 451925 / (\pi \cdot 1386^2 \cdot 167)$$

$$= 58884.09 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) + 451925 / (\pi \cdot 1386^2 \cdot 167)$$

$$= 59779.92 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) - 1433719 / (0.11 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= 72660.80 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1386 / (2 \cdot 167) + 1433719 / (0.19 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= 51944.36 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 2507326 (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.34 / 3))$$

$$= 1299910.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 1299910 / (1386 \cdot 167)$$

$$= 6572.53 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1386 \cdot 167)}$$

$$= 375.330 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 2507326 \cdot 0.1 / (167(400 + 375 + 375))$$

$$= -992.02 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, L < 8Rm (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -2507326 / (4 \cdot 167(400 + 375 + 375)) - 12 \cdot 0.053 \cdot 2507326 \cdot 1386 / (9.4 \cdot 167^2)$$

$$= -11672.59 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 103 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{ 1386 \cdot 167 }, 2 \cdot 1.8)$$

$$= 1150.66 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ 1303 \cdot 167 } + 300) 167^3) / 12$$

$$= 398874272.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 398874272 + 0.17 (170 - 83.5)^2$$

$$= 1682372864.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	1027.7	167.0	83.5	0.2	14330825.0	0.168E+10
Wearplate	300.0	30.0	182.0	0.0	1638000.0	0.198E+07
Web	40.0	325.0	359.5	0.0	4673499.5	0.581E+09
BasePlate	420.0	75.0	559.5	0.0	17624248.0	0.479E+10
Totals	...	...	...	0.2	38266576.0	0.706E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 38266576 / 0.23$$

$$= 169.978 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2507326$$

$$= 510295.1875 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 9538.4814 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 104 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

$$= 1900 - 1303 \cdot \sin(120/2)/1.05$$

$$= 822.428 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 419851.0625 \text{ N-m}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 10104.7412 \text{ kPa}$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(2507326 + 23423) \cdot 420}{4 \cdot 2616 \cdot 129781}}$$

$$= 48.457 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2530748 / (2616 \cdot 420)$$

$$= 2303932.000 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.091 \cdot 2303932$$

$$= 209450.734 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 209451 / 0.023$$

$$= 9083.005 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 105 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10,2026

= 0/0.023
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 400/2
= 200.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{213333120/0.023}$
= 96.183 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1900 - cos(120/2)(1300 + 170 + 30) - 75
= 1075.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199955/194672}$
= 142.386

Slenderness ratio [KL/r]:

= KL/r
= 1 • 1075/96.2
= 11.177

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 307588 / (2 • 2616) • 433 • 1075/2
= 13688.544 N-m

Compressive Allowable, KL/r < Cc (11.2 < 142) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (11.2)^2 / (2 \cdot 142^2)) 194672 /$
 $(5/3 + 3 \cdot (11.2) / (8 \cdot 142) - (11.2^3) / (8 \cdot 142^3))$
= 114427 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 9083/114427 + (13689 • 200/21333328) / 129781
= 0.178

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/0.032
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 400/2
= 200.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 106 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

$$= 1900 - \cos(1500/2)(75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{215429184 / 0.032}$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325 / 82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 307588 / (2 \cdot 2616)$$

$$= 58.790 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 58.8 \cdot 433 \cdot 325$$

$$= 8276.794 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 8277 / 1077146$$

$$= 7681.089 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($3.94 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.94)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (3.94) / (8 \cdot 142) - (3.94)^3 / (8 \cdot 142^3))$$

$$= 116036 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 13297 / 116036 + (8277 \cdot 200 / 215429184) / 129781$$

$$= 0.174$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	283983.50 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m²
BasePlate Allowable Stress	S	129114.48 kPa
Area Available in a Single Bolt	BlArea	0.0014 m²
Saddle Load Q0 (Weight)	Q0	1859743.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	108225.5 N
Maximum Transverse Force	Ft	305846.0 N
Maximum Longitudinal Force	F1	307588.3 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 1432896.6 \text{ N}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,I]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1432896 / (0.0014 \cdot 8)$$

$$= 127265.3 \text{ kPa. Must be less than } 170390.1 \text{ kPa.}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 107 of 220

Saddle Calcs: Operating Case: Step: 18 4:24pm Apr 10, 2026

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $305846 / (0.0014 \cdot 8)$
 = 27164.3 kPa. Must be less than 170390.1 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $307588 / (170390 \cdot 8)$
 = 0.0002 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1900 \cdot 305846 + 0$
 = 581343.12 N-m

Eccentricity (e):
 = R_{mom} / QO
 = $581343 / 1859743$
 = 312.47 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 0.00023, 0]$
 = 0.0002 m²

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 30, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = -1$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 108 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

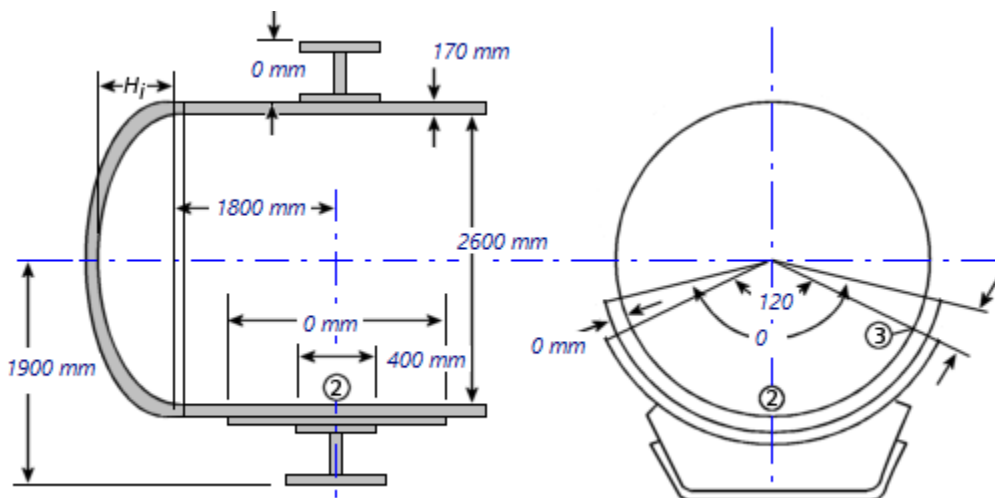
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1385 \cdot 170}, 2 \cdot 1800)$$

$$= 1156.9617 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1385.00	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		224982.88	kPa
Head Allowable Stress used in Calculation		224982.88	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		4971032.50	N
Pressure used in Analysis	P	19.230	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	77471.00	224982.88
Long. Stress at Bottom	of Midspan	79188.37	224982.88



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 109 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10, 2026

Long. Stress at Top of Saddles	104456.83	224982.88
Long. Stress at Bottom of Saddles	63848.41	224982.88
Tangential Shear in Shell	12801.65	133239.86
Circ. Stress at Horn of Saddle	22392.65	337474.28
Circ. Compressive Stress in Shell	1921.55	224982.88

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (16841/2 + 0) \cdot 1900/2598$$

$$= 18474.5 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(4119, 0) \cdot 1900/5400$$

$$= 1449.2 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 4952558 + \max(1449, 18474, 0, 0)$$

$$= 4971032.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.94 \cdot 1.2)^2)$$

$$= 8237.911 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 4994455.00 N

Transverse Shear Load 8420.71 N

Longitudinal Shear Load 4118.74 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2600/2 + 170 + 30)\sin(120/2), 2 \cdot 1385)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -4971032 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.35^2)/$$

$$(2 \cdot 1.8 \cdot 9.4)/(1 + (4 \cdot 1.35)/(3 \cdot 9.4))]$$

$$= -2854678.2 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 4971032 \cdot 9.4/4((1 + 2(1.39^2 - 1.35^2)/(9.4^2))/(1 + (4 \cdot 1.35)/$$

$$(3 \cdot 9.4)) - 4 \cdot 1.8/9.4)$$

$$= 880022.3 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 19.2 \cdot 1385/(2 \cdot 170) - 880022/(\pi \cdot 1385^2 \cdot 170)$$

$$= 77471.00 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 110 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10, 2026

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 19.2 \cdot 1385 / (2 \cdot 170) + 880022 / (\pi \cdot 1385^2 \cdot 170)$$

$$= 79188.37 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 19.2 \cdot 1385 / (2 \cdot 170) - 2854678 / (0.11 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= 104456.83 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 19.2 \cdot 1385 / (2 \cdot 170) + 2854678 / (0.19 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= 63848.41 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 4971032 \cdot (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.35 / 3))$$

$$= 2574597.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 2574597 / (1385 \cdot 170)$$

$$= 12801.65 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1385 \cdot 170}$$

$$= 378.481 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 4971032 \cdot 0.1 / (170(400 + 378 + 378))$$

$$= -1921.55 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -4971032 / (4 \cdot 170(400 + 378 + 378)) - 12 \cdot 0.053 \cdot 4971032 \cdot 1385 / (9.4 \cdot 170^2)$$

$$= -22392.65 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1385 \cdot 170}, 2 \cdot 1.8)$$

$$= 1156.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1300 \cdot 170} + 300)^3) / 12$$

$$= 423076832.000 \text{ mm}^4$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 111 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10,2026

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 423076832 + 0.18 (170 - 85)^2$$

$$= 1701329024.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	1033.4	170.0	85.0	0.2	14932138.0	0.170E+10
Wearplate	300.0	30.0	185.0	0.0	1665000.0	0.262E+07
Web	40.0	325.0	362.5	0.0	4712499.5	0.595E+09
BasePlate	420.0	75.0	562.5	0.0	17718748.0	0.486E+10
Totals	...	...	...	0.2	39028384.0	0.716E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 39028384 / 0.23$$

$$= 170.302 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 4971032$$

$$= 1011712.8750 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 18911.0234 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1900 - 1300 \cdot \sin(120/2) / 1.05$$

$$= 824.909 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 834908.9375 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 19854.5586 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(4971032 + 23423) \cdot 420) / (4 \cdot 2616 \cdot 173320)}$$

$$= 58.906 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 112 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10,2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 4994455 / (2616 \cdot 420)$$

$$= 4546830.500 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.091 \cdot 4546830$$

$$= 413352.938 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 413353 / 0.023$$

$$= 17925.393 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	400.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (213333120 / 0.023) }$$

$$= 96.183 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1900 - \cos(120 / 2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955 / 259980) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 113 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10, 2026

$$= 1 \cdot 1075/96.2$$

$$= 11.177$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 4119 / (2 \cdot 2616) \cdot 433 \cdot 1075/2$$

$$= 183.296 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($11.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152247 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 17925/152247 + (183 \cdot 200/21333328) / 173320$$

$$= 0.119$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1900 - \cos(1500/2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (215429184/0.032) }$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325/82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4119 / (2 \cdot 2616)$$

$$= 0.787 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.79 \cdot 433 \cdot 325$$

$$= 110.830 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 114 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10, 2026

= 111/1077146
= 102.853 kPa

Compressive Allowable, $KL/r < C_c$ ($3.94 < 123$) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (3.94)^2 / (2 \cdot 123^2)) 259980 /$
 $(5/3 + 3 \cdot (3.94) / (8 \cdot 123) - (3.94^3) / (8 \cdot 123^3))$
= 154794 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
= $26242 / 154794 + (111 \cdot 200 / 215429184) / 173320$
= 0.170

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load QO (Weight)	QO	4975981.0 N
Saddle Load QL (Wind/Seismic contribution)	QL	1449.2 N
Maximum Transverse Force	Ft	8420.7 N
Maximum Longitudinal Force	F1	4118.7 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $4119 / (0.0014 \cdot 8)$
= 365.8 kPa. Must be less than 103192.1 kPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $8421 / (0.0014 \cdot 8)$
= 747.9 kPa. Must be less than 103192.1 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (BoltShearAllowable \cdot Nbolts)$
= $4119 / (103192 \cdot 8)$
= 0.0000 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
= $1900 \cdot 8421 + 0$
= 16005.84 N-m

Eccentricity (e):

= $Rmom / QO$
= $16006 / 4975981$
= 3.22 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 115 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10,2026

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.49881E-05, 0]

= 0.0000 m²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min(b + $1.56 \cdot \sqrt{Rm \cdot t}$, 2a)

= min(400 + $1.56 \cdot \sqrt{1385 \cdot 170}$, 2 • 1800)

= 1156.9617 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1385.00	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.34	m
Shell Allowable Stress used in Calculation		224982.88	kPa
Head Allowable Stress used in Calculation		224982.88	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2800300.25	N
Pressure used in Analysis	P	19.230	N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	77836.24	224982.88
Long. Stress at Bottom of Midspan	78823.12	224982.88
Long. Stress at Top of Saddles	92979.45	224982.88
Long. Stress at Bottom of Saddles	70209.88	224982.88
Tangential Shear in Shell	7220.07	133239.86
Circ. Stress at Horn of Saddle	12614.31	337474.28
Circ. Compressive Stress in Shell	1082.45	224982.88

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (16841/2 + 0) • 1900/2598

= 18474.5 N

Longitudinal Saddle Reaction Force [Fwl]:

= max(FI,wind; Sum of X Forces) • B / Ls



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 116 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10, 2026

$$= \max(4119, 0) \cdot 1900/5400$$

$$= 1449.2 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2781826 + \max(1449, 18474, 0, 0)$$

$$= 2800300.2 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766(0 + \pi/4(2.94 \cdot 1.2)^2)$$

$$= 8237.911 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2823723.00	N
Transverse Shear Load	8420.71	N
Longitudinal Shear Load	4118.74	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2600/2 + 170 + 30)\sin(120/2), 2 \cdot 1385)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a[1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -2800300 \cdot 1.8[1 - (1 - 1.8/9.4 + (1.39^2 - 1.34^2)/(2 \cdot 1.8 \cdot 9.4))/(1 + (4 \cdot 1.34)/(3 \cdot 9.4))]$$

$$= -1600647.8 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 2800300 \cdot 9.4/4((1 + 2(1.39^2 - 1.34^2)/(9.4^2))/(1 + (4 \cdot 1.34)/(3 \cdot 9.4)) - 4 \cdot 1.8/9.4)$$

$$= 505705.5 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 19.2 \cdot 1385/(2 \cdot 170) - 505706/(\pi \cdot 1385^2 \cdot 170)$$

$$= 77836.24 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 19.2 \cdot 1385/(2 \cdot 170) + 505706/(\pi \cdot 1385^2 \cdot 170)$$

$$= 78823.12 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 19.2 \cdot 1385/(2 \cdot 170) - 1600647/(0.11 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= 92979.45 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 19.2 \cdot 1385/(2 \cdot 170) + 1600647/(0.19 \cdot \pi \cdot 1385^2 \cdot 170)$$

$$= 70209.88 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 117 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10,2026

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 2800300 (9.4 - 2 \cdot 1.8)/(9.4 + (4 \cdot 1.34/3))$$

$$= 1452060.9 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1452060 / (1385 \cdot 170)$$

$$= 7220.07 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (1385 \cdot 170) }$$

$$= 378.481 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t (b + X_1 + X_2))$$

$$= -0.76 \cdot 2800300 \cdot 0.1 / (170 (400 + 378 + 378))$$

$$= -1082.45 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t (b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -2800300 / (4 \cdot 170 (400 + 378 + 378)) -$$

$$12 \cdot 0.053 \cdot 2800300 \cdot 1385 / (9.4 \cdot 170^2)$$

$$= -12614.31 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B_1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{ (1385 \cdot 170) }, 2 \cdot 1.8)$$

$$= 1156.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (1300 \cdot 170) } + 300) 170^3) / 12$$

$$= 423076832.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 423076832 + 0.18 (170 - 85)^2$$

$$= 1701329024.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	1033.4	170.0	85.0	0.2	14932138.0	0.170E+10
Wearplate	300.0	30.0	185.0	0.0	1665000.0	0.262E+07
Web	40.0	325.0	362.5	0.0	4712499.5	0.595E+09
BasePlate	420.0	75.0	562.5	0.0	17718748.0	0.486E+10
Totals	...	...	...	0.2	39028384.0	0.716E+10

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 118 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10, 2026

$$\begin{aligned}
 &= AY / A \\
 &= 39028384 / 0.23 \\
 &= 170.302 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120 / 2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.2 \cdot 2800300 \\
 &= 569921.8750 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 10653.0283 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1900 - 1300 \cdot \sin(120/2) / 1.05 \\
 &= 824.909 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 470324.0312 \text{ N-m}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 11184.5439 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\
 &= \sqrt{ 3(2800300 + 23423) 420 / (4 \cdot 2616 \cdot 173320) } \\
 &= 44.292 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120 / 2) (1300 + 170 + 30) \\
 &= 2598.076 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 2598 / (7 - 1) \\
 &= 433.013 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 433 \cdot 420 / 2 \\
 &= 0.091 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 2823722 / (2616 \cdot 420) \\
 &= 2570648.750 \text{ N/m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 0.091 \cdot 2570648 \\
 &= 233698.000 \text{ N}
 \end{aligned}$$

$$\text{Area of the Rib and Web [Ar]:}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 119 of 220

Saddle Calcs: Test Case:

Step: 19 4:24pm Apr 10, 2026

= Rib Area + Web Area
= 0.014 + 0.0087
= 0.023 m²

Compressive Stress [Sc]:
= P/Ar
= 233698/0.023
= 10134.507 kPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:
= Saddle Width - Web Thickness
= 400 - 40
= 360.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/0.023
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 400/2
= 200.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{213333120/0.023}$
= 96.183 mm

Length of Outer Rib [L]:
= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1900 - cos(120/2)(1300 + 170 + 30) - 75
= 1075.000 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199955/259980}$
= 123.211

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1075/96.2
= 11.177

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 4119 / (2 • 2616) • 433 • 1075/2
= 183.296 N-m

Compressive Allowable, KL/r < Cc (11.2 < 123) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (11.2)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$
= 152247 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
= Sc/Sca + (Rm • C1 / I) / Sba



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 120 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10,2026

$$= 10135/152247 + (183 \cdot 200/21333328)/173320$$

$$= 0.068$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1900 - \cos(1500/2)(75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(215429184/0.032)}$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325/82.4$$

$$= 3.944$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4119/(2 \cdot 2616)$$

$$= 0.787 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.79 \cdot 433 \cdot 325$$

$$= 110.830 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 111/1077146$$

$$= 102.853 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (3.94 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2/(2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$$

$$= (1 - (3.94)^2/(2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (3.94)/(8 \cdot 123) - (3.94^3)/(8 \cdot 123^3))$$

$$= 154794 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I)/Sba$$

$$= 14836/154794 + (111 \cdot 200/215429184)/173320$$

$$= 0.096$$

Input Data for Base Plate Bolting Calculations:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 121 of 220

Saddle Calcs: Test Case: Step: 19 4:24pm Apr 10, 2026

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m²
Saddle Load Q0 (Weight)	Q0	2805248.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	1449.2 N
Maximum Transverse Force	Ft	8420.7 N
Maximum Longitudinal Force	Fl	4118.7 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4119 / (0.0014 \cdot 8)$$

$$= 365.8 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8421 / (0.0014 \cdot 8)$$

$$= 747.9 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 4119 / (103192 \cdot 8)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1900 \cdot 8421 + 0$$

$$= 16005.84 \text{ N-m}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 16006 / 2805248$$

$$= 5.70 \text{ mm} < Bplen / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.49881\text{E-}05, 0]$$

$$= 0.0000 \text{ m}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 122 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

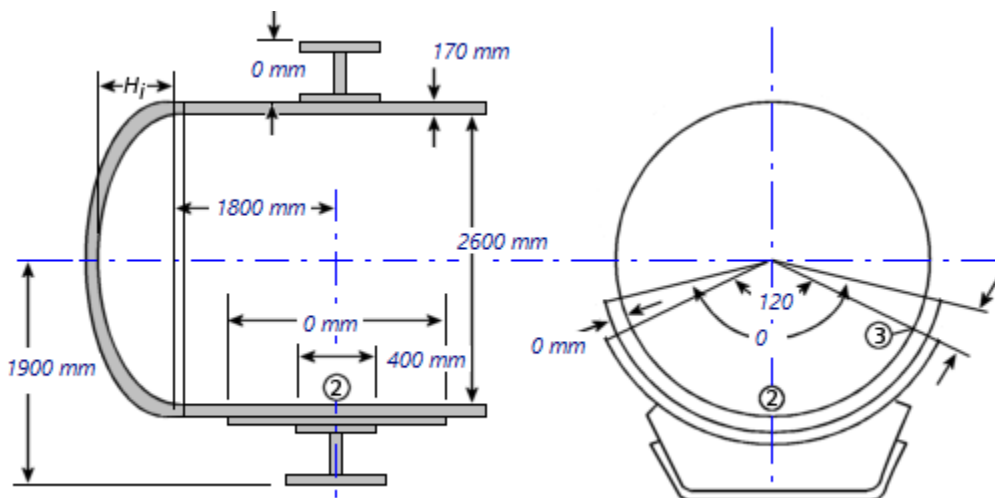
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1800)$$

$$= 1150.6591 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1386.50	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		128860.33	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	4229826.50	N
Pressure used in Analysis	P	-0.101 N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	-1161.22	-165952.55
Long. Stress at Top	of Midspan	1161.22	122834.11



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 123 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Long. Stress at Bottom of Midspan	320.19	122834.11
Long. Stress at Top of Saddles	22169.79	122834.11
Long. Stress at Bottom of Saddles	-12941.46	-165952.55
Long. Stress at Bottom of Saddles	12941.46	122834.11
Tangential Shear in Shell	11074.57	98267.29
Circ. Stress at Horn of Saddle	19691.51	153542.62
Circ. Compressive Stress in Shell	1673.52	122834.11

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (33683/2 + 152053) \cdot 1900/2598$$

$$= 370542.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8237, 260476) \cdot 1900/5400$$

$$= 4615.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(307588, 260476) \cdot 1900/5400$$

$$= 108225.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (307588/2 + 152053) \cdot 1900/2598$$

$$= 671007.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 3558819 + \max(4616, 370543, 108226, 671008)$$

$$= 4229826.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.94 \cdot 1.2)^2)$$

$$= 8237.911 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4253249.50	N
Transverse Shear Load	305847.25	N
Longitudinal Shear Load	307588.31	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2600/2 + 170 + 30) \sin(120/2), 2 \cdot 1386)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -4229826 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.35^2) / (2 \cdot 1.8 \cdot 9.4)) / (1 + (4 \cdot 1.35) / (3 \cdot 9.4))]$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 124 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

= -2429937.8 N-m

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 4229826 \cdot 9.4 / 4 \cdot ((1 + 2(1.39^2 - 1.35^2) / (9.4^2)) / (1 + (4 \cdot 1.35) / (3 \cdot 9.4)) - 4 \cdot 1.8 / 9.4)$$

$$= 747332.5 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -0.1 \cdot 1386 / (2 \cdot 167) - 747332 / (\pi \cdot 1386^2 \cdot 167)$$

$$= -1161.22 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1386 / (2 \cdot 167) + 747332 / (\pi \cdot 1386^2 \cdot 167)$$

$$= 320.19 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -0.1 \cdot 1386 / (2 \cdot 167) - 2429937 / (0.11 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= 22169.79 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1386 / (2 \cdot 167) + 2429937 / (0.19 \cdot \pi \cdot 1386^2 \cdot 167)$$

$$= -12941.46 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 4229826 \cdot (9.4 - 2 \cdot 1.8) / (9.4 + (4 \cdot 1.35 / 3))$$

$$= 2190321.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 2190321 / (1386 \cdot 167)$$

$$= 11074.57 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1386 \cdot 167)}$$

$$= 375.330 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 4229826 \cdot 0.1 / (167(400 + 375 + 375))$$

$$= -1673.52 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -4229826 / (4 \cdot 167(400 + 375 + 375)) - 12 \cdot 0.053 \cdot 4229826 \cdot 1386 / (9.4 \cdot 167^2)$$

$$= -19691.51 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1386 \cdot 167)}, 2 \cdot 1.8)$$

$$= 1150.66 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 125 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw, c	438.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1303 \cdot 167)} + 300)167^3)/12$$

$$= 398874272.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 398874272 + 0.17(170 - 83.5)^2$$

$$= 1682372864.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	1027.7	167.0	83.5	0.2	14330825.0	0.168E+10
Wearplate	300.0	30.0	182.0	0.0	1638000.0	0.198E+07
Web	40.0	325.0	359.5	0.0	4673499.5	0.581E+09
BasePlate	420.0	75.0	559.5	0.0	17624248.0	0.479E+10
Totals	...	...	...	0.2	38266576.0	0.706E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 38266576/0.23$$

$$= 169.978 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 4229826$$

$$= 860861.4375 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 16091.2959 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1900 - 1303 \cdot \sin(120/2) / 1.05$$

$$= 822.428 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 708283.3125 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 17046.5684 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(4229826 + 23423)420 / (4 \cdot 2616 \cdot 129781)}$$

$$= 62.820 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 126 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4253249 / (2616 \cdot 420)$$

$$= 3872054.750 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.091 \cdot 3872054$$

$$= 352008.969 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 352009 / 0.023$$

$$= 15265.161 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(213333120 / 0.023)}$$

$$= 96.183 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 127 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900 - \cos(120/2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 194672)}$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1075 / 96.2$$

$$= 11.177$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 307588 / (2 \cdot 2616) \cdot 433 \cdot 1075 / 2$$

$$= 13688.544 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($11.2 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (11.2) / (8 \cdot 142) - (11.2^3) / (8 \cdot 142^3))$$

$$= 114427 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 15265 / 114427 + (13689 \cdot 200 / 213333328) / 129781$$

$$= 0.232$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1900 - \cos(1500/2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(215429184 / 0.032)}$$

$$= 82.410 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325 / 82.4$$

$$= 3.944$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 128 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10,2026

Unit Force [Force,u]:
 = $F_l / (2 \cdot \text{Baseplate Length})$
 = $307588 / (2 \cdot 2616)$
 = 58.790 N/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force $\cdot e \cdot L$
 = $58.8 \cdot 433 \cdot 325$
 = 8276.794 N-m

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $8277 / 1077146$
 = 7681.089 kPa

Compressive Allowable, $KL/r < C_c$ ($3.94 < 142$) per AISC E2-1 [Sca]:
 = $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (3.94)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (3.94) / (8 \cdot 142) - (3.94^3) / (8 \cdot 142^3))$
 = 116036 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 = $22347 / 116036 + (8277 \cdot 200 / 215429184) / 129781$
 = 0.252

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(400 + 1.56 \cdot \sqrt{1386 \cdot 167}, 2 \cdot 1800)$
 = 1150.6591 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1386.50	mm
Shell Thickness used in this Case	t	170.000	mm
Stiffened Vessel Length per 4.15.6	L	9.40	m
Distance from Saddle to Vessel tangent	a1 or a	1800.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.34	m
Shell Allowable Stress used in Calculation		122834.11	kPa
Head Allowable Stress used in Calculation		129114.78	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 2507326.00 N
 Pressure used in Analysis P -0.101 N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 129 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10,2026

Long. Stress at Top of Midspan	-868.43	-165952.55
Long. Stress at Top of Midspan	868.43	122834.11
Long. Stress at Bottom of Midspan	27.40	122834.11
Long. Stress at Top of Saddles	12908.29	122834.11
Long. Stress at Bottom of Saddles	-7808.16	-165952.55
Long. Stress at Bottom of Saddles	7808.16	122834.11
Tangential Shear in Shell	6572.53	98267.29
Circ. Stress at Horn of Saddle	11672.59	153542.62
Circ. Compressive Stress in Shell	992.02	122834.11

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (33683/2 + 152052) \cdot 1900/2598$$

$$= 370540.2 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}, \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(8237, 260476) \cdot 1900/5400$$

$$= 4615.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}, \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(307588, 260476) \cdot 1900/5400$$

$$= 108225.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (307588/2 + 152052) \cdot 1900/2598$$

$$= 671005.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1836320 + \max(4616, 370540, 108226, 671005)$$

$$= 2507326.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.94 \cdot 1.2)^2)$$

$$= 8237.911 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2530748.50 N

Transverse Shear Load 305846.03 N

Longitudinal Shear Load 307588.31 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + \text{t} + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2600/2 + 170 + 30) \sin(120/2), 2 \cdot 1386)$$

$$= 2598.076 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 130 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -2507326 \cdot 1.8 [1 - (1 - 1.8/9.4 + (1.39^2 - 1.34^2)/(2 \cdot 1.8 \cdot 9.4))/(1 + (4 \cdot 1.34)/(3 \cdot 9.4))] \\
 &= -1433719.5 \text{ N-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 2507326 \cdot 9.4/4 ((1 + 2(1.39^2 - 1.34^2)/(9.4^2))/(1 + (4 \cdot 1.34)/(3 \cdot 9.4)) - 4 \cdot 1.8/9.4) \\
 &= 451924.6 \text{ N-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\
 &= -0.1 \cdot 1386/(2 \cdot 167) - 451925/(\pi \cdot 1386^2 \cdot 167) \\
 &= -868.43 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= -0.1 \cdot 1386/(2 \cdot 167) + 451925/(\pi \cdot 1386^2 \cdot 167) \\
 &= 27.40 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -0.1 \cdot 1386/(2 \cdot 167) - 1433719/(0.11 \cdot \pi \cdot 1386^2 \cdot 167) \\
 &= 12908.29 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -0.1 \cdot 1386/(2 \cdot 167) + 1433719/(0.19 \cdot \pi \cdot 1386^2 \cdot 167) \\
 &= -7808.16 \text{ kPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 2507326 (9.4 - 2 \cdot 1.8)/(9.4 + (4 \cdot 1.34/3)) \\
 &= 1299910.2 \text{ N}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 1299910/(1386 \cdot 167) \\
 &= 6572.53 \text{ kPa}
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1386 \cdot 167)} \\
 &= 375.330 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.76 \cdot 2507326 \cdot 0.1/(167(400 + 375 + 375)) \\
 &= -992.02 \text{ kPa}
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ^*7]:

$$\begin{aligned}
 &= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm/(L \cdot t^2) \\
 &= -2507326/(4 \cdot 167(400 + 375 + 375)) - 12 \cdot 0.053 \cdot 2507326 \cdot 1386/(9.4 \cdot 167^2) \\
 &= -11672.59 \text{ kPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(400 + 1.56 \cdot \sqrt{(1386 \cdot 167)}, 2 \cdot 1.8) \\
 &= 1150.66 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length Bplen 2616.0000 mm



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 131 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10,2026

Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	7	
Rib Thickness	Ribtk	40.0000	mm
Web Thickness	Webtk	40.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	438.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1303 \cdot 167) + 300})^3) / 12$$

$$= 398874272.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 398874272 + 0.17 (170 - 83.5)^2$$

$$= 1682372864.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	1027.7	167.0	83.5	0.2	14330825.0	0.168E+10
Wearplate	300.0	30.0	182.0	0.0	1638000.0	0.198E+07
Web	40.0	325.0	359.5	0.0	4673499.5	0.581E+09
BasePlate	420.0	75.0	559.5	0.0	17624248.0	0.479E+10
Totals	...	...	...	0.2	38266576.0	0.706E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 38266576 / 0.23$$

$$= 169.978 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2507326$$

$$= 510295.1875 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 9538.4814 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1900 - 1303 \cdot \sin(120/2) / 1.05$$

$$= 822.428 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 419851.0625 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 10104.7412 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 132 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(2507326 + 23423)420}{4 \cdot 2616 \cdot 129781}}$$

$$= 48.457 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1300 + 170 + 30)$$

$$= 2598.076 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2598 / (7 - 1)$$

$$= 433.013 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 433 \cdot 420 / 2$$

$$= 0.091 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2530748 / (2616 \cdot 420)$$

$$= 2303932.000 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.091 \cdot 2303932$$

$$= 209450.734 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.014 + 0.0087$$

$$= 0.023 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 209451 / 0.023$$

$$= 9083.005 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	40.0	400.0	...	...	...	0.213E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 40$$

$$= 360.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 0.023$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 133 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{213333120 / 0.023}$$

$$= 96.183 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900 - \cos(120/2) (1300 + 170 + 30) - 75$$

$$= 1075.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955 / 194672}$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1075 / 96.2$$

$$= 11.177$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 307588 / (2 \cdot 2616) \cdot 433 \cdot 1075 / 2$$

$$= 13688.544 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($11.2 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 142^2)) 194672 /$$

$$(5/3 + 3 \cdot (11.2) / (8 \cdot 142) - (11.2^3) / (8 \cdot 142^3))$$

$$= 114427 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 9083 / 114427 + (13689 \cdot 200 / 213333328) / 129781$$

$$= 0.178$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	40.0	360.0	0.0	0.0	0.0	0.213E+09
Web	433.0	40.0	0.0	0.0	0.0	0.231E+07
Totals	...	...	...	...	...	0.215E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.032$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1900 - \cos(1500/2) (75 + 0 + 0) - 0$$

$$= 325.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{215429184 / 0.032}$$

$$= 82.410 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 134 of 220

Saddle Calcs: Ext. Pressure: Step: 20 4:24pm Apr 10, 2026

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 325/82.4$$

$$= 3.944$$

Unit Force $[Force, u]$:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 307588 / (2 \cdot 2616)$$

$$= 58.790 \text{ N/mm}$$

Moment at base of inner Rib $[M_{base, c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 58.8 \cdot 433 \cdot 325$$

$$= 8276.794 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_B, base, c]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 8277/1077146$$

$$= 7681.089 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($3.94 < 142$) per AISC E2-1 $[Sca]$:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.94)^2 / (2 \cdot 142^2)) 194672 /$$

$$(5/3 + 3 \cdot (3.94) / (8 \cdot 142) - (3.94^3) / (8 \cdot 142^3))$$

$$= 116036 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base, c} \cdot C1 / I) / S_{ba}$$

$$= 13297/116036 + (8277 \cdot 200/215429184) / 129781$$

$$= 0.174$$

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 123-C2-NVS

Page 135 of 220

Nozzle Calcs.: Manhole

Nozl: 11 4:24pm Apr 10,2026

Input, Nozzle Desc: Manhole

From: 10

Pressure for Reinforcement Calculations	P	4.9511	N/mm ²
Temperature for Internal Pressure	Temp	370	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	128860.33	kPa
Parent Allowable Stress At Ambient	Sva	137996.80	kPa
Inside Diameter of Hemispherical Head	D	2600.00	mm
Head Finished (Minimum) Thickness	t	98.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138190.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	125.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		270.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	50.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



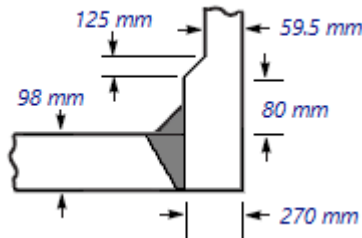
PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 136 of 220

Nozzle Calcs.: Manhole

Nozl: 11 4:24pm Apr 10, 2026



Reinforcement CALCULATION, Description: Manhole

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	19.314 in.
Actual Thickness Used in Calculation	2.343 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 4.95 \cdot 1303 / (2 \cdot 128860 \cdot 1 - 0.2 \cdot 4.95)$
 $= 25.1273 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 248 / (138191 \cdot 1 - 0.6 \cdot 4.95)$
 $= 9.0906 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 192 / 0.5773$
 $= 332.1652 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512 mm
Parallel to Vessel Wall, opening length	d	496.5756 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	237.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138191 / 128860)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138191 / 128860)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



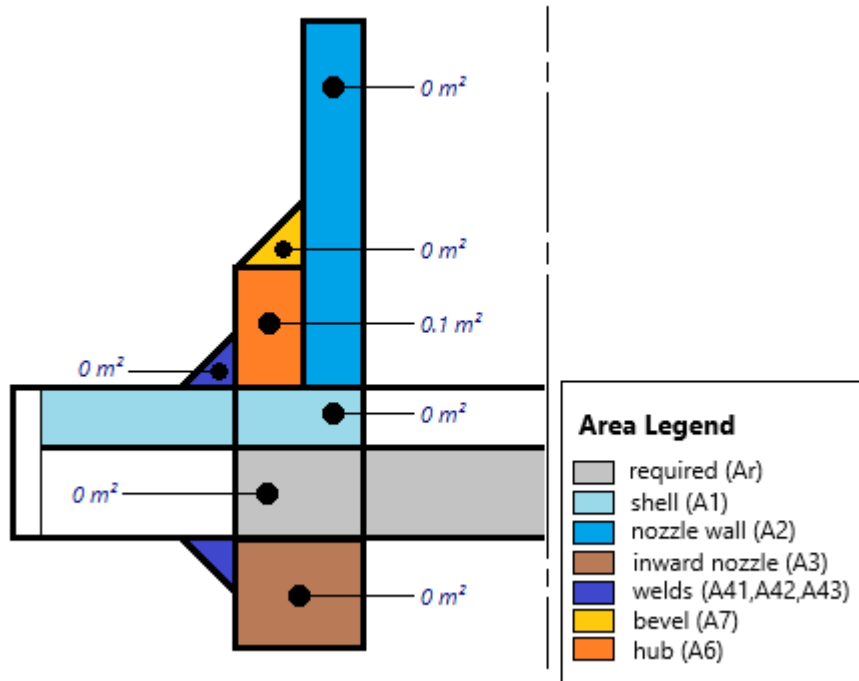
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FileName : 123-C2-NVS

Page 137 of 220

Nozzle Calcs.: Manhole

Nozl: 11 4:24pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.012	NA	NA
Area in Shell	A1	0.035	NA	NA
Area in Nozzle Wall	A2	0.023	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.055	NA	NA
TOTAL AREA AVAILABLE	Atot	0.112	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (497 \cdot 25.1 \cdot 1 + 2 \cdot 248 \cdot 25.1 \cdot 1 \cdot (1 - 1))$$

$$= 0.012 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 497(1 \cdot 95 - 1 \cdot 25.1) - 2 \cdot 248$$

$$(1 \cdot 95 - 1 \cdot 25.1) \cdot (1 - 1)$$

$$= 0.035 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 238)(56.5 - 9.09)1$$

$$= 0.023 \text{ m}^2$$



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FileName : 123-C2-NVS

Page 138 of 220

Nozzle Calcs.: Manhole

Nozl: 11 4:24pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot fr_2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost} \cdot fr_2$$

$$= (50^2 - 0.0025) \cdot 1 + (0^2 - 0) \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(238, 500, 80)) \cdot (251 - 59.5) \cdot 1$$

$$= 0.055 \text{ m}^2 \text{ Includes Bevel Area of } 0.024 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 12.0906 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 12.0906 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 52.1$, $t_r = 8.48$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.17$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 98$, $t_r = 23.5$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.25$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	47 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : $-104 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.95/25.51 = 0.194$$

Weld Size Calculations, Description: Manhole

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$35.3500 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (0.012 - 0.035 + 2 \cdot 248 \cdot 1 \cdot (1 \cdot 95 - 25.1)) 128860)$$

$$= 1607822.38 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 139 of 220

Nozzle Calcs.: Manhole

Nozl: 11 4:24pm Apr 10, 2026

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (0.077 + 0 - 0 \cdot 1) \cdot 128860$$

$$= 9945341.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (0.077 + 0 + 0 + (0.047)) \cdot 128860$$

$$= 16024122.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (0.077 + 0 + 0 + 0 + (0.047)) \cdot 128860$$

$$= 16024122.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 993 \cdot 50 \cdot 0.49 \cdot 128860$$

$$= 4925030. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 372) \cdot (251 - 3) \cdot 0.7 \cdot 138191$$

$$= 28100698. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 993 \cdot (50 - 0) \cdot 0.74 \cdot 138191$$

$$= 7976349. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (4925030 + 28100698) = 33025728 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (4925030 + 0 + 7976349 + 0) = 12901379 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (4925030 + 7976349 + 0) = 12901379 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 33025728N	, must exceed W = 1607822 N	or W1 = 9945341 N
Path 2-2 = 12901379N	, must exceed W = 1607822 N	or W2 = 16024122 N
Path 3-3 = 12901379N	, must exceed W = 1607822 N	or W3 = 16024122 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.95 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 98.5790 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 696.5790 mm

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FileName : 123-C2-NVS

Page 140 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10,2026

Input, Nozzle Desc: Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	4.9511	N/mm ²
Temperature for Internal Pressure	Temp	370	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	138190.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2600.00	mm
Shell Finished (Minimum) Thickness	t	190.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.25	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138190.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	76.2000	mm
Hub Height of Integral Nozzle	h	210.0000	mm
Height of Beveled Transition	L'	95.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		310.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	50.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16			g

The Pressure Design option was Overall MAWP.



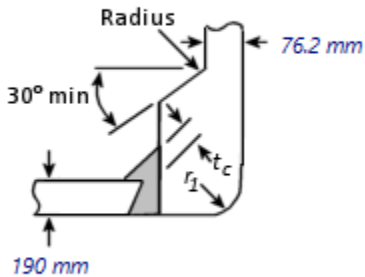
PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 141 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10, 2026



UW-16.1 [g]

Reinforcement CALCULATION, Description: Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	3.000 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1303 / (138191 \cdot 1 - 0.6 \cdot 4.95)$
 $= 47.7069 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 384 / (138191 \cdot 1 - 0.6 \cdot 4.95)$
 $= 14.0594 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 234 / 0.5773$
 $= 404.9536 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000	mm
Parallel to Vessel Wall, opening length	d	768.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	467.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138191 / 138191)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138191 / 138191)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:



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FileName : 123-C2-NVS

Page 142 of 220

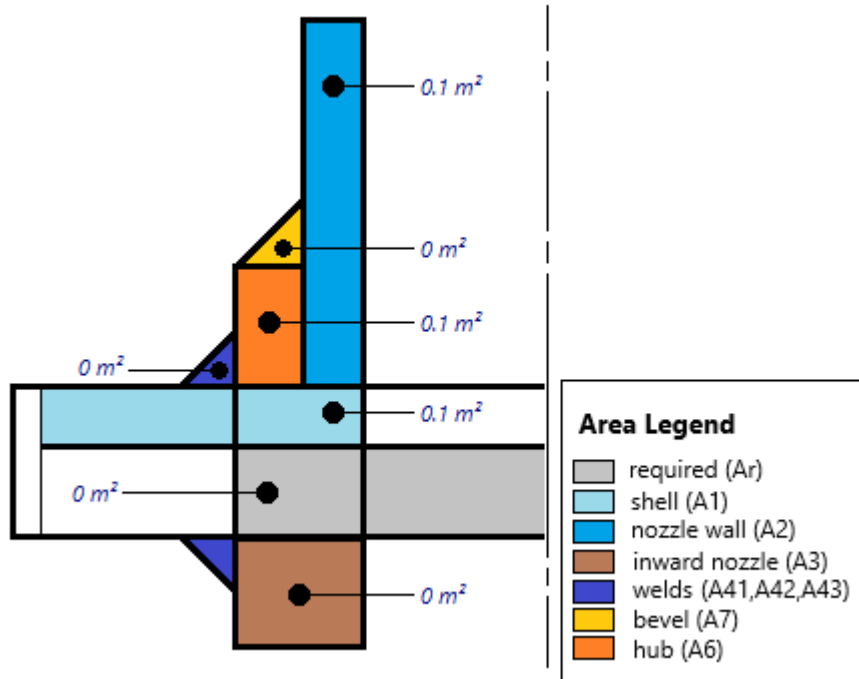
Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10, 2026

= min(fr2, fr4)

= min(1, 1)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.037	NA	NA
Area in Shell	A1	0.107	NA	NA
Area in Nozzle Wall	A2	0.055	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.002	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.120	NA	NA
TOTAL AREA AVAILABLE	Atot	0.285	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)

= (768 • 47.7 • 1 + 2 • 307 • 47.7 • 1 • (1-1))

= 0.037 m²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)

= 768 (1 • 187 - 1 • 47.7) - 2 • 307

(1 • 187 - 1 • 47.7) • (1 - 1)

= 0.107 m²

Area Available in Nozzle Projecting Outward [A2]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 143 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10, 2026

$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr^2$$

$$= (2 \cdot 468) (73.2 - 14.1) \cdot 1$$

$$= 0.055 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr^2 + (W_i - c_{an}/0.707)^2 \cdot fr^2$$

$$= 50^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.002 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr^2$$

$$= (2 \cdot \min(468, 600, 210)) \cdot (310 - 76.2) \cdot 1$$

$$= 0.120 \text{ m}^2 \text{ Includes Bevel Area of } 0.022 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 17.0594 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 50.7069 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 50.7069 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(50.7, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(17.1, 11.3)$$

$$= 17.0594 \text{ mm}$$

Available Nozzle Neck Thickness = 76.2000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17697.6,	Allowable	: 138190.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 340026.7 kPa	Passed
Occasional	: 11855.8,	Allowable	: 183793.7 kPa	Passed
Shear	: 4102.4,	Allowable	: 96733.5 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: Tube inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$35.3500 = 0.7 \cdot W_o \text{ mm}$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.95 N/mm²



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 144 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10, 2026

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 198.8556 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 988.8556 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm

Input Echo, WRC297 Item 1, Description: Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2600.000	mm
Shell Thickness	190.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 138190.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	310.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 138190.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.95	N/mm ²
Radial Load	P	-93715.80	N
Circumferential Shear	Vc	74967.03	N
Longitudinal Shear	Vl	74967.03	N
Circumferential Moment	Mc	168748.14	N-m
Longitudinal Moment	Ml	168748.14	N-m
Torsional Moment	Mt	196872.38	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (15.684) is < 20
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (4.458) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 1382.0 + 2 \cdot 1.65 \cdot \sqrt{(1396.5(190.0 - 3.0))} \\
 &= 3068.379 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2793.000 mm



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FileName : 123-C2-NVS

Page 145 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10,2026

Nozzle Outside Diameter (d) = 1382.000 mm
 Vessel Thickness used (T) = 187.000 mm
 Nozzle Thickness used (t) = 307.000 mm
 T / t = 0.609
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.912$$

Nr/P = 0.074
 Mr/P = 0.140
 M0/P = 0.042
 N0/P = 0.101
 MrD/Mc = 0.278
 NrDL/Mc = 0.109
 M0d/Mc = 0.089
 N0DL/Mc = 0.065
 MrD/Ml = 0.117
 NrDL/Ml = 0.069
 M0D/Ml = 0.039
 N0DL/Ml = 0.127

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	268	268	268	268
Outplane Bending (P)	668	-668	668	-668
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-441	-441	441	441
Outplane Bending (ML)	-813	813	813	-813
Normal Pressure Stress	32183	37134	32183	37134
Outplane Stress Summary	31866	37106	34376	36362

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	193	193	193	193
Inplane Bending (P)	2247	-2247	2247	-2247
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-234	-234	234	234
Inplane Bending (ML)	-2440	2440	2440	-2440
Inplane Pressure Stress	16091	16091	16091	16091
Inplane Stress Summary	15857	16243	21207	11831

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	179	179	-179	-179
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	344	344	344	344
Shear Stress Summary	523	523	165	165

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
--	-----------	-----------	--------------	--------------



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 146 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10,2026

	Outside	Inside	Outside	Inside
Two * Max Shear Stress	31880	37113	34376	36362

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	268	268	268	268
Outplane Bending (P)	668	-668	668	-668
Outplane Membrane (Mc)	-220	-220	220	220
Outplane Bending (Mc)	-1861	1861	1861	-1861
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16091	16091	16091	16091

Outplane Stress Summary	14947	17333	19111	14051
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	193	193	193	193
Inplane Bending (P)	2247	-2247	2247	-2247
Inplane Membrane (Mc)	-379	-379	379	379
Inplane Bending (Mc)	-5819	5819	5819	-5819
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32183	37134	32183	37134

Inplane Stress Summary	28426	40519	40822	29639
------------------------	-------	-------	-------	-------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-179	-179	179	179
Torsional Shear (Mt)	344	344	344	344

Shear Stress Summary	165	165	523	523
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	28426	40519	40829	29653
------------------------	-------	-------	-------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	268	268	268	268
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-441	-441	441	441
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5308	11245	5308	11245



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 147 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10,2026

Hoop Stress Summary	5136	11072	6018	11955
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	89	89	89	89
Axial Bending (P)	613	-613	613	-613
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-717	-717	717	717
Axial Bending (ML)	-634	634	634	-634
Axial Pressure Stress	2654	2654	2654	2654

Axial Stress Summary	2006	2047	4709	2213
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	110	110	-110	-110
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	206	206	206	206

Shear Stress Summary	317	317	96	96
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	5164	11079	6025	11955
------------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	268	268	268	268
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-220	-220	220	220
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5308	11245	5308	11245

Hoop Stress Summary	5357	11293	5798	11734
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	89	89	89	89
Axial Bending (P)	613	-613	613	-613
Axial Membrane (Mc)	-717	-717	717	717
Axial Bending (Mc)	-1730	1730	1730	-1730
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2654	2654	2654	2654

Axial Stress Summary	910	3143	5805	1116
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 148 of 220

Nozzle Calcs.: Tube inlet

Noz1: 12 4:24pm Apr 10, 2026

(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-110	-110	110	110
Shear due to Torsion	206	206	206	206
Shear Stress Summary	96	96	317	317

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5357	11293	6115	11741

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32183	37134	32183	37134	32183	37134	32183	37134
Circ. Pl (SUS)	-173	-173	709	709	-186	-186	572	572
Circ. Q (SUS)	-145	145	1481	-1481	-3572	3572	8066	-8066
Long. Pm (SUS)	16091	16091	16091	16091	16091	16091	16091	16091
Long. Pl (SUS)	-41	-41	427	427	48	48	488	488
Long. Q (SUS)	-193	193	4687	-4687	-1193	1193	2529	-2529
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	179	179	-179	-179	-179	-179	179	179
Shear Q (SUS)	344	344	344	344	344	344	344	344
Pm (SUS)	32183	37134	32183	37134	32183	37134	32183	37134
Pm+Pl (SUS)	32012	36962	32893	37844	31999	36949	32756	37707
Pm+Pl+Q (Total)	31882	37119	34375	36363	28427	40521	40833	29657

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	37134	138190	Passed
Pm+P1 (SUS)	37844	207286	Passed
Pm+P1+Q (TOTAL)	40833	414572	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

[illegible]



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 149 of 220

Nozzle Calcs.: Tube inlet

Nozl: 12 4:24pm Apr 10,2026

Long. P1 (SUS)	-628	-628	806	806	-628	-628	806	806
Long. Q (SUS)	-21	21	1247	-1247	-1117	1117	2343	-2343
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	110	110	-110	-110	-110	-110	110	110
Shear Q (SUS)	206	206	206	206	206	206	206	206
Pm (SUS)	5308	11245	5308	11245	5308	11245	5308	11245
Pm+P1 (SUS)	5138	11073	6021	11955	5359	11294	5801	11734
Pm+P1+Q (Total)	5166	11083	6023	11954	5358	11294	6115	11742

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11245	138190	Passed
Pm+P1 (SUS)	11955	207286	Passed
Pm+P1+Q (TOTAL)	11954	414572	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 150 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

Input, Nozzle Desc: Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	4.9511	N/mm ²
Temperature for Internal Pressure	Temp	370	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	138190.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2600.00	mm
Shell Finished (Minimum) Thickness	t	190.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.25	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138190.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	76.2000	mm
Hub Height of Integral Nozzle	h	210.0000	mm
Height of Beveled Transition	L'	95.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		310.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	522.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	50.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16			g

The Pressure Design option was Overall MAWP.



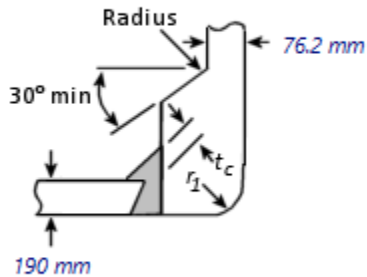
PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 151 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10, 2026



UW-16.1 [g]

Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	3.000 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1303 / (138191 \cdot 1 - 0.6 \cdot 4.95)$
 $= 47.7069 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 384 / (138191 \cdot 1 - 0.6 \cdot 4.95)$
 $= 14.0594 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 234 / 0.5773$
 $= 404.9536 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000 mm
Parallel to Vessel Wall, opening length	d	768.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	467.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138191 / 138191)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138191 / 138191)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 152 of 220

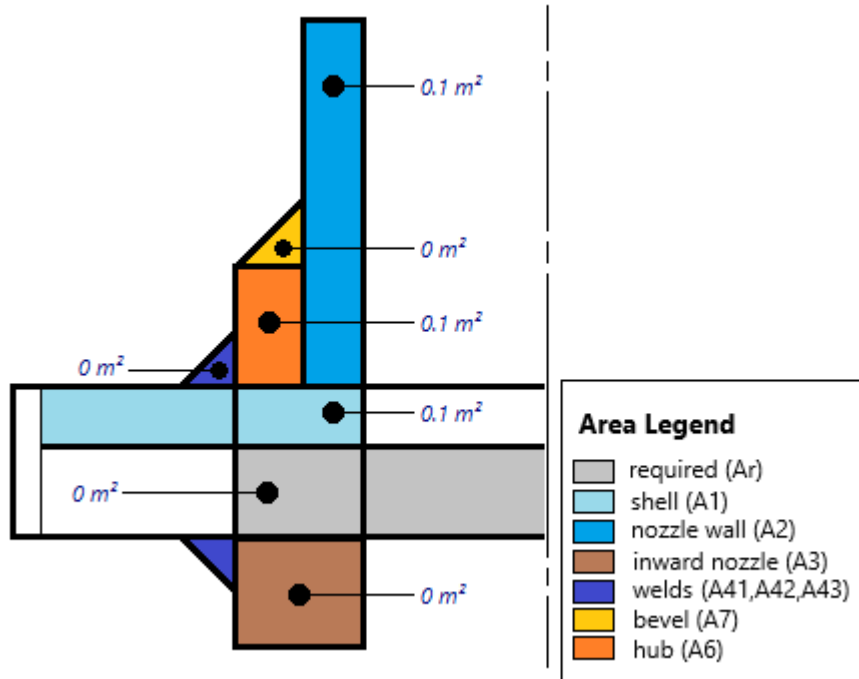
Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10, 2026

= min(fr2, fr4)

= min(1, 1)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.037	NA	NA
Area in Shell	A1	0.107	NA	NA
Area in Nozzle Wall	A2	0.055	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.002	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.120	NA	NA
TOTAL AREA AVAILABLE	Atot	0.285	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)

= (768 • 47.7 • 1 + 2 • 307 • 47.7 • 1 • (1-1))

= 0.037 m²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)

= 768 (1 • 187 - 1 • 47.7) - 2 • 307

(1 • 187 - 1 • 47.7) • (1 - 1)

= 0.107 m²

Area Available in Nozzle Projecting Outward [A2]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 153 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10, 2026

$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr^2$$

$$= (2 \cdot 468) (73.2 - 14.1) \cdot 1$$

$$= 0.055 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr^2 + (W_i - c_{an}/0.707)^2 \cdot fr^2$$

$$= 50^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.002 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr^2$$

$$= (2 \cdot \min(468, 522, 210)) \cdot (310 - 76.2) \cdot 1$$

$$= 0.120 \text{ m}^2 \text{ Includes Bevel Area of } 0.022 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 17.0594 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 50.7069 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 50.7069 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(50.7, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(17.1, 11.3)$$

$$= 17.0594 \text{ mm}$$

Available Nozzle Neck Thickness = 76.2000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17697.6,	Allowable	: 138190.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 340026.7 kPa	Passed
Occasional	: 11855.8,	Allowable	: 183793.7 kPa	Passed
Shear	: 4102.4,	Allowable	: 96733.5 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$35.3500 = 0.7 \cdot W_o \text{ mm}$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.95 N/mm²



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 154 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 198.8556 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 910.8557 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2600.000	mm
Shell Thickness	190.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 138190.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	310.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 138190.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.95	N/mm ²
Radial Load	P	-93715.80	N
Circumferential Shear	Vc	74967.03	N
Longitudinal Shear	Vl	74967.03	N
Circumferential Moment	Mc	168748.14	N-m
Longitudinal Moment	Ml	168748.14	N-m
Torsional Moment	Mt	196872.38	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (15.684) is < 20
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.458) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1382.0 + 2 \cdot 1.65 \cdot \sqrt{1396.5(190.0 - 3.0)}$$

$$= 3068.379 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2793.000 mm
Nozzle Outside Diameter	(d) =	1382.000 mm
Vessel Thickness used	(T) =	187.000 mm
Nozzle Thickness used	(t) =	307.000 mm
T / t	=	0.609



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 155 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.912$$

Nr/P	=	0.074
Mr/P	=	0.140
MØ/P	=	0.042
NØ/P	=	0.101
MrD/Mc	=	0.278
NrDL/Mc	=	0.109
MØd/Mc	=	0.089
NØDL/Mc	=	0.065
MrD/Ml	=	0.117
NrDL/Ml	=	0.069
MØD/Ml	=	0.039
NØDL/Ml	=	0.127

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	268	268	268	268
Outplane Bending (P)	668	-668	668	-668
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-441	-441	441	441
Outplane Bending (ML)	-813	813	813	-813
Normal Pressure Stress	32183	37134	32183	37134
Outplane Stress Summary	31866	37106	34376	36362

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	193	193	193	193
Inplane Bending (P)	2247	-2247	2247	-2247
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-234	-234	234	234
Inplane Bending (ML)	-2440	2440	2440	-2440
Inplane Pressure Stress	16091	16091	16091	16091
Inplane Stress Summary	15857	16243	21207	11831

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	179	179	-179	-179
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	344	344	344	344
Shear Stress Summary	523	523	165	165

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	31880	37113	34376	36362



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 156 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	268	268	268	268
Outplane Bending (P)	668	-668	668	-668
Outplane Membrane (Mc)	-220	-220	220	220
Outplane Bending (Mc)	-1861	1861	1861	-1861
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16091	16091	16091	16091
Outplane Stress Summary	14947	17333	19111	14051

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	193	193	193	193
Inplane Bending (P)	2247	-2247	2247	-2247
Inplane Membrane (Mc)	-379	-379	379	379
Inplane Bending (Mc)	-5819	5819	5819	-5819
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32183	37134	32183	37134
Inplane Stress Summary	28426	40519	40822	29639

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-179	-179	179	179
Torsional Shear (Mt)	344	344	344	344
Shear Stress Summary	165	165	523	523

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	28426	40519	40829	29653

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	268	268	268	268
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-441	-441	441	441
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5308	11245	5308	11245
Hoop Stress Summary	5136	11072	6018	11955

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 157 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	89	89	89	89
Axial Bending (P)	613	-613	613	-613
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-717	-717	717	717
Axial Bending (ML)	-634	634	634	-634
Axial Pressure Stress	2654	2654	2654	2654

Axial Stress Summary	2006	2047	4709	2213
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
-----------------------------------	------------	-----------	------------	-----------

Shear due to (Vc)	110	110	-110	-110
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	206	206	206	206

Shear Stress Summary	317	317	96	96
----------------------	-----	-----	----	----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
---	------------	-----------	------------	-----------

Two * Max Shear Stress	5164	11079	6025	11955
------------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------	----------------	------------------	-----------------

Hoop Membrane (P)	268	268	268	268
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-220	-220	220	220
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5308	11245	5308	11245

Hoop Stress Summary	5357	11293	5798	11734
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------	----------------	------------------	-----------------

Axial Membrane (P)	89	89	89	89
Axial Bending (P)	613	-613	613	-613
Axial Membrane (Mc)	-717	-717	717	717
Axial Bending (Mc)	-1730	1730	1730	-1730
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2654	2654	2654	2654

Axial Stress Summary	910	3143	5805	1116
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
--------------------------------------	------------	-----------	------------	-----------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-110	-110	110	110



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 158 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

Shear due to Torsion	206	206	206	206
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Shear Stress Summary	96	96	317	317
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5357	11293	6115	11741

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32183	37134	32183	37134	32183	37134	32183	37134
Circ. Pl (SUS)	-173	-173	709	709	-186	-186	572	572
Circ. Q (SUS)	-145	145	1481	-1481	-3572	3572	8066	-8066
Long. Pm (SUS)	16091	16091	16091	16091	16091	16091	16091	16091
Long. Pl (SUS)	-41	-41	427	427	48	48	488	488
Long. Q (SUS)	-193	193	4687	-4687	-1193	1193	2529	-2529
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	179	179	-179	-179	-179	-179	179	179
Shear Q (SUS)	344	344	344	344	344	344	344	344
Pm (SUS)	32183	37134	32183	37134	32183	37134	32183	37134
Pm+Pl (SUS)	32012	36962	32893	37844	31999	36949	32756	37707
Pm+Pl+Q (Total)	31882	37119	34375	36363	28427	40521	40833	29657

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	37134	138190	Passed
Pm+Pl (SUS)	37844	207286	Passed
Pm+Pl+Q (TOTAL)	40833	414572	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5308	11245	5308	11245	5308	11245	5308	11245
Circ. Pl (SUS)	-173	-173	709	709	48	48	488	488
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2654	2654	2654	2654	2654	2654	2654	2654
Long. Pl (SUS)	-628	-628	806	806	-628	-628	806	806
Long. Q (SUS)	-21	21	1247	-1247	-1117	1117	2343	-2343
Shear Pm (SUS)	0	0	0	0	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 159 of 220

Nozzle Calcs.: Tube outlet

Nozl: 13 4:24pm Apr 10,2026

Shear P1 (SUS)	110	110	-110	-110	-110	-110	110	110
Shear Q (SUS)	206	206	206	206	206	206	206	206
Pm (SUS)	5308	11245	5308	11245	5308	11245	5308	11245
Pm+P1 (SUS)	5138	11073	6021	11955	5359	11294	5801	11734
Pm+P1+Q (Total)	5166	11083	6023	11954	5358	11294	6115	11742

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11245	138190	Passed
Pm+P1 (SUS)	11955	207286	Passed
Pm+P1+Q (TOTAL)	11954	414572	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 160 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10,2026

Input, Nozzle Desc: Shell Outlet

From: 40

Pressure for Reinforcement Calculations	P	14.3055	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-266 2	
Parent Allowable Stress at Temperature	Sv	122834.11	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa

Inside Diameter of Cylindrical Shell	D	2600.00	mm
Design Length of Section	L	9833.3643	mm
Shell Finished (Minimum) Thickness	t	170.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7.52	m
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	122834.11	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		12.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	74.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	155.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



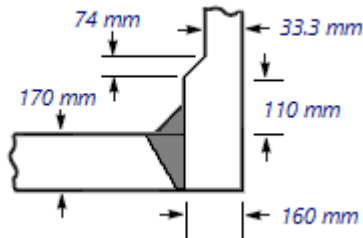
PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 161 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10, 2026



Reinforcement CALCULATION, Description: Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	1.312 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.3 \cdot 1303 / (122834 \cdot 1 - 0.6 \cdot 14.3)$
 $= 163.1408 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 14.3 \cdot 162 / (122834 \cdot 1 + 0.4 \cdot 14.3)$
 $= 18.0178 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1395 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 127 / 0.5773$
 $= 219.4079 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	657.8500 mm
Parallel to Vessel Wall	Rn+tn+t	328.9250 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	259.8120 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 122834)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 122834)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$



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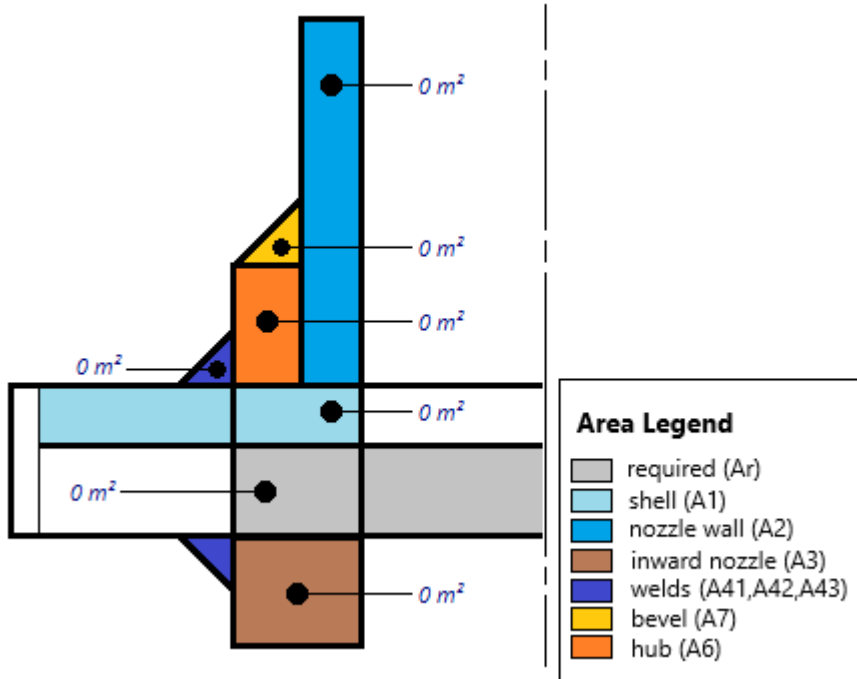
FileName : 123-C2-NVS

Page 162 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10, 2026

= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.043	0.002	NA
Area in Shell	A1	0.002	0.060	NA
Area in Nozzle Wall	A2	0.006	0.015	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.037	0.037	NA
TOTAL AREA AVAILABLE	Atot	0.045	0.113	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (263 \cdot 163 \cdot 1 + 2 \cdot 157 \cdot 163 \cdot 1 \cdot (1 - 1))$
 $= 0.043 \text{ m}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 395 (1 \cdot 167 - 1 \cdot 163) - 2 \cdot 157$
 $(1 \cdot 167 - 1 \cdot 163) \cdot (1 - 1)$
 $= 0.002 \text{ m}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp) (tn - trn) fr2$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 163 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10, 2026

$$= (2 \cdot 260) (30.3 - 18) 1$$

$$= 0.006 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(260, 400, 110)) \cdot (160 - 33.3) \cdot 1$$

$$= 0.037 \text{ m}^2 \text{ Includes Bevel Area of } 0.009 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 21.0178 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 166.1408 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 166.1408 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0751 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(166, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(21, 11.3)$$

$$= 21.0178 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 33.325 = 29.159 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 46794.9,	Allowable	: 122834.1 kPa	Passed
Expansion	: 0.0,	Allowable	: 279234.6 kPa	Passed
Occasional	: 27832.3,	Allowable	: 163369.4 kPa	Passed
Shear	: 12924.7,	Allowable	: 85983.9 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 29.2, tr = 16.1, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.62$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	2 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 164 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10, 2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 14.31/25.51 = 0.561

Weld Size Calculations, Description: Shell Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (0.043 - 0.0015 + 2 \cdot 157 \cdot 1 \cdot (1 \cdot 167 - 163)) \cdot 122834)$
 $= 5235967.50 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/7.07)^2 \cdot f_{r2}) \cdot S_v$
 $= (0.044 + 0.00022 - 0 \cdot 1) \cdot 122834$
 $= 5387671.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (0.044 + 0 + 0.00022 + (0.052)) \cdot 122834$
 $= 11828672.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (0.044 + 0 + 0.00022 + 0 + (0.052)) \cdot 122834$
 $= 11828672.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 324 \cdot 15 \cdot 0.49 \cdot 122834$
 $= 459260. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 147) \cdot (160 - 3) \cdot 0.7 \cdot 122834$
 $= 6224008. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 324 \cdot (155 - 0) \cdot 0.74 \cdot 122834$
 $= 7166952. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (459260 + 6224008) = 6683268 \text{ N}$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 165 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10, 2026

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (459260 + 0 + 7166951 + 0) = 7626211 N

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (459260 + 7166951 + 0) = 7626211 N

Summary of Failure Path Calculations:

Path 1-1 = 6683268 N , must exceed W = 5235967 N or W1 = 5387671 N

Path 2-2 = 7626211 N , must exceed W = 5235967 N or W2 = 11828672 N

Path 3-3 = 7626211 N , must exceed W = 5235967 N or W3 = 11828672 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.3 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 10.1239 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 580.1239 mm

Input Echo, WRC297 Item 1, Description: Shell Outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2600.000	mm
Shell Thickness	170.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-266 2	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 122834.11	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	577.200	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 122834.11	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	14.31	N/mm ²
Radial Load	P	-37488.32	N
Circumferential Shear	Vc	29986.73	N
Longitudinal Shear	Vl	29986.73	N
Circumferential Moment	Mc	26996.47	N-m
Longitudinal Moment	Ml	26996.47	N-m
Torsional Moment	Mt	31499.25	N-m

Include Axial Pressure Thrust No

Include Pressure Stress Indices per Div. 2 No



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 166 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14

4:24pm Apr 10, 2026

Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (17.294) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_v (3.395) is < 5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (3.608) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 577.2 + 2 \cdot 1.65 \cdot \sqrt{1386.5(170.0 - 3.0)}$$

$$= 2165.133 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2773.000 mm
Nozzle Outside Diameter	(d) =	577.200 mm
Vessel Thickness used	(T) =	167.000 mm
Nozzle Thickness used	(t) =	157.000 mm
T / t	=	1.064
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.848$$

Nr/P	=	0.140
Mr/P	=	0.204
M0/P	=	0.072
N0/P	=	0.130
MrD/Mc	=	0.280
NrDL/Mc	=	0.149
M0d/Mc	=	0.114
N0DL/Mc	=	-0.062
MrD/Ml	=	0.213
NrDL/Ml	=	0.124
M0D/Ml	=	0.086
N0DL/Ml	=	0.096

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	172	172	172	172
Outplane Bending (P)	579	-579	579	-579
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-158	-158	158	158
Outplane Bending (ML)	-861	861	861	-861
Normal Pressure Stress	104887	119193	104887	119193
Outplane Stress Summary	104618	119490	106659	118083

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	186	186	186	186
Inplane Bending (P)	1647	-1647	1647	-1647
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-206	-206	206	206



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 167 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10,2026

Inplane Bending (ML)	-2137	2137	2137	-2137
Inplane Pressure Stress	52440	52440	52440	52440
Inplane Stress Summary	51930	52909	56618	49048

Vessel Stresses (kPa):

LONGITUDINAL PLANE		Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	193	193	-193	-193	
Outplane Shear (Vl)	0	0	0	0	
Outplane Shear (Mt)	358	358	358	358	
Shear Stress Summary	551	551	165	165	

Vessel Stresses (kPa):

LONGITUDINAL PLANE		Au	Al	Bu	B1
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	104618	119490	106659	118083	

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	172	172	172	172	
Outplane Bending (P)	579	-579	579	-579	
Outplane Membrane (Mc)	96	96	-96	-96	
Outplane Bending (Mc)	-1137	1137	1137	-1137	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	52440	52440	52440	52440	
Outplane Stress Summary	52150	53267	54232	50799	

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	186	186	186	186	
Inplane Bending (P)	1647	-1647	1647	-1647	
Inplane Membrane (Mc)	-248	-248	248	248	
Inplane Bending (Mc)	-2812	2812	2812	-2812	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	104887	119193	104887	119193	
Inplane Stress Summary	103660	120296	109782	115167	

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	
Outplane Shear (Vl)	-193	-193	193	193	
Torsional Shear (Mt)	358	358	358	358	
Shear Stress Summary	165	165	551	551	

Vessel Stresses (kPa):



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 168 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10,2026

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	103660	120296	109782	115167
------------------------	--------	--------	--------	--------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	172	172	172	172
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-158	-158	158	158
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9011	26171	9011	26171

Hoop Stress Summary	9025	26185	9342	26502
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	179	179	179	179
Axial Bending (P)	1220	-1220	1220	-1220
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1489	-1489	1489	1489
Axial Bending (ML)	-1709	1709	1709	-1709
Axial Pressure Stress	4502	4502	4502	4502

Axial Stress Summary	2702	3681	9100	3240
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	206	206	-206	-206
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	379	379	379	379

Shear Stress Summary	586	586	172	172
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	9073	26199	9431	26502
------------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	172	172	172	172
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	96	96	-96	-96
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9011	26171	9011	26171



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 169 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14 4:24pm Apr 10,2026

Hoop Stress Summary	9280	26440	9087	26247
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	179	179	179	179
Axial Bending (P)	1220	-1220	1220	-1220
Axial Membrane (Mc)	-1489	-1489	1489	1489
Axial Bending (Mc)	-2337	2337	2337	-2337
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4502	4502	4502	4502

Axial Stress Summary	2075	4309	9728	2613
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-206	-206	206	206
Shear due to Torsion	379	379	379	379

Shear Stress Summary	172	172	586	586
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	9280	26440	10073	26261
------------------------	------	-------	-------	-------

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	104887	119193	104887	119193	104887	119193	104887	119193
Circ. Pl (SUS)	14	14	330	330	-62	-62	434	434
Circ. Q (SUS)	-282	282	1440	-1440	-1165	1165	4459	-4459
Long. Pm (SUS)	52440	52440	52440	52440	52440	52440	52440	52440
Long. Pl (SUS)	-20	-20	392	392	268	268	76	76
Long. Q (SUS)	-490	490	3784	-3784	-558	558	1716	-1716
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	193	193	-193	-193	-193	-193	193	193
Shear Q (SUS)	358	358	358	358	358	358	358	358
Pm (SUS)	104887	119193	104887	119193	104887	119193	104887	119193
Pm+Pl (SUS)	104901	119207	105217	119523	104825	119131	105321	119627
Pm+Pl+Q (Total)	104624	119493	106657	118083	103660	120296	109785	115172

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119193	122834	Passed



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 170 of 220

Nozzle Calcs.: Shell Outlet

Nozl: 14

4:24pm Apr 10, 2026

Pm+Pl (SUS)	119627	184251	Passed
Pm+Pl+Q (TOTAL)	120296	368502	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9011	26171	9011	26171	9011	26171	9011	26171
Circ. Pl (SUS)	14	14	330	330	268	268	76	76
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4502	4502	4502	4502	4502	4502	4502	4502
Long. Pl (SUS)	-1310	-1310	1668	1668	-1310	-1310	1668	1668
Long. Q (SUS)	-489	489	2929	-2929	-1117	1117	3557	-3557
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	206	206	-206	-206	-206	-206	206	206
Shear Q (SUS)	379	379	379	379	379	379	379	379
Pm (SUS)	9011	26171	9011	26171	9011	26171	9011	26171
Pm+Pl (SUS)	9032	26186	9354	26503	9285	26440	9101	26249
Pm+Pl+Q (Total)	9078	26200	9431	26502	9283	26440	10073	26261

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26171	122834	Passed
Pm+Pl (SUS)	26503	184251	Passed
Pm+Pl+Q (TOTAL)	26502	368502	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 123-C2-NVS

Page 171 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10,2026

Input, Nozzle Desc: Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	14.3055	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-266 2	
Parent Allowable Stress at Temperature	Sv	122834.11	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa

Inside Diameter of Cylindrical Shell	D	2600.00	mm
Design Length of Section	L	9833.3643	mm
Shell Finished (Minimum) Thickness	t	170.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7.52	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	122834.11	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		12.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	74.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	155.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



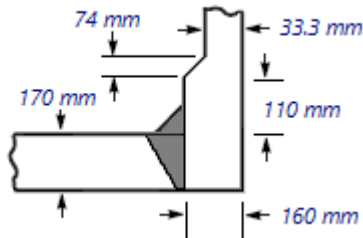
PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 172 of 220

Nozzle Calcs.: Shell Inlet

Noz1: 15 4:24pm Apr 10, 2026



Reinforcement CALCULATION, Description: Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	1.312 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.3 \cdot 1303 / (122834 \cdot 1 - 0.6 \cdot 14.3)$
 $= 163.1408 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 14.3 \cdot 162 / (122834 \cdot 1 + 0.4 \cdot 14.3)$
 $= 18.0178 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1395 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 127 / 0.5773$
 $= 219.4079 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	657.8500 mm
Parallel to Vessel Wall	Rn+tn+t	328.9250 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	259.8120 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 122834)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 122834)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$



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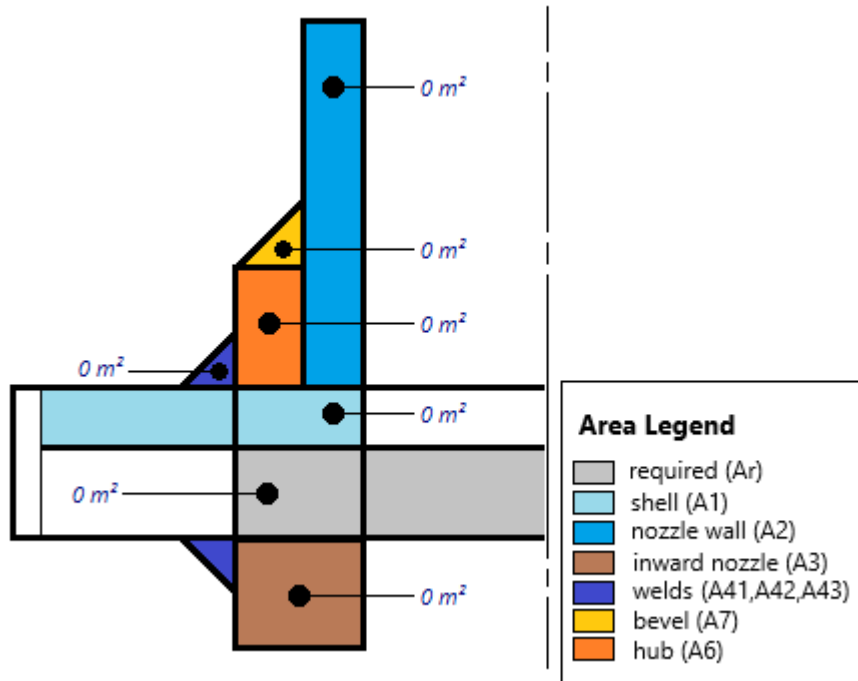
FileName : 123-C2-NVS

Page 173 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10, 2026

= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.043	0.002	NA
Area in Shell	A1	0.002	0.060	NA
Area in Nozzle Wall	A2	0.006	0.015	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.037	0.037	NA
TOTAL AREA AVAILABLE	Atot	0.045	0.113	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (263 \cdot 163 \cdot 1 + 2 \cdot 157 \cdot 163 \cdot 1 \cdot (1 - 1))$
 $= 0.043 \text{ m}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 395(1 \cdot 167 - 1 \cdot 163) - 2 \cdot 157$
 $(1 \cdot 167 - 1 \cdot 163) \cdot (1 - 1)$
 $= 0.002 \text{ m}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 174 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10, 2026

$$= (2 \cdot 260)(30.3 - 18)1$$

$$= 0.006 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(260, 400, 110)) \cdot (160 - 33.3) \cdot 1$$

$$= 0.037 \text{ m}^2 \text{ Includes Bevel Area of } 0.009 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 21.0178 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 166.1408 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 166.1408 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0751 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(166, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(21, 11.3)$$

$$= 21.0178 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 33.325 = 29.159 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 40813.7,	Allowable	: 122834.1 kPa	Passed
Expansion	: 0.0,	Allowable	: 285215.8 kPa	Passed
Occasional	: 27832.3,	Allowable	: 163369.4 kPa	Passed
Shear	: 9446.4,	Allowable	: 85983.9 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 29.2, tr = 16.1, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.62$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	2 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 175 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10, 2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 14.31/25.51 = 0.561

Weld Size Calculations, Description: Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (0.043 - 0.0015 + 2 \cdot 157 \cdot 1 \cdot (1 \cdot 167 - 163)) \cdot 122834)$
 $= 5235967.50 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (0.044 + 0.00022 - 0 \cdot 1) \cdot 122834$
 $= 5387671.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (0.044 + 0 + 0.00022 + (0.052)) \cdot 122834$
 $= 11828672.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (0.044 + 0 + 0.00022 + 0 + (0.052)) \cdot 122834$
 $= 11828672.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 324 \cdot 15 \cdot 0.49 \cdot 122834$
 $= 459260. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 147) \cdot (160 - 3) \cdot 0.7 \cdot 122834$
 $= 6224008. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 324 \cdot (155 - 0) \cdot 0.74 \cdot 122834$
 $= 7166952. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (459260 + 6224008) = 6683268 \text{ N}$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 176 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10,2026

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (459260 + 0 + 7166951 + 0) = 7626211 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (459260 + 7166951 + 0) = 7626211 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 6683268 N , must exceed W = 5235967 N or W1 = 5387671 N
 Path 2-2 = 7626211 N , must exceed W = 5235967 N or W2 = 11828672 N
 Path 3-3 = 7626211 N , must exceed W = 5235967 N or W3 = 11828672 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.3 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 10.1239 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 580.1239 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 81.90 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 328.92 mm

Input Echo, WRC297 Item 1, Description: Shell Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2600.000	mm
Shell Thickness	170.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-266 2	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 122834.11	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	577.200	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 122834.11	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	14.31	N/mm ²
Radial Load	P	-31241.91	N
Circumferential Shear	Vc	24995.49	N
Longitudinal Shear	Vl	24995.49	N
Circumferential Moment	Mc	18746.51	N-m
Longitudinal Moment	Ml	18746.51	N-m



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 177 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10, 2026

Torsional Moment Mt 21875.76 N-m

Include Axial Pressure Thrust No

Include Pressure Stress Indices per Div. 2 No

Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (17.294) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_v (3.395) is < 5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (3.608) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 577.2 + 2 \cdot 1.65 \cdot \sqrt{1386.5(170.0 - 3.0)}$$

$$= 2165.133 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2773.000 mm

Nozzle Outside Diameter (d) = 577.200 mm

Vessel Thickness used (T) = 167.000 mm

Nozzle Thickness used (t) = 157.000 mm

T / t = 1.064

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.848$$

Nr/P = 0.140

Mr/P = 0.204

M0/P = 0.072

N0/P = 0.130

MrD/Mc = 0.280

NrDL/Mc = 0.149

M0d/Mc = 0.114

N0DL/Mc = -0.062

MrD/Ml = 0.213

NrDL/Ml = 0.124

M0D/Ml = 0.086

N0DL/Ml = 0.096

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Membrane (P)	144	144	144	144
------------------------	-----	-----	-----	-----

Outplane Bending (P)	482	-482	482	-482
-----------------------	-----	------	-----	------

Outplane Membrane (Mc)	0	0	0	0
------------------------	---	---	---	---

Outplane Bending (Mc)	0	0	0	0
-----------------------	---	---	---	---

Outplane Membrane (ML)	-110	-110	110	110
------------------------	------	------	-----	-----

Outplane Bending (ML)	-599	599	599	-599
-----------------------	------	-----	-----	------

Normal Pressure Stress	104887	119193	104887	119193
------------------------	--------	--------	--------	--------

Outplane Stress Summary	104804	119345	106225	118366
-------------------------	--------	--------	--------	--------

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	151	151	151	151
-----------------------	-----	-----	-----	-----



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 178 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10,2026

Inplane Bending (P)	1372	-1372	1372	-1372
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-137	-137	137	137
Inplane Bending (ML)	-1482	1482	1482	-1482
Inplane Pressure Stress	52440	52440	52440	52440
Inplane Stress Summary	52343	52564	55584	49875

Vessel Stresses (kPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	158	158	-158	-158	
Outplane Shear (Vl)	0	0	0	0	
Outplane Shear (Mt)	248	248	248	248	
Shear Stress Summary	406	406	89	89	

Vessel Stresses (kPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	104804	119345	106225	118366	

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	144	144	144	144	
Outplane Bending (P)	482	-482	482	-482	
Outplane Membrane (Mc)	68	68	-68	-68	
Outplane Bending (Mc)	-792	792	792	-792	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	52440	52440	52440	52440	
Outplane Stress Summary	52343	52964	53791	51240	

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	151	151	151	151	
Inplane Bending (P)	1372	-1372	1372	-1372	
Inplane Membrane (Mc)	-172	-172	172	172	
Inplane Bending (Mc)	-1951	1951	1951	-1951	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	104887	119193	104887	119193	
Inplane Stress Summary	104287	119752	108534	116194	

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	
Outplane Shear (Vl)	-158	-158	158	158	
Torsional Shear (Mt)	248	248	248	248	



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 179 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10,2026

Shear Stress Summary	89	89	406	406
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	104287	119752	108534	116194
------------------------	--------	--------	--------	--------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	144	144	144	144
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-110	-110	110	110
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9011	26171	9011	26171

Hoop Stress Summary	9045	26206	9266	26427
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	144	144	144	144
Axial Bending (P)	1020	-1020	1020	-1020
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1034	-1034	1034	1034
Axial Bending (ML)	-1185	1185	1185	-1185
Axial Pressure Stress	4502	4502	4502	4502

Axial Stress Summary	3447	3778	7887	3474
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	172	172	-172	-172
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	261	261	261	261

Shear Stress Summary	434	434	89	89
----------------------	-----	-----	----	----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	9073	26213	9266	26427
------------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	144	144	144	144
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	68	68	-68	-68
Hoop Bending (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 180 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10,2026

Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9011	26171	9011	26171

Hoop Stress Summary	9224	26385	9087	26247
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	144	144	144	144
Axial Bending (P)	1020	-1020	1020	-1020
Axial Membrane (Mc)	-1034	-1034	1034	1034
Axial Bending (Mc)	-1620	1620	1620	-1620
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4502	4502	4502	4502

Axial Stress Summary	3012	4212	8321	3040
----------------------	------	------	------	------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-172	-172	172	172
Shear due to Torsion	261	261	261	261

Shear Stress Summary	89	89	434	434
----------------------	----	----	-----	-----

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	9224	26385	9280	26254
------------------------	------	-------	------	-------

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	104887	119193	104887	119193	104887	119193	104887	119193
Circ. Pl (SUS)	34	34	254	254	-21	-21	323	323
Circ. Q (SUS)	-117	117	1081	-1081	-579	579	3323	-3323
Long. Pm (SUS)	52440	52440	52440	52440	52440	52440	52440	52440
Long. Pl (SUS)	14	14	288	288	212	212	76	76
Long. Q (SUS)	-110	110	2854	-2854	-310	310	1274	-1274
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	158	158	-158	-158	-158	-158	158	158
Shear Q (SUS)	248	248	248	248	248	248	248	248
Pm (SUS)	104887	119193	104887	119193	104887	119193	104887	119193
Pm+Pl (SUS)	104921	119227	105141	119447	104866	119172	105210	119516
Pm+Pl+Q (Total)	104807	119346	106222	118366	104287	119751	108536	116195

Vessel Stress Summation Comparison (kPa):



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 181 of 220

Nozzle Calcs.: Shell Inlet

Nozl: 15 4:24pm Apr 10,2026

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119193	122834	Passed
Pm+Pl (SUS)	119516	184251	Passed
Pm+Pl+Q (TOTAL)	119751	368502	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9011	26171	9011	26171	9011	26171	9011	26171
Circ. Pl (SUS)	34	34	254	254	212	212	76	76
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4502	4502	4502	4502	4502	4502	4502	4502
Long. Pl (SUS)	-890	-890	1178	1178	-890	-890	1178	1178
Long. Q (SUS)	-165	165	2205	-2205	-600	600	2640	-2640
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	172	172	-172	-172	-172	-172	172	172
Shear Q (SUS)	261	261	261	261	261	261	261	261
Pm (SUS)	9011	26171	9011	26171	9011	26171	9011	26171
Pm+Pl (SUS)	9050	26206	9273	26426	9228	26384	9095	26248
Pm+Pl+Q (Total)	9078	26213	9270	26425	9224	26383	9281	26255

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26171	122834	Passed
Pm+Pl (SUS)	26426	184251	Passed
Pm+Pl+Q (TOTAL)	26425	368502	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 123-C2-NVS

Page 182 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	14.28 N/mm ²
Shell Maximum Operating Pressure	Psox,max	15.93 N/mm ²
Shell Minimum Operating Pressure	Psox,min	0.00 N/mm ²
Shell Thickness	ts	170.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2600.000 mm
Shell Temperature for Internal Pressure	Ts	343.00 °C
Shell Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03506
Shell Allowable Stress at Temperature	Ss	122834.11 kPa
Shell Allowable Stress at Ambient		137989.50 kPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	18.00 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	12.55 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 N/mm ²
Channel Thickness	tc	190.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2600.000 mm
Channel Design Temperature	TEMPC	370.00 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	138190.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	5328
Tube Wall Thickness	et	2.7690 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	9000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8485.00 mm
Design Temperature of the Tubes		370.00 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		113991.31 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	184985.91 kPa
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1255.670 mm
Largest Center-to-Center Tube Distance	Ul	101.6000 mm



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 183 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10,2026

Length of Expanded Portion of Tube	ltx	300.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	4.8 m ²
Perimeter of Tube Layout	Cp	8067.5 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	370.00 °C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

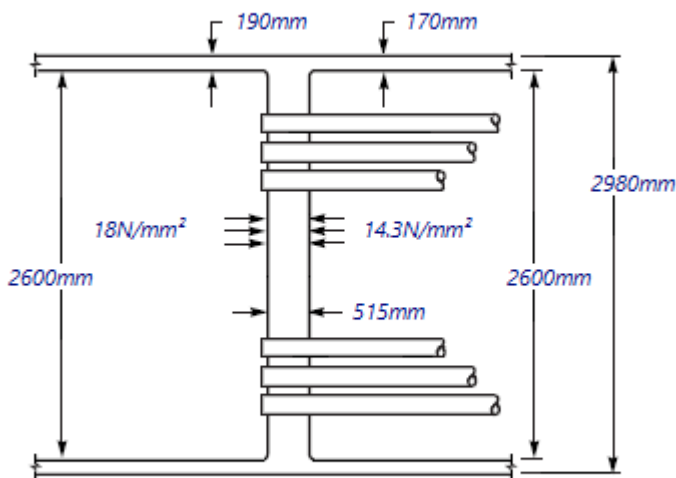
Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	138190.73 kPa
Tubesheet Allowable Stress at Ambient	Tt	147988.73 kPa
Thickness of Tubesheet	h	515.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2980.000 mm

Area of the Untubed Lanes	AL	0.198 m ²
---------------------------	----	----------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C> 0.3%



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 184 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Elastic Mod. at Design Temperature	343.0 °C	0.17829E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20133E+06 N/mm ²

Channel - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 °C	0.18626E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	370.0 °C	0.18626E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 °C	0.18626E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (18 \cdot (25.4/2 - 0)) / (113991 \cdot 1 + 0.4 \cdot 18)$$

$$= 1.8863 + 0.0000 = 1.8863 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 9741 / \sqrt{(12.7 \cdot 2.77)}$$

$$= 1642.576$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 2 \cdot (Do/t)0.94, Ch = 0.55 / (Do/t)$$

$$= 0.55 / (25.4/2.77)$$

$$= 0.0600$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.06 \cdot 186264 (2.77/25.4)$$

$$= 1947895.125 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 1947895.1 / 184985.9$$

$$= 10.530$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = Fy$$

$$= 184985.906 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [184986] \geq Fy [184986], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = Fy / FS$$

$$= 184985.906 / 1.667$$

$$= 110969.352 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 111 (2.77/25.4)$$

$$= 24.196 \text{ N/mm}^2$$

Summary of Tube Required Thickness Results:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 185 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Total Required Thickness including Corrosion all.	1.8863 mm
Allowable Internal Pressure at Corroded thickness	27.23 N/mm ²
Required Internal Design Pressure	18.00 N/mm ²
Allowable External Pressure at Corroded thickness	24.20 N/mm ²
Required External Design Pressure	14.28 N/mm ²
Required Thickness due to Shell Side pressure	1.6343 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1256 + 25.4$$

$$= 2536.740 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 300 / 515$$

$$= 0.583 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t((Et/E) \cdot (StT/S) \cdot (\rho)); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.77 (186264 / 186264) \cdot (113991 / 138191) \cdot (0.58); 25.4 - 2 \cdot 2.77)$$

$$= 22.7389 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2)) }$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.7) / 32.4$$

$$= 0.2980$$

$h/p = 16.220472$; $\mu^* = 0.297992$; $E^* = 51878 \text{ N/mm}^2$

$E^*/E = 0.278520$; $v^* = 0.355216$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 186 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Note: As h/p (16.220) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right) \\ = \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2600 + 170)170} \right) \\ = 0.0026 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\ = 0.0026 \cdot 178285 \cdot 170^3 / (6(1 - 0.3^2)) \\ = 424972352.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\ = 6(2600 + 170) \cdot 424972352 / 515^3 \cdot (1 + 515 \cdot 0.0026 + (515 \cdot 0.0026)^2 / 2) \\ = 170376.1 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\ = 2600^2 / (4 \cdot 178285 \cdot 170) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 170))) \\ = 0.47912\text{E-}04 \text{ mm/kPa}$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ = 1.02 \cdot 424972352 \cdot 0.0026 \cdot 0.47912\text{E-}04 (1 + 515 \cdot 0.0026) \\ = 130696.8438 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right) \\ = \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2600 + 190)190} \right) \\ = 0.0025 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\ = 0.0025 \cdot 186264 \cdot 190^3 / (6(1 - 0.3^2)) \\ = 584217216.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ = 6(2600 + 190 / \cos(0)) \cdot 584217216 / 515^3 \cdot (1 + 515 \cdot 0.0025 \cdot (1 + 1) / 2 + (515 \cdot 0.0025)^2 / 2 \cdot 1) \\ = 222853.4 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ = 2600^2 / (4 \cdot 186264 \cdot 190) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 190))) \\ = 0.41081\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ = 1.02 \cdot 584217216 [0.41081\text{E-}04 (1 + 0.0025 \cdot 1 \cdot 515) - 0.000 (1/515 + 1 \cdot 0.0025/2)] \\ = 140379.3906 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 187 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980/2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 51878 \cdot (170376 + 222853 + 186264 \cdot \ln(1.17))$$

$$= 5.2602$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 2537^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 0 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 18]$$

$$= -370204.6562 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -370205 + 140379 \cdot 18 - 130697 \cdot 0$$

$$= 2156771.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (Ps - Pt) / (1 + F)$$

$$= (2156771) - 2537^2 / 32 \cdot 5.26 (0 - 18) / (1 + 5.26)$$

$$= 3386201.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 3386201 + 2537^2 / 64 \cdot (3 + 0.36) \cdot (0 - 18)$$

$$= -2686613.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|3386201|; |-2686613|)$$

$$= 3386201.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 3386201 / (0.3 (515 - 5)^2)$$

$$= 262116.6406 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138191$$

$$= 276381.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 3389027 / (0.3 (500 - 5)^2)$$

$$= 276186.8125 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 500 + 0 + 0$$

$$= 500.0044 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 188 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS}) \\ = \max(500; 482) \\ = 500.0044 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | \text{Ps} - \text{Pt} | \\ = | 0 - 18 | \\ = 18.001 \text{ N/mm}^2$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot \text{S}; 0.533 \cdot \text{Sy}) \\ = 4 \cdot 0.2 \cdot 515 / 2537 \cdot \min(0.8 \cdot 138191; 0.533 \cdot 245585) \\ = 17955.195 \text{ kPa}$$

Note: As $\text{Fs} - \text{Pt} > \text{Tau_limit}$ & Ap , Cp are given, the more accurate formula for Tau is used.

Average shear stress at the outer edge of perforated region [τ]:

$$= (1 / (4 \cdot \mu)) \cdot (1/h \{ (4 \cdot \text{Ap}) / \text{Cp} \}) \cdot | \text{Ps} - \text{Pt} | \\ = (1 / (4 \cdot 0.2)) \cdot (1 / 515 \{ (4 \cdot 4.78) / 8067 \}) \cdot | 0 - 18 | \\ = 103538.4297 \text{ kPa}$$

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot \text{S}; 0.533 \cdot \text{Sy})) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\ = 1 / (4 \cdot 0.2 \cdot 110553) \cdot (4 \cdot 4.78 / 8067) \cdot | 0 - 18 | + 0 \\ = 482.3251 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 170.000mm

Min. Shell len. adjacent to the tubesheet is: 1196.695 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= \text{Ds}^2 / (4 \cdot \text{ts} (\text{Ds} + \text{ts})) \cdot \text{Ps} \\ = 2600^2 / (4 \cdot 170 (2600 + 170)) \cdot 0 \\ = 0.0000 \text{ kPa}$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot \text{ks} / \text{ts}^2 \cdot [\beta \text{S} \cdot \text{P's} + 6(1 - \nu^*) / \text{E}^* \cdot \text{Do} / h^3 \cdot (1 + h \cdot \beta \text{S} / 2) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\ = 6 \cdot 95534224 / 170^2 \cdot [0.0026 \cdot 0 + 6(1 - 0.36) / 51878 \cdot 2537 / 515^3 \cdot (1 + 515 \cdot 0.0026 / 2) \cdot (3386201 + 2537^2 / 32 \cdot (0 - 18))] \\ = -48038.047 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_s, m | + | \sigma_s, b | \text{ (Should be } \leq 1.5 \cdot \text{Ss)} \\ = 48038.047 \text{ kPa (Should be } \leq 184251.17)$$

Note: For a given Channel thickness of: 190.000 mm

Min. Channel len. adjacent to the tubesheet is: 1265.132mm

Axial Channel Membrane Stress [σ_c, m]:

$$= \text{Dc}^2 / (4 \cdot \text{tc} (\text{Dc} + \text{tc})) \cdot \cos(\text{theta}) \cdot \text{Pt} \\ = 2600^2 / (4 \cdot 190 (2600 + 190) \cdot \cos(0)) \cdot 18 \\ = 57385.398 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot \text{kc} / \text{tc}^2 \cdot [(\delta_c \cdot \text{Cm} \Delta \cdot \beta_c - \zeta_c \cdot \text{Cm} \Theta) \cdot \text{Pt} - 6((1 - \nu^*) / \text{E}^*) \cdot \text{Do} / h^3 \cdot (\text{Cm} \Theta + h \cdot \beta_c / 2 \cdot \text{Cm} \Delta) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\ = 6 \cdot 131332640 / 190^2 \cdot [(0.41081 \text{E}-04 \cdot 1 \cdot 0.0025 - 0.000 \cdot 1) \cdot 18 - 6((1 - 0.36) / 51878) \cdot 2537 / 515^3 \cdot (1 +$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 189 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 0.0025 \cdot 515/2 \cdot 1 \cdot (3386201 + 2537^2/32 \cdot (0 - 18))]$$

$$= 230892.953 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 288278.344 \text{ kPa (Should be } \leq 207286.09)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{1.5 \cdot S_c / \sigma_c}$$

$$= 186264 \cdot \sqrt{1.5 \cdot 138191 / 288278}$$

$$= 157946.0 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 3369345 / (0.3 \cdot (515 - 5)^2)$$

$$= 260811.9 \text{ kPa (Should be } \leq 276381.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 500 + 0 + 0 = 500 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(500; 482) = 500 \text{ mm}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 113991$$

$$= 22440.715 \text{ N}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 22440.7148 \text{ N}$$

Weld Strength Factor [f_w]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 113991 / 113991$$

$$= 1.000$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 16195.8193 \text{ N}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 190 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 3.371 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 22441 is >= tube strength 22441 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	260811.9 <=	276381.5 kPa	Ok
Tubesheet shear stress	103538.4 <=	110552.6 kPa	Ok
Stress in Shell at Tubesheet	48038.0 <=	184251.2 kPa	Ok
Stress in Channel at Tubesheet	288278.3 <=	491193.2 kPa	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	500.004	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D2 un-corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1256 + 25.4$$

$$= 2536.740 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Ds / Do \text{ (Configurations a, b, c)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

Ratio [pc]:

$$= Dc / Do \text{ (Configurations a, e, f)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 191 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 300/515$$

$$= 0.583 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.77 (186264/186264) \cdot (113991/138191) \cdot (0.58); 25.4 - 2 \cdot 2.77)$$

$$= 22.7389 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2)) }$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.7) / 32.4$$

$$= 0.2980$$

h/p = 16.220472 ; μ* = 0.297992 ; E* = 51878 N/mm²

E*/E = 0.278520 ; ν* = 0.355216

Note: As h/p (16.220) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (2600 + 170) 170 }$$

$$= 0.0026 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0026 \cdot 178285 \cdot 170^3 / (6 (1 - 0.3^2))$$

$$= 424972352.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (2600 + 170) \cdot 424972352/515^3 \cdot (1 + 515 \cdot 0.0026 + (515 \cdot 0.0026)^2 / 2)$$

$$= 170376.1 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 2600^2 / (4 \cdot 178285 \cdot 170) \cdot (1 - 2600 \cdot 0.3 / (2 (2600 + 170)))$$

$$= 0.47912\text{E-}04 \text{ mm/kPa}$$

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 424972352 \cdot 0.0026 \cdot 0.47912\text{E-}04 (1 + 515 \cdot 0.0026)$$

$$= 130696.8438 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (2600 + 190) 190 }$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6 (1 - \nu_s^2))$$

$$= 0.0025 \cdot 186264 \cdot 190^3 / (6 (1 - 0.3^2))$$

$$= 584217216.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 192 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2600 + 190 / \cos(0)) \cdot 584217216 / 515^3 \cdot (1 + 515 \cdot 0.0025 \cdot (1 + 1) / 2 + (515 \cdot 0.0025)^2 / 2 \cdot 1)$$

$$= 222853.4 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc / (2(Dc + tc)))$$

$$= 2600^2 / (4 \cdot 186264 \cdot 190) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 190)))$$

$$= 0.41081E-04 \text{ mm/kPa}$$

Channel coefficient [ζ_{ac}]:

$$\zeta_{ac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot kc [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 584217216 [0.41081E-04 (1 + 0.0025 \cdot 1 \cdot 515) - 0.000 (1 / 515 + 1 \cdot 0.0025 / 2)]$$

$$= 140379.3906 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 2980 / 2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 51878 \cdot (170376 + 222853 + 186264 \cdot \ln(1.17))$$

$$= 5.2602$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= Do^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 2537^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 14.3 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 0]$$

$$= 293695.7188 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 293696 + 140379 \cdot 0 - 130697 \cdot 14.3$$

$$= -1572763.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - Do^2 / 32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= ((-1572763) - 2537^2 / 32 \cdot 5.26 (14.3 - 0)) / (1 + 5.26)$$

$$= -2664299.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= -2664299 + 2537^2 / 64 \cdot (3 + 0.36) \cdot (14.3 - 0)$$

$$= 2153468.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp |; | Mo |)$$

$$= \max(| -2664299 |; | 2153468 |)$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 193 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 2664299.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 2664299 / (0.3 (515 - 5)^2) \\ &= 206236.1094 \text{ kPa} \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138191 \\ &= 276381.47 \text{ kPa} \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 2686243 / (0.31 (438 - 5)^2) \\ &= 275374.9688 \text{ kPa} \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 438 + 0 + 0 \\ &= 437.7720 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(438 ; 383) \\ &= 437.7720 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |14.3 - 0| \\ &= 14.281 \text{ N/mm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 515 / 2537 \cdot \min(0.8 \cdot 138191 ; 0.533 \cdot 245585) \\ &= 17955.195 \text{ kPa} \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.2 \cdot 110553) \cdot (4 \cdot 4.78 / 8067) \cdot |14.3 - 0| + 0 \\ &= 382.6446 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 170.000mm

Min. Shell len. adjacent to the tubesheet is: 1196.695 mm

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 2600^2 / (4 \cdot 170 (2600 + 170)) \cdot 14.3 \\ &= 51249.1016 \text{ kPa} \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 95534224 / 170^2 \cdot [0.0026 \cdot 0.68 + 6(1 - 0.36) / 51878 \cdot \\ &\quad 2537 / 515^3 \cdot (1 + 515 \cdot 0.0026 / 2) \cdot (-2664299 + 2537^2 / 32 \cdot \\ &\quad (14.3 - 0))] \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 194 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

= 202551.828 kPa

Shell Membrane + Bending Stress [σ_s]:

= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq 1.5 \cdot S_s$)

= 253800.938 kPa (Should be ≤ 184251.17)

Note: For a given Channel thickness of: 190.000 mm

Min. Channel len. adjacent to the tubesheet is: 1265.132mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$

= $2600^2 / (4 \cdot 190 (2600 + 190) \cdot \cos(0)) \cdot 0$

= 0.000 kPa

Axial Channel Bending Stress [$\sigma_{c,b}$]:

= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$

$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$

$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$

= $6 \cdot 131332640 / 190^2 \cdot [(0.41081E-04 \cdot 1 \cdot 0.0025 - 0.000 \cdot 1) \cdot 0 -$

$6((1 - 0.36) / 51878) \cdot 2537 / 515^3 \cdot (1 +$

$0.0025 \cdot 515 / 2 \cdot 1) \cdot (-2664299 + 2537^2 / 32 \cdot (14.3 - 0))]$

= -45843.590 kPa

Channel Membrane + Bending Stress [σ_c]:

= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq 1.5 \cdot S_c$)

= 45843.590 kPa (Should be ≤ 207286.09)

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

= $E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$

= $178285 \cdot \sqrt{(1.5 \cdot 122834 / 253801)}$

= 151905.6 N/mm²

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

= $6 \cdot M / (\mu \cdot (h - h'g)^2)$

= $6 \cdot 2653365 / (0.3 (515 - 5)^2)$

= 205389.8 kPa (Should be ≤ 276381.5)



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

= $h + C_{ats} + C_{atc} = 438 + 0 + 0 = 438$ mm

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

= $\max(H_{reqB}, H_{reqS}) = \max(438; 383) = 438$ mm

Tube Metal Area [A_t]:

= $\pi (d_o - t) t$

= $\pi (25.4 - 2.77) 2.77$

= 0.000 m²

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

= $\pi \cdot t (d_o - t) S$

= $\pi \cdot 2.77 (25.4 - 2.77) 113991$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 195 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

= 22440.715 N

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)

= $0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$

= 22440.7148 N

Weld Strength Factor [fw]:

= S / S_w where S_w = min(S and S_t)

= 113991/113991

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)

= $0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$

= 16195.8193 N

Minimum Length of Weld [ar]:

= $2 \cdot \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o}$

= $2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1) - 0.75 \cdot 25.4}$

= 3.371 mm

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 22441 is >= tube strength 22441 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 un-corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	205389.8	<= 276381.5 kPa	Ok
Tubesheet shear stress	82140.5	<= 110552.6 kPa	Ok
Stress in Shell at Tubesheet	253800.9	<= 368800.8 kPa	Ok
Stress in Channel at Tubesheet	45843.6	<= 207286.1 kPa	Ok

Thickness results for loadcase D2 un-corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	437.772	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= 2 • ro + dt

= 2 • 1256 + 25.4



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 196 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

= 2536.740 mm

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2600 / 2537$$

$$= 1.0249$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 300 / 515$$

$$= 0.583 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t_t (E_t T / E) \cdot (StT / S) \cdot (\rho); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.77 (186264 / 186264) \cdot (113991 / 138191) \cdot (0.58); 25.4 - 2 \cdot 2.77)$$

$$= 22.7389 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2))}$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.7) / 32.4$$

$$= 0.2980$$

$h/p = 16.220472$; $\mu^* = 0.297992$; $E^* = 51878 \text{ N/mm}^2$

$E^*/E = 0.278520$; $\nu^* = 0.355216$

Note: As h/p (16.220) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4} / \sqrt{(D_s + t_s)t_s})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(2600 + 170)170})$$

$$= 0.0026 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0026 \cdot 178285 \cdot 170^3 / (6(1 - 0.3^2))$$

$$= 424972352.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2600 + 170) \cdot 424972352 / 515^3 \cdot (1 + 515 \cdot 0.0026 + (515 \cdot 0.0026)^2 / 2)$$

$$= 170376.1 \text{ N/mm}^2$$

Shell coefficient [δ_s]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 197 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2600^2 / (4 \cdot 178285 \cdot 170) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 170)))$$

$$= 0.47912E-04 \text{ mm/kPa}$$

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 424972352 \cdot 0.0026 \cdot 0.47912E-04 (1 + 515 \cdot 0.0026)$$

$$= 130696.8438 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(2600 + 190)190})$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.0025 \cdot 186264 \cdot 190^3 / (6(1 - 0.3^2))$$

$$= 584217216.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2600 + 190 / \cos(0)) \cdot 584217216 / 515^3 \cdot (1 + 515 \cdot 0.0025 \cdot (1 + 1) / 2 + (515 \cdot 0.0025)^2 / 2 \cdot 1)$$

$$= 222853.4 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2600^2 / (4 \cdot 186264 \cdot 190) \cdot (1 - 2600 \cdot 0.3 / (2(2600 + 190)))$$

$$= 0.41081E-04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 584217216 [0.41081E-04 (1 + 0.0025 \cdot 1 \cdot 515) - 0.000 (1/515 + 1 \cdot 0.0025/2)]$$

$$= 140379.3906 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 51878 \cdot (170376 + 222853 + 186264 \cdot \ln(1.17))$$

$$= 5.2602$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2537^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 14.3 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 18]$$

$$= -76508.9219 \text{ N/mm}^2 \cdot \text{mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 198 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + \omega c \cdot Pt - \omega s \cdot Ps \\
 &= -76509 + 140379 \cdot 18 - 130697 \cdot 14.3 \\
 &= 584007.2500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F) \\
 &= ((584007) - 2537^2/32 \cdot 5.26 (14.3 - 18)) / (1 + 5.26) \\
 &= 721902.7500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt) \\
 &= 721903 + 2537^2/64 \cdot (3 + 0.36) \cdot (14.3 - 18) \\
 &= -533145.4375 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp | ; | Mo |) \\
 &= \max(| 721903 | ; | -533145 |) \\
 &= 721902.7500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\
 &= 6 \cdot 721903 / (0.3 (515 - 5)^2) \\
 &= 55880.5195 \text{ kPa}
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 138191 \\
 &= 276381.47 \text{ kPa}
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\
 &= 6 \cdot 736058 / (0.36 (217 - 5)^2) \\
 &= 276320.1875 \text{ kPa}
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 217 + 0 + 0 \\
 &= 216.6207 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(217 ; 99.7) \\
 &= 216.6207 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= | Ps - Pt | \\
 &= | 14.3 - 18 | \\
 &= 3.720 \text{ N/mm}^2
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.2 \cdot 515 / 2537 \cdot \min(0.8 \cdot 138191 ; 0.533 \cdot 245585) \\
 &= 17955.195 \text{ kPa}
 \end{aligned}$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 199 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$\begin{aligned}
 &= 1/(4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p/C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\
 &= 1/(4 \cdot 0.2 \cdot 110553) \cdot (4 \cdot 4.78/8067) \cdot |14.3 - 18| + 0 \\
 &= 99.6805 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 170.000mm

Min. Shell len. adjacent to the tubesheet is: 1196.695 mm

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\
 &= 2600^2 / (4 \cdot 170 (2600 + 170)) \cdot 14.3 \\
 &= 51249.1016 \text{ kPa}
 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot \\
 &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\
 &= 6 \cdot 95534224 / 170^2 \cdot [0.0026 \cdot 0.68 + 6(1 - 0.36) / 51878 \cdot \\
 &\quad 2537 / 515^3 \cdot (1 + 515 \cdot 0.0026 / 2) \cdot (721903 + 2537^2 / 32 \cdot \\
 &\quad (14.3 - 18))] \\
 &= 154513.781 \text{ kPa}
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned}
 &= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s) \\
 &= 205762.875 \text{ kPa (Should be } \leq 184251.17)
 \end{aligned}$$

Note: For a given Channel thickness of: 190.000 mm

Min. Channel len. adjacent to the tubesheet is: 1265.132mm

Axial Channel Membrane Stress [σ_c, m]:

$$\begin{aligned}
 &= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t \\
 &= 2600^2 / (4 \cdot 190 (2600 + 190) \cdot \cos(0)) \cdot 18 \\
 &= 57385.398 \text{ kPa}
 \end{aligned}$$

Axial Channel Bending Stress [σ_c, b]:

$$\begin{aligned}
 &= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - \\
 &\quad 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + \\
 &\quad h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\
 &= 6 \cdot 131332640 / 190^2 \cdot [(0.41081E-04 \cdot 1 \cdot 0.0025 - 0.000 \cdot 1) \cdot 18 - \\
 &\quad 6((1 - 0.36) / 51878) \cdot 2537 / 515^3 \cdot (1 + \\
 &\quad 0.0025 \cdot 515 / 2 \cdot 1) \cdot (721903 + 2537^2 / 32 \cdot (14.3 - 18))] \\
 &= 185049.406 \text{ kPa}
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 &= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c) \\
 &= 242434.812 \text{ kPa (Should be } \leq 207286.09)
 \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$\begin{aligned}
 &= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)} \\
 &= 178285 \cdot \sqrt{(1.5 \cdot 122834 / 205763)} \\
 &= 168708.5 \text{ N/mm}^2
 \end{aligned}$$

Modified Elasticity [E_c]:

$$\begin{aligned}
 &= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)} \\
 &= 186264 \cdot \sqrt{(1.5 \cdot 138191 / 242435)} \\
 &= 172233.3 \text{ N/mm}^2
 \end{aligned}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 200 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 6 \cdot 720480 / (0.3 \cdot (515 - 5)^2)$$

$$= 55770.4 \text{ kPa (Should be } \leq 276381.5 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:
= h + Cats + Catc = 217 + 0 + 0 = 217 mm

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:
= max(HreqB; HreqS) = max(217 ; 99.7) = 217 mm

Tube Metal Area [At]:
= $\pi (d_o - t) t$
= $\pi (25.4 - 2.77) 2.77$
= 0.000 m²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 2.77 (25.4 - 2.77) 113991$
= 22440.715 N

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$
= 22440.7148 N

Weld Strength Factor [fw]:
= S / S_w where S_w = min(S and S_t)
= 113991/113991
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$
= 16195.8193 N

Minimum Length of Weld [ar]:
= $2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
= $2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$
= 3.371 mm

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm
Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 22441 is >= tube strength 22441 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	55770.4 <=	276381.5 kPa	Ok
Tubesheet shear stress	21397.9 <=	110552.6 kPa	Ok
Stress in Shell at Tubesheet	205762.9 <=	368800.8 kPa	Ok



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 201 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10,2026

Stress in Channel at Tubesheet 242434.8 <= 491193.2 kPa Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	216.621	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1256 + 25.4$$

$$= 2536.740 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2606 / 2537$$

$$= 1.0273$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2606 / 2537$$

$$= 1.0273$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 300 / 499$$

$$= 0.601 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 2.77 (186264 / 186264) \cdot (113991 / 138191) \cdot (0.6); 25.4 - 2 \cdot 2.77)$$

$$= 22.6536 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2)) }$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.7) / 32.4$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 202 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 0.3006$$

$$h/p = 15.716536; \mu^* = 0.300626; E^* = 52521 \text{ N/mm}^2$$

$$E^*/E = 0.281972; \nu^* = 0.353398$$

Note: As h/p (15.717) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2606 + 167)167} \right)$$

$$= 0.0027 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0027 \cdot 178285 \cdot 167^3 / (6(1 - 0.3^2))$$

$$= 406251040.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2606 + 167) \cdot 406251040 / 499^3 \cdot (1 + 499 \cdot 0.0027 + (499 \cdot 0.0027)^2 / 2)$$

$$= 175241.8 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2606^2 / (4 \cdot 178285 \cdot 167) \cdot (1 - 2606 \cdot 0.3 / (2(2606 + 167)))$$

$$= 0.48988\text{E-}04 \text{ mm/kPa}$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 406251040 \cdot 0.0027 \cdot 0.48988\text{E-}04 (1 + 499 \cdot 0.0027)$$

$$= 127406.9922 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2606 + 187)187} \right)$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0025 \cdot 186264 \cdot 187^3 / (6(1 - 0.3^2))$$

$$= 561126784.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2606 + 187 / \cos(0)) \cdot 561126784 / 499^3 \cdot (1 + 499 \cdot 0.0025 \cdot (1 + 1) / 2 + (499 \cdot 0.0025)^2 / 2 \cdot 1)$$

$$= 230285.3 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2606^2 / (4 \cdot 186264 \cdot 187) \cdot (1 - 2606 \cdot 0.3 / (2(2606 + 187)))$$

$$= 0.41924\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 203 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 1.03 \cdot 561126784 [0.41924E-04 (1 + 0.0025 \cdot 1 \cdot 499) - 0.000 (1/499 + 1 \cdot 0.0025/2)]$$

$$= 137080.1562 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980/2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.35) / 52521 \cdot (175242 + 230285 + 186264 \cdot \ln(1.17))$$

$$= 5.3618$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 2537^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 0 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 18]$$

$$= -406278.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -406278 + 137080 \cdot 18 - 127407 \cdot 0$$

$$= 2061307.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((2061307) - 2537^2 / 32 \cdot 5.36 (0 - 18)) / (1 + 5.36)$$

$$= 3374930.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 3374930 + 2537^2 / 64 \cdot (3 + 0.35) \cdot (0 - 18)$$

$$= -2694595.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 3374930 | ; | -2694595 |)$$

$$= 3374930.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 3374930 / (0.3 (499 - 0)^2)$$

$$= 270497.7188 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138191$$

$$= 276381.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 3376212 / (0.3 (494 - 0)^2)$$

$$= 275645.4688 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 204 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$\begin{aligned} &= h + \text{Cats} + \text{Catc} \\ &= 494 + 3 + 13 \\ &= 509.6600 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(\text{HreqB}; \text{HreqS}) \\ &= \max(510; 498) \\ &= 509.6600 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned} &= | \text{Ps} - \text{Pt} | \\ &= | 0 - 18 | \\ &= 18.001 \text{ N/mm}^2 \end{aligned}$$

Shear Stress check [τlimit]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\ &= 4 \cdot 0.2 \cdot 499 / 2537 \cdot \min(0.8 \cdot 138191; 0.533 \cdot 245585) \\ &= 17397.367 \text{ kPa} \end{aligned}$$

Note: As $F_s - P_t > \tau_{\text{limit}}$ & A_p, C_p are given, the more accurate formula for τ is used.

Average shear stress at the outer edge of perforated region [τ]:

$$\begin{aligned} &= (1 / (4 \cdot \mu)) \cdot (1 / h \{ (4 \cdot A_p) / C_p \}) \cdot | \text{Ps} - \text{Pt} | \\ &= (1 / (4 \cdot 0.2)) \cdot (1 / 499 \{ (4 \cdot 4.78) / 8067 \}) \cdot | 0 - 18 | \\ &= 106858.2969 \text{ kPa} \end{aligned}$$

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot A_p / C_p) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\ &= 1 / (4 \cdot 0.2 \cdot 110553) \cdot (4 \cdot 4.78 / 8067) \cdot | 0 - 18 | + 16 \\ &= 498.3251 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 167.000mm

Min. Shell len. adjacent to the tubesheet is: 1187.457 mm

Axial Shell Membrane Stress [σs,m]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 2606^2 / (4 \cdot 167 (2606 + 167)) \cdot 0 \\ &= 0.0000 \text{ kPa} \end{aligned}$$

Axial Shell Bending Stress [σs,b]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot \\ &\quad \text{Do} / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + \text{Do}^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 91325656 / 167^2 \cdot [0.0027 \cdot 0 + 6(1 - 0.35) / 52521 \cdot \\ &\quad 2537 / 499^3 \cdot (1 + 499 \cdot 0.0027 / 2) \cdot (3374930 + 2537^2 / 32 \cdot \\ &\quad (0 - 18))] \\ &= -53811.449 \text{ kPa} \end{aligned}$$

Shell Membrane + Bending Stress [σs]:

$$\begin{aligned} &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s) \\ &= 53811.449 \text{ kPa (Should be } \leq 184251.17) \end{aligned}$$

Note: For a given Channel thickness of: 187.000 mm

Min. Channel len. adjacent to the tubesheet is: 1256.552mm

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned} &= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t \\ &= 2606^2 / (4 \cdot 187 (2606 + 187) \cdot \cos(0)) \cdot 18 \\ &= 58512.516 \text{ kPa} \end{aligned}$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 205 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$\begin{aligned}
 & 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \Theta + h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t)) \\
 & = 6 \cdot 126141888 / 187^2 \cdot [(0.41924E-04 \cdot 1 \cdot 0.0025 - 0.000 \cdot 1) \cdot 18 - 6((1 - 0.35) / 52521) \cdot 2537 / 499^3 \cdot (1 + 0.0025 \cdot 499 / 2 \cdot 1) \cdot (3374930 + 2537^2 / 32 \cdot (0 - 18))] \\
 & = 240635.859 \text{ kPa}
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 & = |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c) \\
 & = 299148.406 \text{ kPa (Should be } \leq 207286.09)
 \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$\begin{aligned}
 & = E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)} \\
 & = 186264 \cdot \sqrt{(1.5 \cdot 138191 / 299148)} \\
 & = 155049.8 \text{ N/mm}^2
 \end{aligned}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$\begin{aligned}
 & = 6 \cdot M / (\mu^* (h - h'g)^2) \\
 & = 6 \cdot 3355157 / (0.3 (499 - 0)^2) \\
 & = 268912.9 \text{ kPa (Should be } \leq 276381.5)
 \end{aligned}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 494 + 3 + 13 = 510 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(510; 498) = 510 \text{ mm}$$

Tube Metal Area [A_t]:

$$\begin{aligned}
 & = \pi (d_o - t) t \\
 & = \pi (25.4 - 2.77) 2.77 \\
 & = 0.000 \text{ m}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned}
 & = \pi \cdot t (d_o - t) S \\
 & = \pi \cdot 2.77 (25.4 - 2.77) 113991 \\
 & = 22440.715 \text{ N}
 \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned}
 & = 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 & = 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991 \\
 & = 22440.7148 \text{ N}
 \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned}
 & = S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 & = 113991 / 113991 \\
 & = 1.000
 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 206 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 16195.8193 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\frac{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) f_w \cdot f_d}{0.75 \cdot do}} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{\frac{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1}{0.75 \cdot 25.4}} - 0.75 \cdot 25.4 \right)$$

$$= 3.371 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 22441 is $\geq$ tube strength 22441 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	268912.9 $\leq$	276381.5 kPa	Ok
Tubesheet shear stress	106858.3 $\leq$	110552.6 kPa	Ok
Stress in Shell at Tubesheet	53811.4 $\leq$	184251.2 kPa	Ok
Stress in Channel at Tubesheet	299148.4 $\leq$	491193.2 kPa	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	509.660	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1256 + 25.4$$

$$= 2536.740 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Ds / Do \text{ (Configurations a, b, c)}$$

$$= 2606 / 2537$$

$$= 1.0273$$

Ratio [pc]:

$$= Dc / Do \text{ (Configurations a, e, f)}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 207 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 2606/2537$$

$$= 1.0273$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 300/499$$

$$= 0.601 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 2.77 (186264/186264) \cdot (113991/138191) \cdot (0.6); 25.4 - 2 \cdot 2.77)$$

$$= 22.6536 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2)) }$$

$$= 32.3912 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 22.7) / 32.4$$

$$= 0.3006$$

$$h/p = 15.716536 ; \mu^* = 0.300626 ; E^* = 52521 \text{ N/mm}^2$$

$$E^*/E = 0.281972 ; \nu^* = 0.353398$$

Note: As h/p (15.717) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2606 + 167) 167) }$$

$$= 0.0027 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0027 \cdot 178285 \cdot 167^3 / (6(1 - 0.3^2))$$

$$= 406251040.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2606 + 167) \cdot 406251040/499^3 \cdot (1 + 499 \cdot 0.0027 + (499 \cdot 0.0027)^2 / 2)$$

$$= 175241.8 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2606^2 / (4 \cdot 178285 \cdot 167) \cdot (1 - 2606 \cdot 0.3 / (2(2606 + 167)))$$

$$= 0.48988\text{E-}04 \text{ mm/kPa}$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 406251040 \cdot 0.0027 \cdot 0.48988\text{E-}04 (1 + 499 \cdot 0.0027)$$

$$= 127406.9922 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2606 + 187) 187) }$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 208 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 0.0025 \cdot 186264 \cdot 187^3 / (6(1 - 0.3^2))$$

$$= 561126784.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2606 + 187 / \cos(0)) \cdot 561126784 / 499^3 \cdot (1 + 499 \cdot 0.0025 \cdot (1 + 1) / 2 + (499 \cdot 0.0025)^2 / 2 \cdot 1)$$

$$= 230285.3 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2606^2 / (4 \cdot 186264 \cdot 187) \cdot (1 - 2606 \cdot 0.3 / (2(2606 + 187)))$$

$$= 0.41924\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 561126784 [0.41924\text{E-}04 (1 + 0.0025 \cdot 1 \cdot 499) - 0.000 (1/499 + 1 \cdot 0.0025/2)]$$

$$= 137080.1562 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.35) / 52521 \cdot (175242 + 230285 + 186264 \cdot \ln(1.17))$$

$$= 5.3618$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot Pt]$$

$$= 2537^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 14.3 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 0]$$

$$= 322314.3125 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= 322314 + 137080 \cdot 0 - 127407 \cdot 14.3$$

$$= -1497163.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^* - D_o^2 / 32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= (-1497163) - 2537^2 / 32 \cdot 5.36 (14.3 - 0) / (1 + 5.36)$$

$$= -2655731.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + D_o^2 / 64 \cdot (3 + v^*) \cdot (Ps - Pt)$$

$$= -2655731 + 2537^2 / 64 \cdot (3 + 0.35) \cdot (14.3 - 0)$$

$$= 2159426.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 209 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|-2655731|; |2159426|)$$

$$= 2655731.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 2655731 / (0.3 \cdot (499 - 0)^2)$$

$$= 212854.5312 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138191$$

$$= 276381.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 2678022 / (0.31 \cdot (431 - 0)^2)$$

$$= 275314.6875 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 431 + 3 + 13$$

$$= 447.1308 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(447; 399)$$

$$= 447.1308 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= |P_s - P_t|$$

$$= |14.3 - 0|$$

$$= 14.281 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 499 / 2537 \cdot \min(0.8 \cdot 138191; 0.533 \cdot 245585)$$

$$= 17397.367 \text{ kPa}$$

| Ps - Pt | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110553) \cdot (4 \cdot 4.78 / 8067) \cdot |14.3 - 0| + 16$$

$$= 398.6447 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 167.000mm

Min. Shell len. adjacent to the tubesheet is: 1187.457 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2606^2 / (4 \cdot 167 (2606 + 167)) \cdot 14.3$$

$$= 52354.1016 \text{ kPa}$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 210 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$\begin{aligned} & Do/h^3 \cdot (1 + h \cdot \beta S/2) \cdot (Mp + Do^2/32 \cdot (Ps - Pt)) \\ &= 6 \cdot 91325656/167^2 \cdot [0.0027 \cdot 0.69 + 6(1 - 0.35)/52521 \cdot \\ & \quad 2537/499^3 \cdot (1 + 499 \cdot 0.0027/2) \cdot (-2655731 + 2537^2/32 \cdot \\ & \quad (14.3 - 0))] \\ &= 210776.453 \text{ kPa} \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned} &= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s) \\ &= 263130.562 \text{ kPa (Should be } \leq 184251.17) \end{aligned}$$

Note: For a given Channel thickness of: 187.000 mm

Min. Channel len. adjacent to the tubesheet is: 1256.552mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$\begin{aligned} &= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t \\ &= 2606^2 / (4 \cdot 187 (2606 + 187) \cdot \cos(0)) \cdot 0 \\ &= 0.000 \text{ kPa} \end{aligned}$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$\begin{aligned} &= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - \\ & \quad 6((1 - \nu^*) / E^*) \cdot Do/h^3 \cdot (C_m \theta + \\ & \quad h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))] \\ &= 6 \cdot 126141888/187^2 \cdot [(0.41924E-04 \cdot 1 \cdot 0.0025 - 0.000 \cdot 1) \cdot 0 - \\ & \quad 6((1 - 0.35)/52521) \cdot 2537/499^3 \cdot (1 + \\ & \quad 0.0025 \cdot 499/2 \cdot 1) \cdot (-2655731 + 2537^2/32 \cdot (14.3 - 0))] \\ &= -51060.109 \text{ kPa} \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned} &= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c) \\ &= 51060.109 \text{ kPa (Should be } \leq 207286.09) \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$\begin{aligned} &= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)} \\ &= 178285 \cdot \sqrt{(1.5 \cdot 122834/263131)} \\ &= 149188.3 \text{ N/mm}^2 \end{aligned}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h'g)^2) \\ &= 6 \cdot 2643071 / (0.3 (499 - 0)^2) \\ &= 211839.9 \text{ kPa (Should be } \leq 276381.5) \end{aligned}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 431 + 3 + 13 = 447 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(447; 399) = 447 \text{ mm}$$

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.77) 2.77 \\ &= 0.000 \text{ m}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 211 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10,2026

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 113991$$

$$= 22440.715 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 22440.7148 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 113991/113991$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 16195.8193 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.371 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 22441 is $\geq$ tube strength 22441 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	211839.9 <=	276381.5 kPa	Ok
Tubesheet shear stress	84774.3 <=	110552.6 kPa	Ok
Stress in Shell at Tubesheet	263130.6 <=	368800.8 kPa	Ok
Stress in Channel at Tubesheet	51060.1 <=	207286.1 kPa	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	447.131	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 212 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1256 + 25.4 \\ &= 2536.740 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2606/2537 \\ &= 1.0273 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2606/2537 \\ &= 1.0273 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 300/499 \\ &= 0.601 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t(E_t/E) \cdot (S_t/S) \cdot (\rho); dt - 2t) \\ &= \max(25.4 - 2 \cdot 2.77 (186264/186264) \cdot (113991/138191) \cdot (0.6); 25.4 - 2 \cdot 2.77) \\ &= 22.6536 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2537 \cdot 31.8) / (\pi \cdot 2537^2)) } \\ &= 32.3912 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.4 - 22.7) / 32.4 \\ &= 0.3006 \end{aligned}$$

$h/p = 15.716536$; $\mu^* = 0.300626$; $E^* = 52521 \text{ N/mm}^2$

$E^*/E = 0.281972$; $v^* = 0.353398$

Note: As h/p (15.717) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2606 + 167) 167) } \\ &= 0.0027 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.0027 \cdot 178285 \cdot 167^3 / (6(1 - 0.3^2)) \\ &= 406251040.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 213 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 6(2606 + 167) \cdot 406251040/499^3 \cdot (1 + 499 \cdot 0.0027 + (499 \cdot 0.0027)^2/2)$$

$$= 175241.8 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2606^2 / (4 \cdot 178285 \cdot 167) \cdot (1 - 2606 \cdot 0.3 / (2(2606 + 167)))$$

$$= 0.48988\text{E-}04 \text{ mm/kPa}$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 406251040 \cdot 0.0027 \cdot 0.48988\text{E-}04 (1 + 499 \cdot 0.0027)$$

$$= 127406.9922 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(2606 + 187)187})$$

$$= 0.0025 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.0025 \cdot 186264 \cdot 187^3 / (6(1 - 0.3^2))$$

$$= 561126784.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$$

$$= 6(2606 + 187 / \cos(0)) \cdot 561126784/499^3 \cdot (1 + 499 \cdot 0.0025 \cdot (1 + 1)/2 + (499 \cdot 0.0025)^2/2 \cdot 1)$$

$$= 230285.3 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2606^2 / (4 \cdot 186264 \cdot 187) \cdot (1 - 2606 \cdot 0.3 / (2(2606 + 187)))$$

$$= 0.41924\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 561126784 [0.41924\text{E-}04 (1 + 0.0025 \cdot 1 \cdot 499) - 0.000 (1/499 + 1 \cdot 0.0025/2)]$$

$$= 137080.1562 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980/2537$$

$$= 1.1747$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.35) / 52521 \cdot (175242 + 230285 + 186264 \cdot \ln(1.17))$$

$$= 5.3618$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 214 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 2537^2/16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 14.3 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 18]$$

$$= -83964.1797 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -83964 + 137080 \cdot 18 - 127407 \cdot 14.3$$

$$= 564144.1250 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((564144) - 2537^2/32 \cdot 5.36 (14.3 - 18)) / (1 + 5.36)$$

$$= 719199.3750 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$$

$$= 719199 + 2537^2/64 \cdot (3 + 0.35) \cdot (14.3 - 18)$$

$$= -535168.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 719199 | ; | -535169 |)$$

$$= 719199.3750 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 719199 / (0.3 (499 - 0)^2)$$

$$= 57643.2031 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138191$$

$$= 276381.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 735895 / (0.36 (212 - 0)^2)$$

$$= 276369.0625 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 212 + 3 + 13$$

$$= 227.5787 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(228 ; 116)$$

$$= 227.5787 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 14.3 - 18 |$$

$$= 3.720 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.2 \cdot 499/2537 \cdot \min(0.8 \cdot 138191 ; 0.533 \cdot 245585)$$

$$= 17397.367 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 215 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110553) \cdot (4 \cdot 4.78 / 8067) \cdot |14.3 - 18| + 16$$

$$= 115.6805 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 167.000mm

Min. Shell len. adjacent to the tubesheet is: 1187.457 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2606^2 / (4 \cdot 167 (2606 + 167)) \cdot 14.3$$

$$= 52354.1016 \text{ kPa}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 91325656 / 167^2 \cdot [0.0027 \cdot 0.69 + 6(1 - 0.35) / 52521 \cdot 2537 / 499^3 \cdot (1 + 499 \cdot 0.0027 / 2) \cdot (719199 + 2537^2 / 32 \cdot (14.3 - 18))]$$

$$= 156965.031 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 209319.141 \text{ kPa (Should be } \leq 184251.17)$$

Note: For a given Channel thickness of: 187.000 mm

Min. Channel len. adjacent to the tubesheet is: 1256.552mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2606^2 / (4 \cdot 187 (2606 + 187) \cdot \cos(0)) \cdot 18$$

$$= 58512.516 \text{ kPa}$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 126141888 / 187^2 \cdot [(0.41924E-04 \cdot 1 \cdot 0.0025 - 0.000 \cdot 1) \cdot 18 - 6((1 - 0.35) / 52521) \cdot 2537 / 499^3 \cdot (1 + 0.0025 \cdot 499 / 2 \cdot 1) \cdot (719199 + 2537^2 / 32 \cdot (14.3 - 18))]$$

$$= 189575.750 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 248088.281 \text{ kPa (Should be } \leq 207286.09)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{ (1.5 \cdot S_s / \sigma_s) }$$

$$= 178285 \cdot \sqrt{ (1.5 \cdot 122834 / 209319) }$$

$$= 167269.2 \text{ N/mm}^2$$

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{ (1.5 \cdot S_c / \sigma_c) }$$

$$= 186264 \cdot \sqrt{ (1.5 \cdot 138191 / 248088) }$$



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 216 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

$$= 170259.5 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 717378 / (0.3 \cdot (499 - 0)^2)$$

$$= 57497.2 \text{ kPa (Should be } \leq 276381.5 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + Cats + Catc = 212 + 3 + 13 = 228 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) = \max(228; 116) = 228 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 113991$$

$$= 22440.715 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 22440.7148 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 113991 / 113991$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 16195.8193 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do}$$

$$= 2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) \cdot 1 \cdot 1) - 0.75 \cdot 25.4}$$

$$= 3.371 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6853 mm

Minimum Required Groove Weld Leg [agr]: 1.6853 mm

Tube-Tubesheet Jt allowable, 22441 is $\geq$ tube strength 22441 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 217 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10,2026

Tubesheet bend. stress	57497.2	<=	276381.5 kPa	Ok
Tubesheet shear stress	22084.0	<=	110552.6 kPa	Ok
Stress in Shell at Tubesheet	209319.1	<=	368800.8 kPa	Ok
Stress in Channel at Tubesheet	248088.3	<=	491193.2 kPa	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	227.579	515.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.685	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.685	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	500.004 ...	260812	276381	103538 110553	Ps+Pt D1	Ok
D2uc	437.772 ...	205390	276381	82141 110553	Ps+Pt D2	Ok
D3uc	216.621 ...	55770	276381	21398 110553	Ps+Pt D3	Ok
D1c	509.660 ...	268913	276381	106858 110553	Ps+Pt-c D1	Ok
D2c	447.131 ...	211840	276381	84774 110553	Ps+Pt-c D2	Ok
D3c	227.579 ...	57497	276381	22084 110553	Ps+Pt-c D3	Ok
Max:	509.6600 ... mm		0.973	0.967 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	48038	184251	288278	491193	Ok
D2uc	...	253801	368801	45844	207286	Ok
D3uc	...	205763	368801	242435	491193	Ok
D1c	...	53811	184251	299148	491193	Ok
D2c	...	263131	368801	51060	207286	Ok
D3c	...	209319	368801	248088	491193	Ok
Max	...		0.713	0.609		

Note:

Because there was no net pressure differential across the tubesheet due to
the specified pressures psd,min and ptd,min; load case D4 was not processed as
the resulting tubesheet bending stress would be zero.



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 218 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10,2026

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	509.660	515.000	Ok
Tube thickness :	1.886	2.769	Ok
Tube-tubesheet fillet weld leg :	1.685	3.000	Ok
Tube-tubesheet groove weld leg :	1.685	3.000	Ok

Given Shell Thickness: 170.0000 mm

Given Channel Thickness: 190.0000 mm

Min Shell length of thk, (170.000) adj. to tubesheet: 1196.695 mm

Min Channel length of thk, (190.000) adj. to tubesheet: 1265.132 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18.392	1.000	18.543	1.000
Tubesheet Shear Stress	18.623	1.000	18.543	0.996
Shell Stress (Axial, Junction)	18.623	1.013	18.543	1.000
Tubesheet-Channel Junction	18.623	1.013	18.543	1.000
Tube Pressure Stress	27.230	1.000	24.196	1.000
Minimum MAWP	18.392		18.543	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	20.129	1.000	20.294	1.000
Tubesheet Shear Stress	20.583	1.000	20.583	1.000
Shell Stress (Axial, Junction)	20.583	1.023	20.583	1.014
Tubesheet-Channel Junction	20.583	1.023	20.583	1.014
Tube Pressure Stress	28.185	1.000	26.812	1.000
Minimum MAPnc	20.129		20.294	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(515/4, \min(515, 170)) \\
 &= 170.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(14.3/20.3, 1) \\
 &= 0.704
 \end{aligned}$$

Warning:

The minimum metal temperature could not be computed because the



PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 219 of 220

ASME TS Calc: DESIGN CASE

Case: 1 4:24p Apr 10, 2026

*governing thickness for this component was greater than 100.000 mm.
This component must be impact tested.*

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(515/4, min(515, 190))
= 190.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(18/20.1, 1)
= 0.894

*Warning:
The minimum metal temperature could not be computed because the
governing thickness for this component was greater than 100.000 mm.
This component must be impact tested.*

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PV Elite 28 Licensee: SL User

FileName : 123-C2-NVS

Page 220 of 220

Problems/Failures Summary:

Step: 31 4:24pm Apr 10, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Insufficient BasePlate bolting area!**

**** Error: Baseplate Bolt Shear Stress Failure!**

Please review all reports carefully!

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